GE VERNOVA

ADMS

16.25.1

Viewer Configuration Editor User Guide

Copyright

Change History

Date	Change Description
May 31, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Oct 26, 2023	ADMS 16.2.0: • Updated Visibility.
Feb 22, 2024	ADMS 16.2.0: • Updated Feeder Line Halos.
Mar 20, 2024	ADMS 16.5.0: • Updated Traced Line.
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May 20, 2024	ADMS 16.8.0: • Updated Structures.
Jun 10, 2024	ADMS 16.9.0: • Updated Toolbar.
Aug 14, 2024	ADMS 16.11.0: • Updated Structures.
Feb 28, 2025	ADMS 16.17.0: • Updated Visibility.

Contents

1. Introduction **5**
1.1. Note about Non-Linear Zooming 5
2. General Usage **6**
2.1. Starting the Viewer Configuration Editor 6
2.2. After Starting the Application 6
2.3. Applying Changes to the ADMS Client 7
2.4. Configuring Controls for Different Client Roles 7
3. General Properties **9**
3.1. General Settings 10
3.2. GeoContainer Schematics 13
3.3. Font Parameters 15
3.4. Miscellaneous Settings 15
3.5. Origins of Label Placement 16
3.6. Print Options 17
3.7. Wildcard Search 17
3.8. DNAF Functions 23
3.9. Client Roles 23
3.10. Visibility 24
3.11. Title Options 25
3.12. Externally Owned Device 26
3.13. Externally Owned DLL 26
3.14. TOP/BOP/NOP Colors 26
4. Lines **28**
4.1. Feeder Line Halos 28
4.1.1. Configuring Line Halos 32
4.1.2. Visualizing Line Halos 34
4.2. Feeder Color by Index 36
4.3. Feeder Voltage Color 36
4.4. Underground Loop Colors 37
4.5. Phase Colors 37
4.6. Feeder Line Styles 38
4.7. De-Energized and Abnormal Lines 39
5. Function Keys **42**
5.1. Key and Mouse Functions 42
5.2. Changing the Scroll Feature 44
6. Bitmaps **45**
6.1. Bitmap Definition 47
6.1.1. Annotations 49
6.1.2. Safety Documents 49
6.1.3. Tags 52

6.1.4. Structures ... 53
6.2. Adding New Object Types to the Bitmap Tab ... 54
6.3. Mapping New Object Types with Bitmaps ... 55
6.4. Bitmap Folder Location ... 56
6.4.1. Changing the Default Bitmap Folder Location ... 56
7. Object Properties ... 59
7.1. Label Colors ... 59
7.2. Label Offsets and Sizing ... 60
7.3. Temporary Switch and Jumper State ... 61
7.4. Assessment Location Visibility ... 62
7.5. Object/Labels Visibility Grid ... 62
8. Toolbar ... 64
8.1. Toolbar Options ... 65
8.2. Adding a New Object to the Geographic_Openspace (Common) Toolbar ... 69
9. Right-Click Menu ... 71
9.1. Menu Items ... 72
9.2. Adding Single-Click Tag Options to the Switch Right-Click Menu ... 73
10. Attributes ... 76
10.1. Attribute Properties ... 77
11. Geographic Background ... 82
11.1. Adding a New Background Layer to the Background Layers List ... 85
11.2. Adding a New Background Layer Label to the Geographic_Openspace (Common) Toolbar ... 87
12. OMS Symbology ... 88
12.1. Decorations ... 88
12.2. Customer Symbology ... 89
12.2.1. Layers ... 89
12.2.2. Background Shapes ... 89
12.2.3. Preview ... 90
13. Check Before Operate ... 92
14. Traced Line ... 96
15. Structures ... 98
15.1. Interaction of Structure Label Justification and Rotation ... 100
16. Command String Definition ... 102
16.1. Basic Commands ... 102
16.1.1. OMS Commands ... 110
16.2. Basic Parameters ... 116
16.3. Environment Fields ... 122
16.4. DNOM Query Fields ... 127
16.5. Find Popup Parameters ... 127

1. Introduction

The Viewer Configuration Editor is a support tool provided with ADMS. This tool allows an administrator to configure the appearance and effects of geographic network displays in order to meet the needs of individual utilities.

Some of the aspects that can be configured using the Editor include:

- Appearance of overhead feeders/phases and underground feeders/phases (for example, line color, width, violation halo color, etc.)
- Function keys (association of a keyboard shortcut with a command)
- Appearance of the geographic background (for example, water, roads, boundaries, etc.)
- Appearance of the Geographic_OpenSpace (common) toolbar and device toolbars (show, hide, add buttons, etc.)
- General parameters (default zoom level, font size, etc.)
- Context menu functionality (object right-click menu options)

The Viewer Configuration Editor provides considerable flexibility for designing the way the network view appears to users.

The output of the Viewer Configuration Editor is an XML file (`viewerconfig.xml`). This is a global file that is managed locally and then distributed to all clients, which ensures that they see the same view of the network.

1.1. Note about Non-Linear Zooming

The default zoom level for the geographic network view display is 1.00.

Zooming in on a specific part of the network shows less of the electrical network but with greater detail. Zooming in involves increasing the zoom level, or zoom factor, to a number greater than 1.

Zooming out shows a greater area of the electrical network with potentially less detail (or potentially more clutter). Zooming out involves decreasing the zoom level, or zoom factor, to a number less than1 (but greater than 0).

ADMS leverages the concept of non-linear zooming to facilitate an improved user interface. The Viewer Configuration Editor allows an administrator to define a minimum zoom level for which particular objects are not visible. This allows the objects on the display to be de-cluttered.

In addition, the administrator can define a maximum scaling level to limit the growth of particular objects for cases when the user needs to tightly zoom in. For example, this can prevent the user's screen from becoming filled with an image of a giant switch.

2. General Usage

2.1. Starting the Viewer Configuration Editor

To start the Viewer Configuration Editor: from the Windows Start menu, navigate to **All Programs> Eterra> Distribution> Tools> ConfigurationTools> Viewer Configuration Editor**.

2.2. After Starting the Application

When it starts, the Viewer Configuration Editor opens the default configuration file `viewerconfig.xml`. The default file name is defined in the shortcut that starts the Viewer Configuration Editor. If you need to open another configuration file during startup, modify the file name accordingly.

The File menu and the toolbar include the following commands:

- **New:** Creates a new, empty viewer configuration file.
- **Open:** Opens an alternate or existing configuration file (usually found in the `D:\Eterra\Distribution\FantasyIsland\config\client folder`).
- **Read Global File and Mapper File:** Opens the `Global.csv` file (usually located at `D:\Eterra\Distribution\FantasyIsland\ToolsWorkspace\Converter\input`).

  This process retrieves the switch subtype names that are referenced on the Bitmaps tab. The switch subtype names are saved in the `viewerconfig.xml` file.

  This process is only necessary when the switch subtype names have changed in the `Global.csv` file.
- **Save:** Saves changes to the currently open configuration file.
- **Save As:** Saves all changes to another configuration file.

  The Viewer Configuration Editor allows you to save the configuration file under different names. However, when it starts, it automatically opens the default configuration file defined in the Viewer Configuration Editor's shortcut.
- **Exit:** Closes the application.

The Tools menu includes the following command:

- **Convert BMP Images to PNG:** Opens a Browse for Folder window to select a folder with images where the Viewer Configuration Editor then creates a PNG image for each BMP image that does not have a PNG file with the same name.

  > **Note**
  >
  > The Viewer Configuration Editor only creates PNG images for BMP images that are used on

the Bitmaps tab.

2.3. Applying Changes to the ADMS Client

After changing configurations using the Editor, the configuration file must be saved.

It is important to restart the ADMS client for the changes to take effect, since the client reads the `viewerconfig.xml` file only on startup.

> **Important**
>
> You can load a new `viewerconfig.xml` configuration file without restarting the client using the NETVIEW://RT/UPDATEVC command. This is particularly useful for the testing purposes.

By default, the client uses the `viewerconfig.xml` file located at `D:\Eterra\Distribution\FantasyIsland\config\client\`. To apply another configuration file to the local ADMS client, the new XML file's name and path must be specified in the client's dynamics file (refer to the *Net Dynamics Configuration Editor User Guide*).

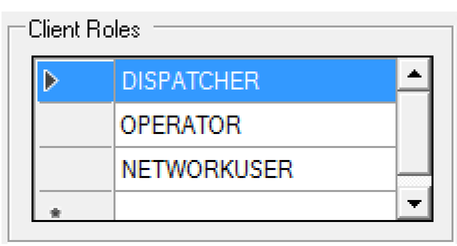
For guidance about distributing the updated `viewerconfig.xml` file to all clients, refer to the *File Updater Software Installation and Maintenance Guide.*

2.4. Configuring Controls for Different Client Roles

The Viewer Configuration Editor can define different toolbars, context menus, and attribute panels for different user types. This allows different controls and data panels to be used for operators, dispatchers, analysts, and users of Outage Management System and Distribution Network Analysis Functions.

To customize controls and data panels for specific users:

1. On the General tab, define available client roles using the Client Roles list.

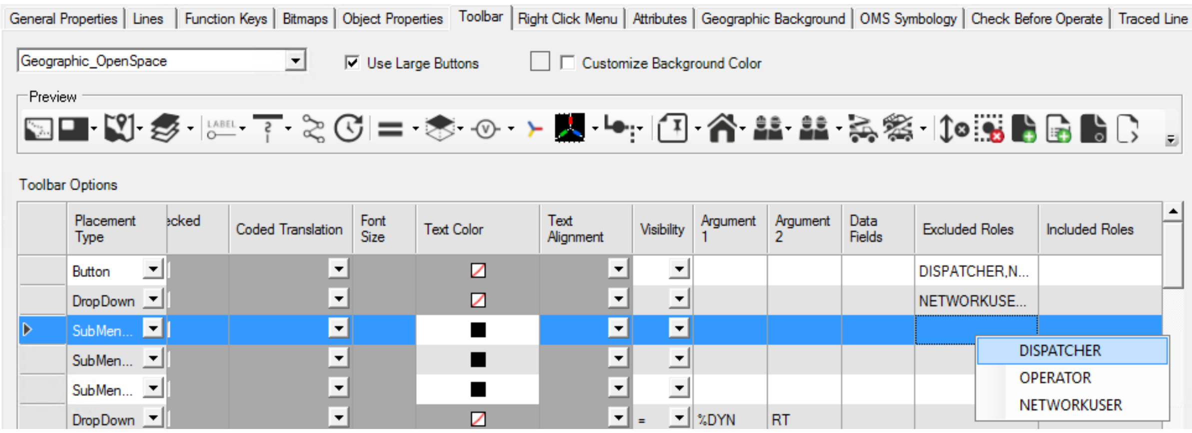
   

   > **Caution**
   >
   > Role names in the Viewer Configuration Editor and Grid Tabular Builder must be consistent.

2. On the Toolbar, Right-Click Menu, and Attributes tabs, specify the roles for which menu items will not be shown in the Excluded Roles field. The items will be shown for all other roles.

3. On the Toolbar, Right-Click Menu, and Attributes tabs, specify the roles for which menu items will be shown in the Included Roles field. The items will not be shown for all other roles.

> **Note**
>
> You can specify either included or excluded roles for an item.



4. Configure the ADMS client shortcut to pass the CLIENTTYPE "<ClientRole>" parameter during startup. For example:

```
D:\Eterra\browser\WebFGViewer.exe " /NETCONF "%IDMS_ROOT%\config\client\netconf_localhost.xml"
-EMPSCADA -CLIENTTYPE "DISPATCHER"
```

5. Start the ADMS client.

   The client shows controls and data panels according to the client role configurations.

3. General Properties

The General Properties tab is used to configure various properties on the geographic network view.

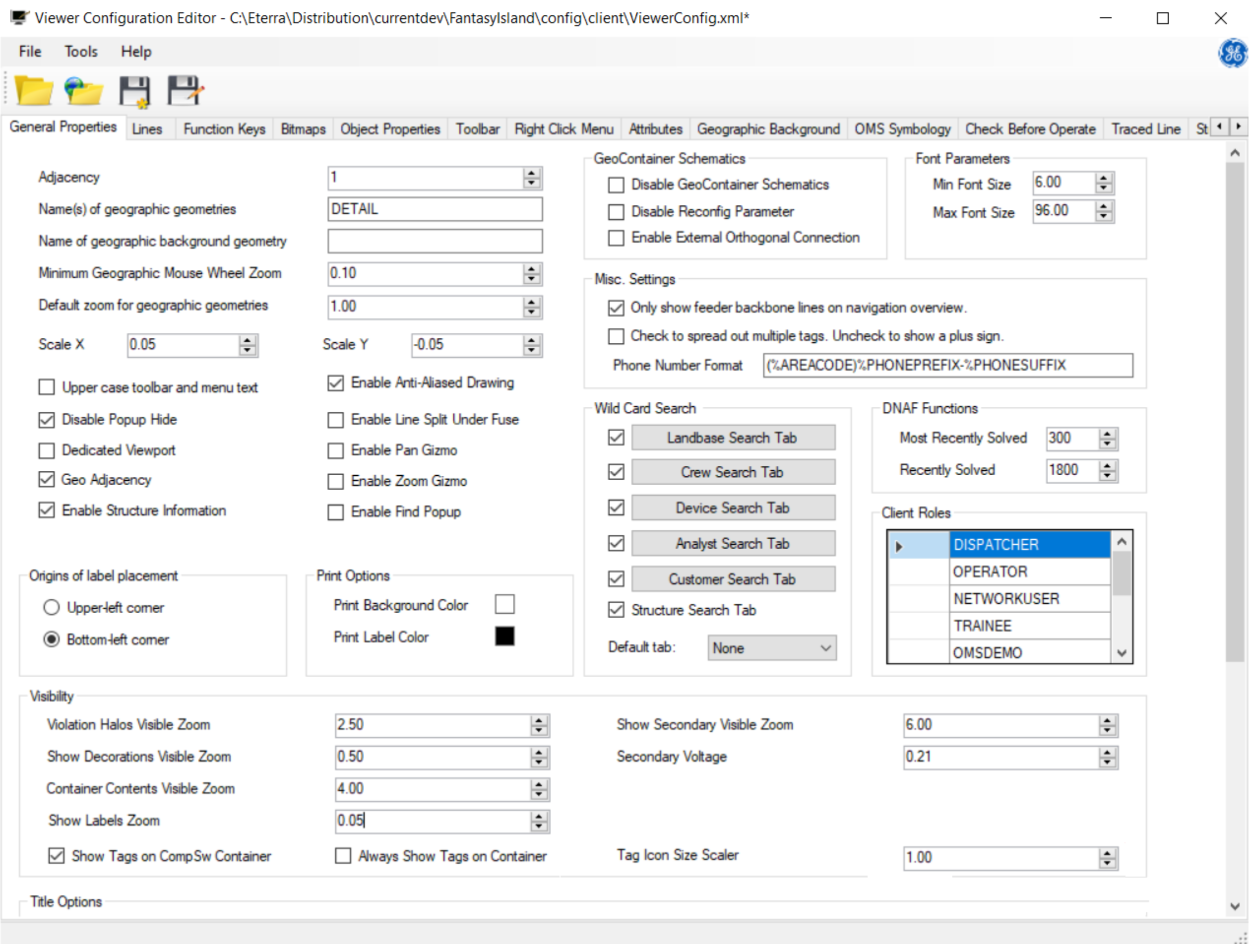
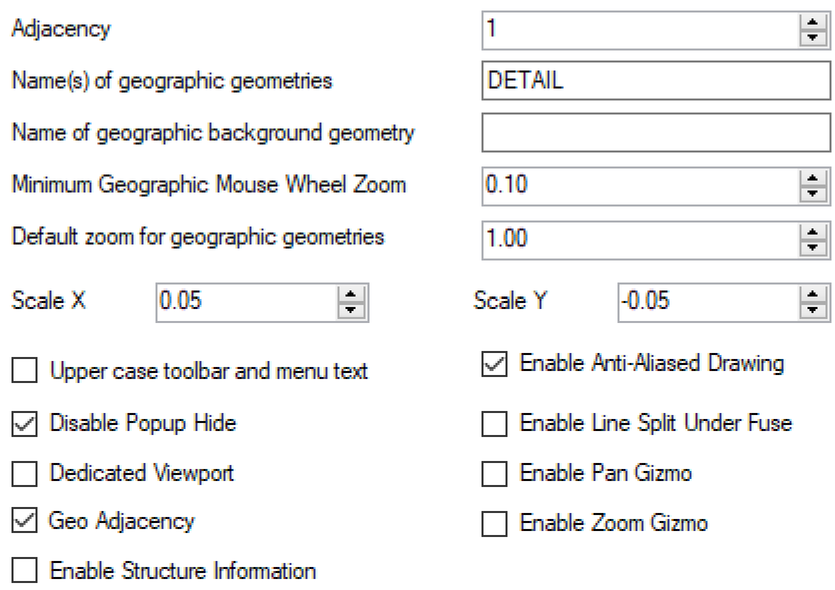


Figure 1. General Properties Tab

3.1. General Settings



The General Settings group includes the following options:

- **Adjacency:** When calling up the display of an electrical substation, it is possible to show the substations that are electrically connected to it.
    - To show only the selected substation when calling up its corresponding display, use an adjacency of 0.
    - At adjacency level 1, the display shows the selected substation, all substations connected by nominally open switches, and immediate parent and child stations. Additionally, Navigation builder traverses through all parent and child stations of the main station, adding their corresponding parent and child stations until no further child or parent station appears, providing a comprehensive view of directly connected substations.
    - To show substations that are indirectly connected to the substation for which the corresponding display has been called up, use an adjacency of 2 or higher.

  The higher the adjacency, the more adjacent substations are shown when calling up a substation display (note that higher numbers negatively affect display call-up performance).

- **Name(s) of Geographic Geometries:** This field sets the name of the geographic geometry that is used in the display call-up URLs. The default name is DETAIL. For example:

  ```
  NETVIEW://RT/VIEW/DETAIL/RAVEN
  ```

  This command calls up the real-time geographic network view centered on the RAVEN station. Another common setting used by customers is "Geographic".

- **Name of the Geographic Background Geometry:** Specifies the geometry where the geographic background should be displayed (when different from the "geographic background").

When this field is empty, the geographic background is displayed on the geographic geometry.

- **Minimum Geographic Mouse Wheel Zoom:** Specifies the minimum zoom level that can be set with the mouse wheel.

  This parameter is used to prevent zooming out with the mouse wheel to a level where the center point is lost.

- **Default Zoom for Geographic Geometries:** Specifies the default zoom level for the geographic network view display.
- **Scale X / Scale Y:** These properties allow you to change the coefficients used for the conversion of the world coordinates into pixel coordinates. It is recommended that you keep the default values.

  These values set the number of pixels per GIS unit at zoom-level 1.0. A value of 0.05 indicates that 20 GIS units (typically in feet) are equal to a pixel at a nominal zoom level.

  A negative value in Scale Y flips the coordinate direction relative to GIS so that as the actual up direction coordinate value gets larger, the up direction coordinate value gets smaller.

  > **Note**
  >
  > Scale values should be configured at the beginning of a project. Changing these values later requires evaluating and changing all sizes and zoom level settings for 30 or 40 different object types and labels.

- **Upper Case Toolbar and Menu Text:** If this option is selected, all toolbar and menu text is displayed in uppercase. If not selected, the text is displayed as it is input in the VCE.
- **Disable Pop-Up Hide:** Use this checkbox to enable or disable the standard Windows Minimize button for ADMS pop-up displays.

  Selecting this checkbox prevents pop-up displays from being minimized.

- **Dedicated Viewport:** Controls how new viewports are opened from the Navigation Overview display.

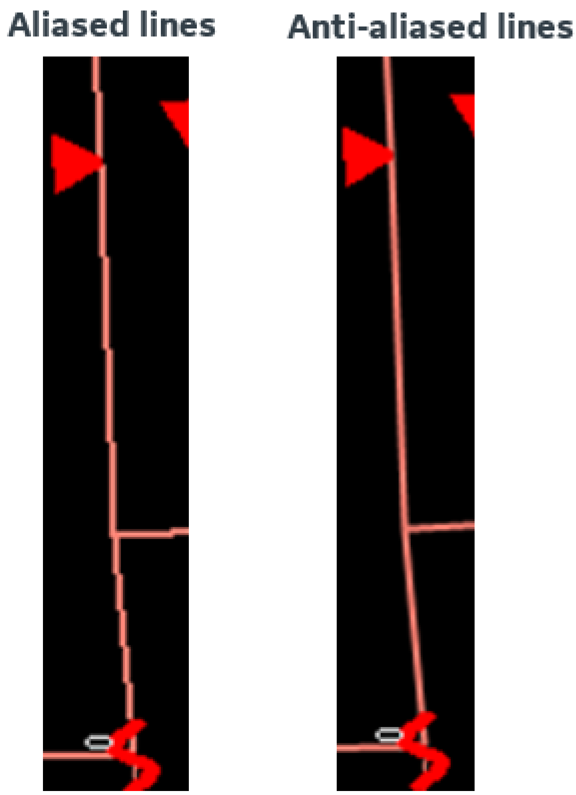
  When the Dedicated Viewport option is not selected, each time an area on the Navigation Overview display is selected, the relevant geographic network view is opened in a new viewport.

  When the Dedicated Viewport option is selected, a new viewport opens only the first time when an area on the Navigation Overview display is selected. All subsequent clicks update the existing geographic network view in the existing viewport.

- **Geo Adjacency:** Allows enabling or disabling the Geographic Adjacency feature by clicking on the Navigation Overview. This feature is disabled when this option is not selected. The Geo Adjacent feature can also be disabled on the Navigation Overview by using the `NOGEOADJACENCY` parameter in a NETVIEW command.
- **Enable Structure Information:** Allows enabling and disabling the Structure Information

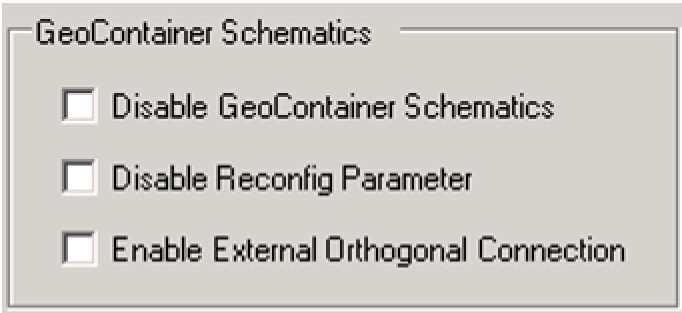
feature. This feature is disabled when this option is not selected. When enabled, users will have access to request Structure Information for poles and a wide range of equipment. It also enables a new halo option that can be used to show equipment that shares a structure with equipment from a different feeder.

- **Enable Anti-Aliased Drawing:** If this option is selected, all lines are drawn anti-aliased.



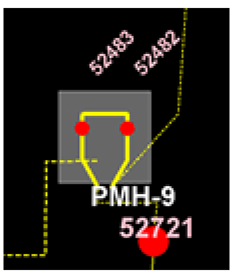
- **Enable Line Split Under Fuse:** Select this option to place a visible gap in the line behind fuse-switch symbols. This improves the clarity of the display.
- **Enable Pan Gizmo:** Allows the Pan Gizmo to appear on geographic network displays to enable navigating with mouse clicks.
- **Enable Zoom Gizmo:** Allows the Zoom Gizmo to appear on geographic network displays to enable zoom controlling with mouse clicks.

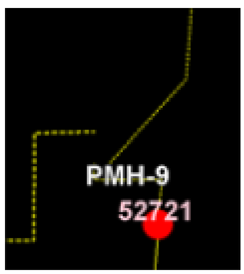
3.2. GeoContainer Schematics



The GeoContainer Schematics group includes the following options:

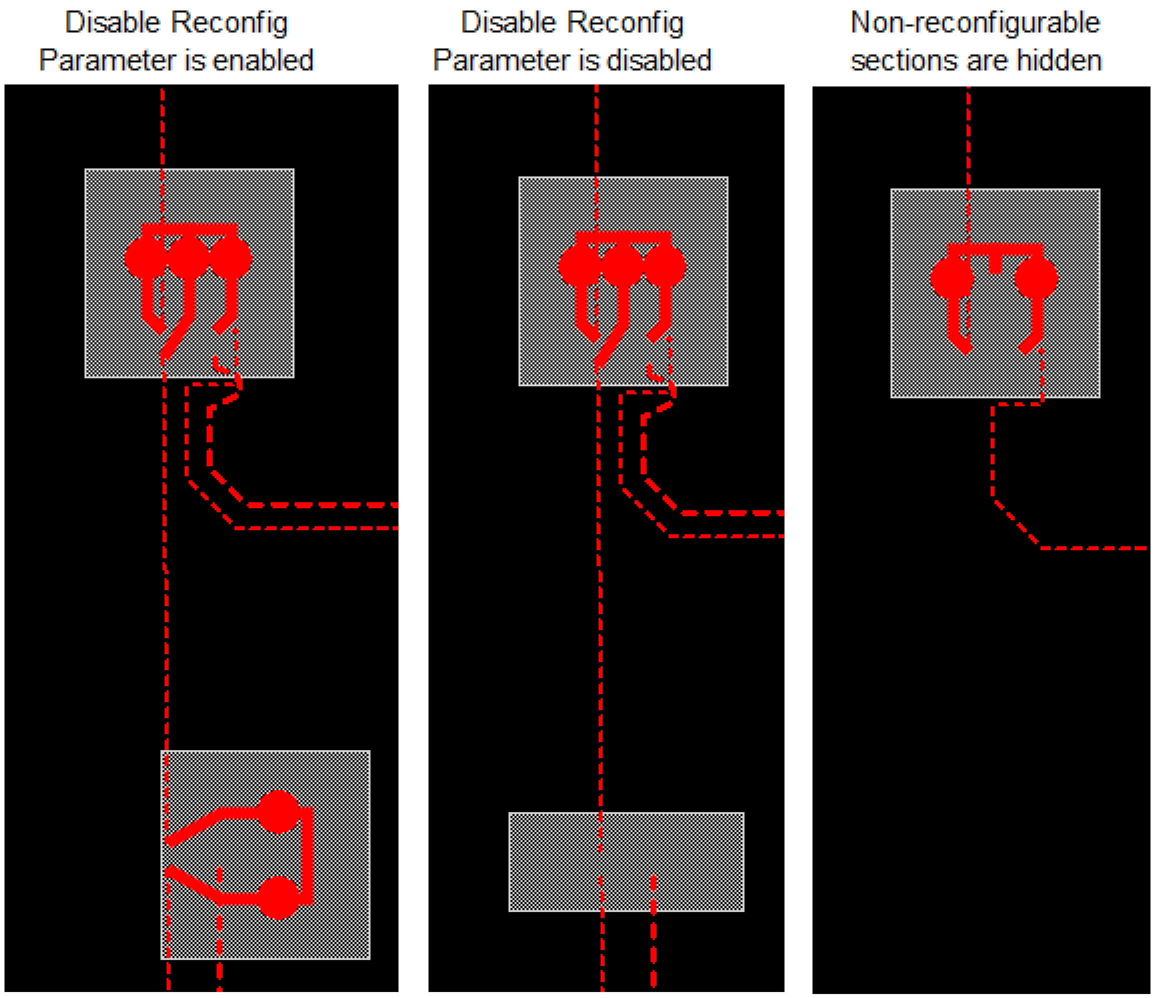
- **Disable GeoContainer Schematics:** Select this option to hide the container placements on the geographic view.

- **Disable Reconfig Parameter:** This option configures the display of non-reconfigurable switches when a user zooms in and the switch cabinets explode to show a schematic rendering of the contents.

  If the Disable Reconfig Parameter option is enabled, all switches are displayed on exploded cabinet views. If the option is disabled, only reconfigurable switches are displayed.



> **Note**
>
> "Reconfigurable sections" is a display concept that is used by operators to display only those devices and lines that are capable of changing the configuration of the feeder main (backbone). To define "reconfigurable" devices, the algorithm closes all normally open switches in the system to see where loop conditions arise. All the devices and conductors that are inline of the loop are deemed to be reconfigurable. All devices beyond the loop, such as dead-end and lateral lines and devices, are deemed to be non-reconfigurable. Displaying non-reconfigurable sections on the geographic view is configured using the "Show/Hide Non-Reconfigurable Sections" button on the geographic (common) toolbar.

> **Note**
>
> Fuses (reconfigurable or non-reconfigurable) are never displayed on exploded cabinet views.

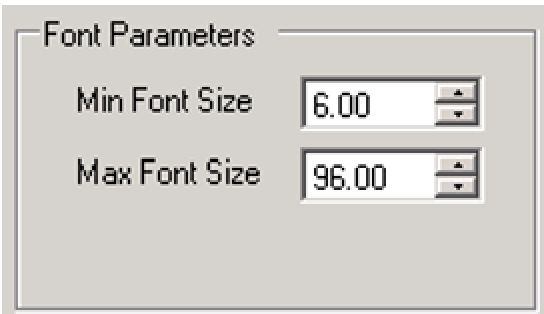
- **Enable External Orthogonal Connection:** For vaults or cabinets that contain a multitude of "reconfigurable" switches, the GeoContainer schematics can become quite cluttered.

Selecting this option tells the software to only use orthogonal lines within the boundaries of the vault or cabinet before connecting to the external lines.

With this option not selected, the software uses slanted lines to allow for a more spread-out layout.

It is recommended that a set of complicated vaults be tested against the options to determine which layout is best for the majority of vaults within the model.

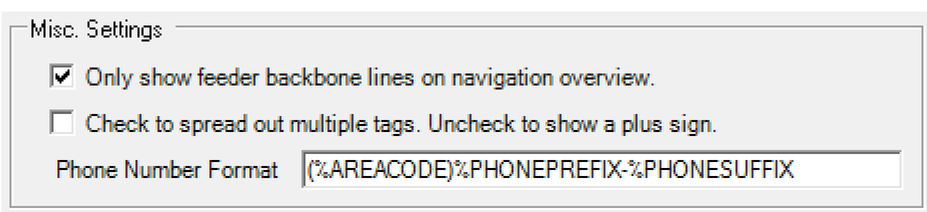
3.3. Font Parameters



The Font Parameters group includes the following options:

- **Min Font Size:** This is the minimum font size for all labels. In other words, when zooming out, the font never gets smaller than what is set in this field.
- **Max Font Size:** This is the maximum font size for all labels. In other words, when zooming in, the font never gets larger than what is set in this field.

3.4. Miscellaneous Settings



The Miscellaneous Settings group includes the following options:

- **Only show feeder backbone lines on navigation overview**

  The navigation overview (F6 or Ctrl+N) normally shows the network laterals if they are currently being displayed on the geographic network view.

  Selecting this option suppresses the display of laterals, even if the laterals are being displayed on the associated geographic network view.

This option is recommended to improve UI performance. The display can become too cluttered in dense areas, which can make the overview difficult to use. (If the utility services primarily rural areas, then it might be useful to deselect this option.)

- **Check to spread out multiple tags. Uncheck to show a plus sign**

  In cases where multiple tags are applied to a single device, the default display mode is to show a single tag symbol with a plus sign ("+"), indicating that multiple tags have been applied. The color of the tag symbol is set to the highest-priority tag that is currently applied.

  Selecting this option causes each of the different types of tags that are currently applied to be shown with separate symbols (below image).

- **Phone Number Format**

  Specifies the customer phone number format. The format can include the AREACODE, PHONENUMBER, PHONEPREFIX, and PHONESUFFIX query fields. The default format is (%AREACODE)%PHONEPREFIX-%PHONESUFFIX. For example, (555)589-0773. The formatted phone number can be queried using the AREACODEPHONENUMBER query field.

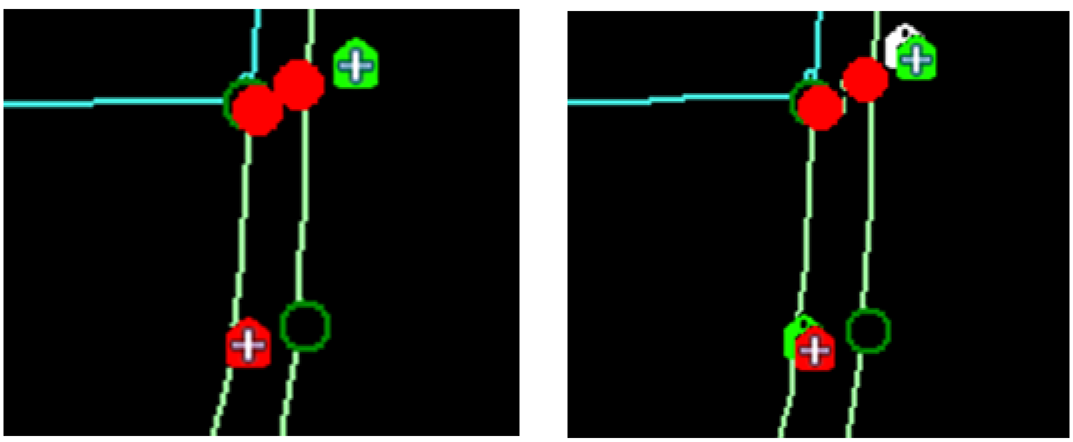
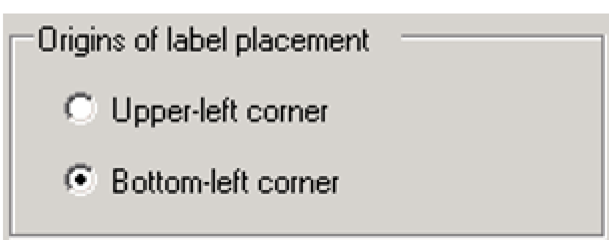


Figure 2. Standard Tag Display and Spread-Out Tags

3.5. Origins of Label Placement



Normally, the labels extend out to the right (if label rotations do not come into play). The label placement options allow you to choose between anchoring a label to the device by the upper left corner or the lower left corner.

- **Upper Left Corner:** Anchors the upper left corner of the label to the device. This places the label down and to the right of the device.
- **Bottom Left Corner:** Anchors the bottom left corner of the label to the device. This places the label up and to the right of the device.

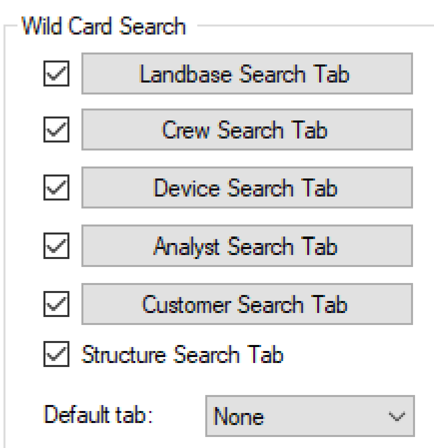
3.6. Print Options

The Print Options group includes the following options:

- **Print Background Color:** This setting configures the background color for the hard-copy printing of network diagrams. Click a color patch to display the color selection palette. The usual colors are black and white.
- **Print Label Color:** This setting configures the color for printing labels. Click a color patch to display the color selection palette. The usual colors are black and white.

> **Note**
>
> When printing in black and white, gray-scale colors for various labels can disappear. This feature allows all label colors to appear equally when printing in black and white.

3.7. Wildcard Search

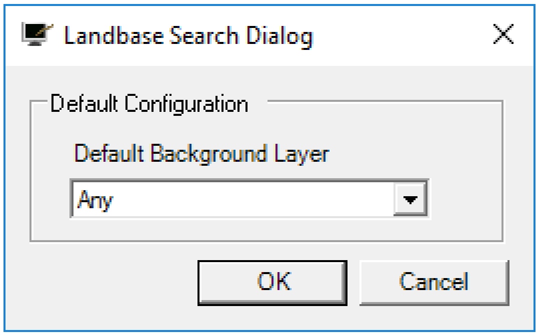


Note

> You can also configure the Find pop-up display using command line parameters. For more information, see Find Popup Parameters.

The Wildcard Search group includes the following search options:

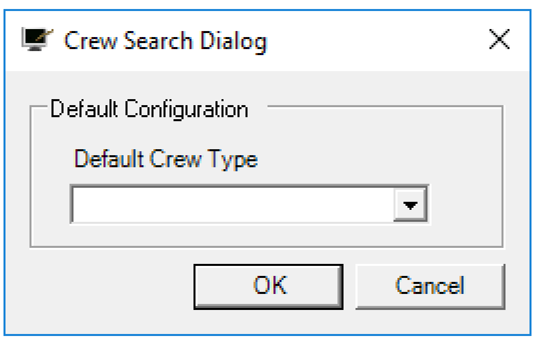
- **Landbase Search Tab:** The Landbase Search Tab checkbox configures the Landbase Search tab visibility on the Find pop-up window. Clicking the Landbase Search Tab button opens the Landbase Search dialog box for configuring search options on the Landbase Search tab.

  You can configure the following:

  - **Default Background Layer:** Specifies the default value for the Background Layer list box.

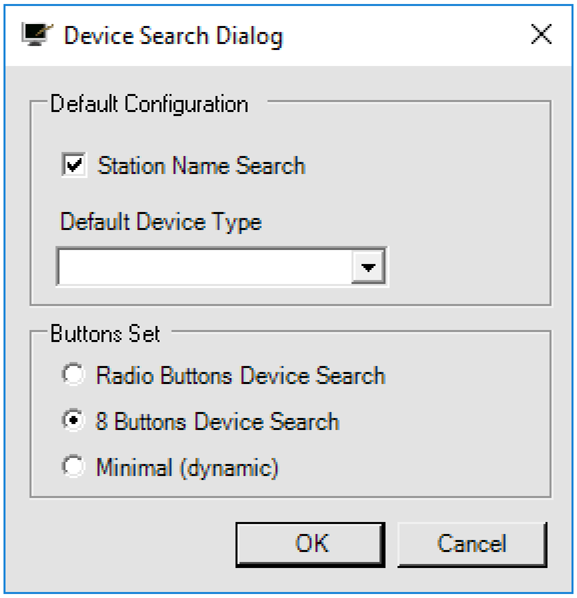
- **Crew Search Tab:** The Crew Search Tab checkbox configures the Crew Search tab visibility on the Find pop-up window. Clicking the Crew Search Tab button opens the Crew Search dialog box for configuring search options on the Crew Search tab.

  You can configure the following:

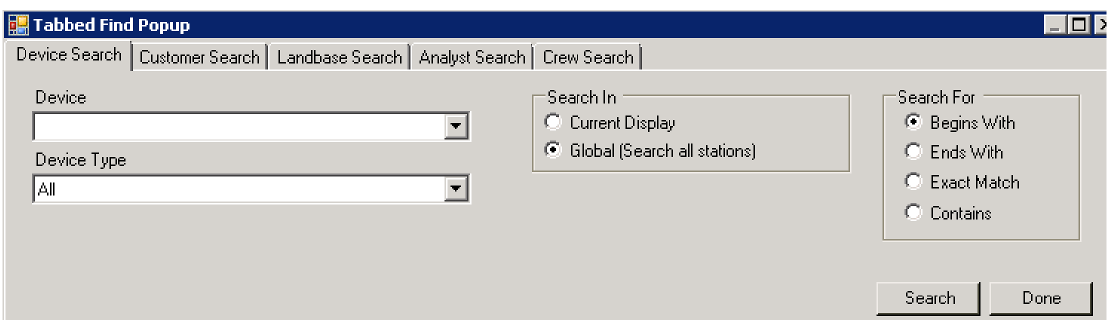
  - **Default Crew Type:** Specifies the default crew type for the Crew Type list box.

- **Device Search Tab:** The Device Search Tab checkbox configures the Device Search tab visibility on the Find pop-up window. Clicking the Device Search Tab button opens the Device Search dialog box where you can configure search options on the Device Search tab.

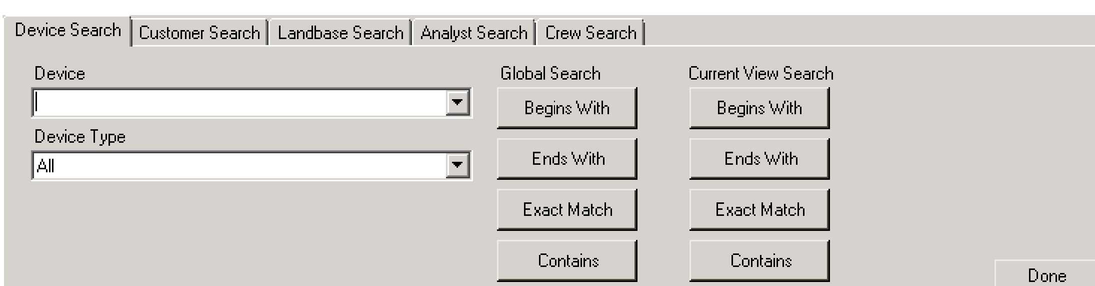


You can configure the following:

- **Station Name Search:** Controls the display of the Station field on the Device Search tab.
- **Default Crew Type:** Specifies the default device type for the Device Type list box.
- **Buttons Set:** Configures how buttons appear on the Device Search tab.
  - **Radio Buttons Device Search:** The Device Search shows radio buttons like the Analyst Search tab.

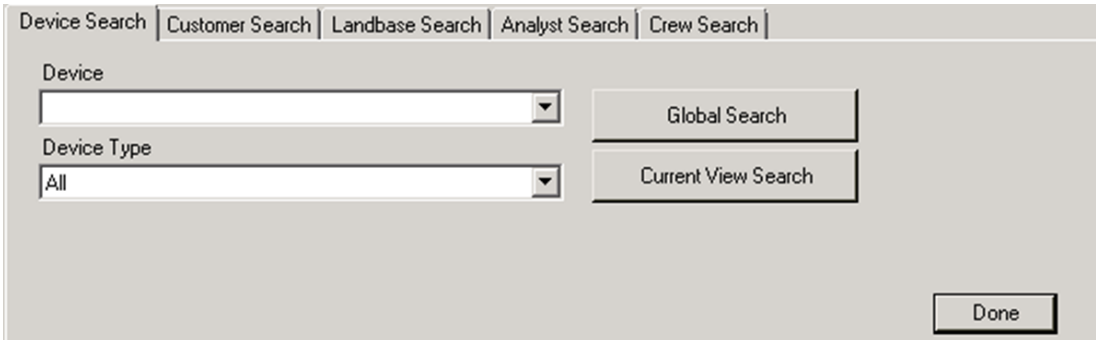


  - **8 Buttons Device Search:** The Device Search shows eight targeted buttons (four under Global Search and four under Current View Search).

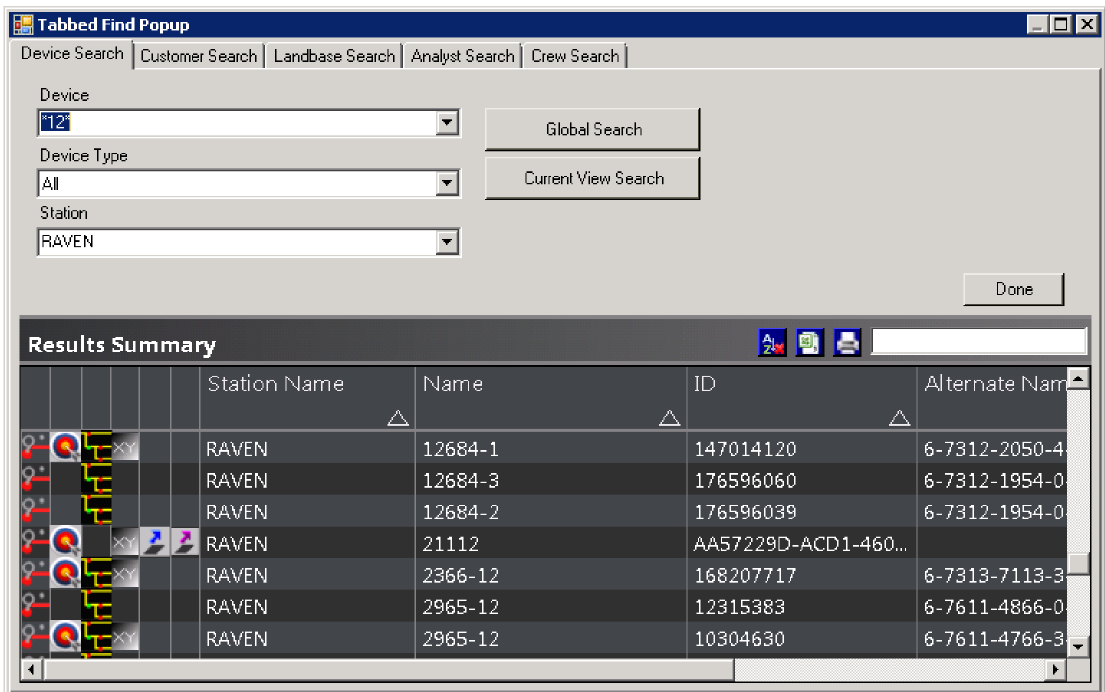


  - **Minimal (Dynamic):** The Device Search shows only two buttons: Global Search and

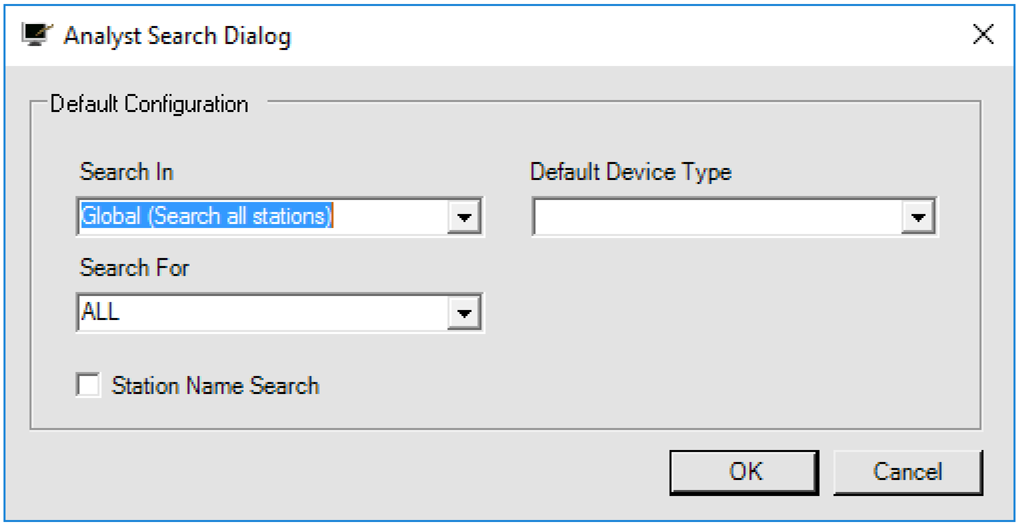
Current View Search.



When the Minimal button set is used, the wildcard search using the asterisk symbol ("*") is enabled. You can add the asterisk before or after the search text. All components whose name starts or ends with the searched text will be displayed.



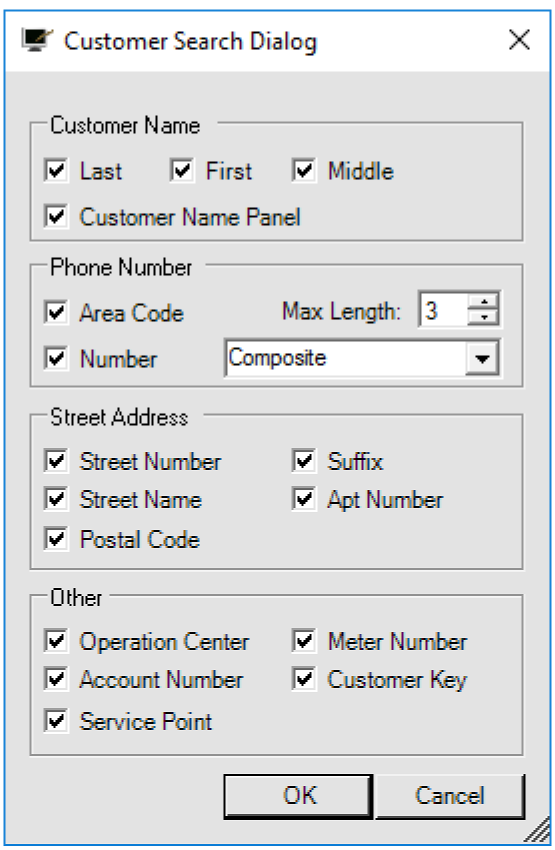
- **Analyst Search Tab:** The Analyst Search Tab checkbox configures the Analyst Search tab visibility on the Find pop-up window. Clicking the Analyst Search Tab button opens the Analyst Search dialog box to configure the default search options on the Device Search tab on the Find pop-up window.



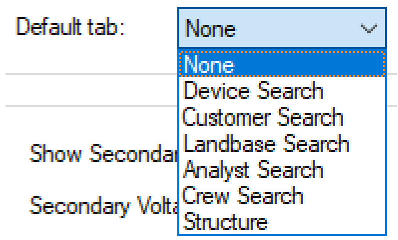
Tip

Removing the Analyst Search tab simplifies the Find function.

- **Customer Search Tab:** The Customer Search Tab checkbox configures the Customer Search tab visibility on the Find pop-up window. Clicking the Customer Search Tab button opens the Customer Search dialog box where you can configure the fields displayed on the Customer Search tab.

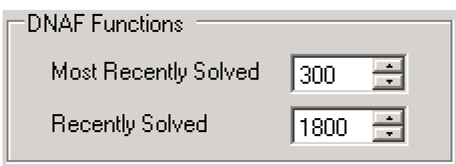


- ◦ **Phone Number Type:** Specifies how the phone number is defined in the customer model (Dedicated or Composite). Set to Dedicated if the phone number is defined by a dedicated PhoneNumber field. Set to Composite if the phone number is composed of AreaCode, PhonePrefix, and PhoneSuffix fields.
- **Structure Search Tab:** The Structure Search Tab checkbox configures the Customer Search tab visibility on the Find pop-up window.
- **Default:** Allows specifying the default tab to be opened when the Find pop-up opens.



3.8. DNAF Functions

The DNAF Functions section allows you to modify the default time range for recently solved and most recently solved cases. This affects the visual presentation of these cases on the DNAF Solution display.



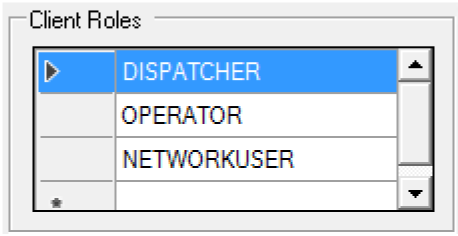
The DNAF Solution display presents the station solution time in different coloring, as explained below.

- Most Recently Solved cases (within 300 seconds) are shown in green.
- Recently Solved cases, which are solved within 300 to 1800 seconds, are shown in light sea green.
- All other cases are shown in blue.

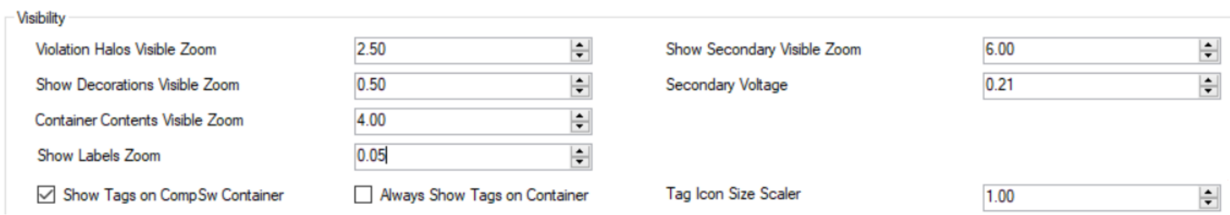
To change the default time, enter numeric values (in seconds) in the Most Recently Solved and Recently Solved fields.

3.9. Client Roles

The Client Roles section defines available client roles. Using client roles, toolbars, context menus, and attribute panels can be configured to have different layouts for different users.



3.10. Visibility



The Visibility group includes the following settings:

- **Violation Halos Visible Zoom:** The zoom level at which overload and voltage violation halos switch between line and device based views on the geographic network view. When zoomed below this level, halos only appear on line sections. When zoomed above this level, halos only appear on devices.

  Note that displaying violation halos is controlled by buttons on the geographic (common) toolbar.

- **Show Decorations Visible Zoom:** The zoom level at which additional icons relating to a network device (for example, tags, controllable, and telemetered symbols) become visible.
- **Container Contents Visible Zoom:** The zoom level at which the internal contents of container objects, such as vaults and cabinets, become visible on the geographic network view.

  At zoom levels below the selected threshold, the container objects are displayed with their standard symbols.

- **Show Labels Zoom:** The zoom level at which labels become visible on the geographic network view.
- **Show Secondary Visible Zoom:** The zoom level at which secondary lines (for example, low voltage lines between distribution transformers and customers) become visible on the geographic network view.

  Use this setting together with the Visible Threshold Non-Main Zoom settings to de-clutter the display from laterals and secondary lines at certain zoom levels.

- **Secondary Voltage:** The upper voltage threshold for secondary lines.
- **Show Tags on CompSw Container:** The **Show Tag on CompSw Container** option controls the display of tag decorations for CompSw objects.

For CompSw objects, there are three placement scenarios:

  - Both the CompSw object and its individual switches have placements.
  - Only the CompSw object has a placement (individual switches do not).
  - Individual switches have placements (the CompSw object does not).

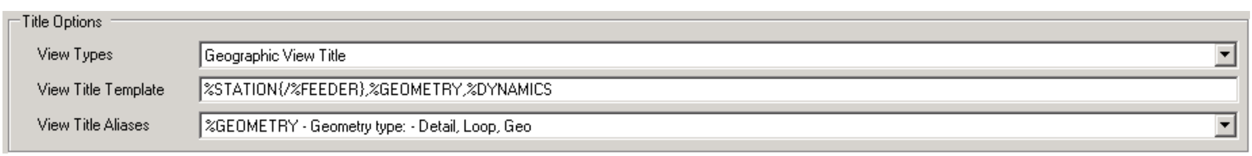
When this option is selected, the tag decoration appears next to the CompSw placement icon.

Separately, selecting the **Always Show Tags on Container** option displays tag decorations next to container objects, including vaults, cabinets, and UI containers.

- **Always Show Tags on Container:** The Always Show Tags on Container checkbox ensures that tags on container are consistently visible, even when zoomed in close on the geographic network view. The default value is set to False.
- **Tag icon Size Scaler:** The Tag Icon Size Scaler controls the size of tag icons in the geographic view. Increasing the value improves tag visibility when other symbols overlap near tagged devices. The default value is one, which reflects the standard/normal behavior. The tag icon size is relative to size device icon it is associated with.

3.11. Title Options

These settings are used to configure the title bar for each type of display.



To configure the title bars for each type of view, select the view from the drop-down menu; this shows the current title bar template. The template can be edited manually.

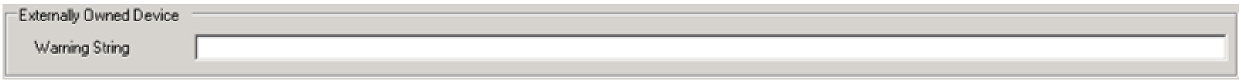
- **View Types:** Allows selecting a display type from the drop-down list.
- **View Title Template:** Allows defining the display title name template.
- **View Title Aliases:** Shows the definitions of all available title aliases.

The example in below image shows a title bar for a real-time geographic network view for the Raven station. The view title template for this display type was changed to:

Test_Name,%STATION{/%FEEDER},%GEOMETRY,%DYNAMICS

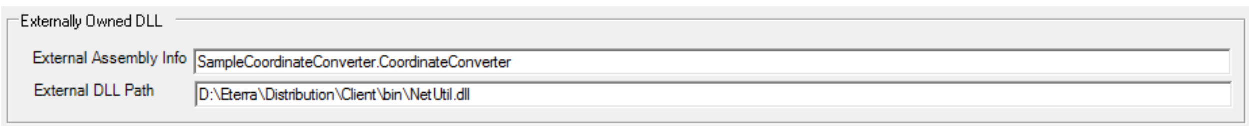
Figure 3. Network View Title Bar

3.12. Externally Owned Device



The Warning String text is shown in a Warning pop-up when the TOGGLEWARNING command is issued for switches modeled as external devices. The warning pop-up provides options to continue with the toggle action or cancel it.

3.13. Externally Owned DLL

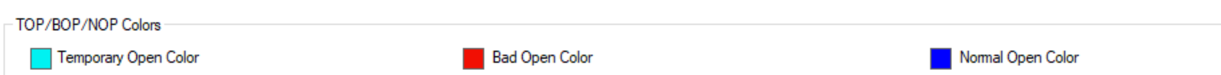


This section specifies parameters for converting system coordinates (X, Y) into projected coordinate system (PCS) coordinates (latitude, longitude). Coordinate conversion is required for navigation from the network view to a URL with a certain latitude and longitude (for example, Google Maps).

- **External Assembly Info:** Name of the class that converts system X/Y coordinates into latitude and longitude coordinates. The value format is namespace.class. For example, SampleCoordinateConverter.CoordinateConverter.
- **External DLL Path:** Name of the assembly (DLL file) that contains a class for converting system X/Y coordinates into latitude and longitude coordinates. The default assembly, `NetUtil.dll` is located in the /client/bin folder.

> **Note**
>
> For US customers, the default coordinate converter can be used. For non-US customers, a custom converter must be developed.

3.14. TOP/BOP/NOP Colors



This section defines colors for the Normal Open, Temporary Open, and Bad Open switch specific open state classification indicators on the network view.

- **Temporary Open Color:** Defines the color for the Temporary Open indicator.
- **Bad Open Color:** Defines the color for the Bad Open indicator.
- **Normal Open Color:** Defines the color for the Normal Open indicator.

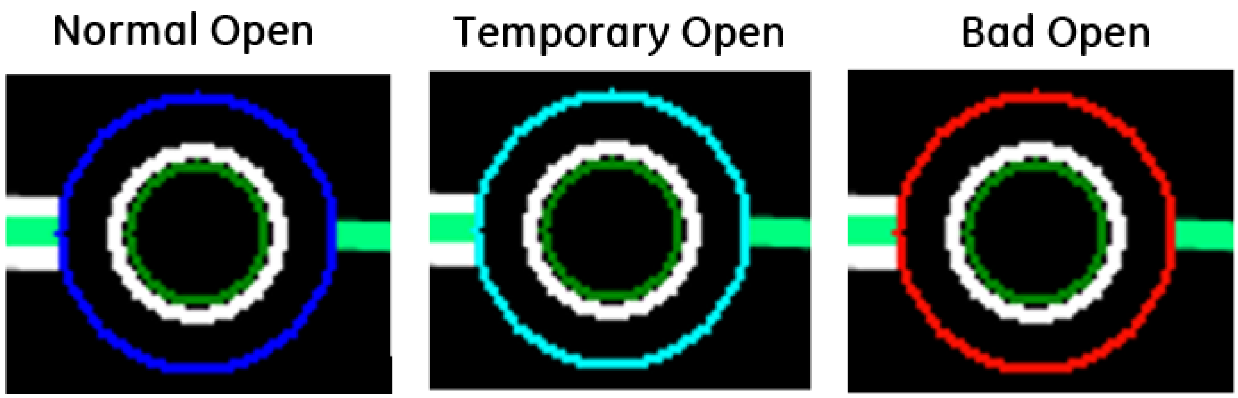


Figure 4. Switch Specific Open State Classification Indicators

4. Lines

The Lines tab allows you to configure how lines are displayed on the network view (below image).

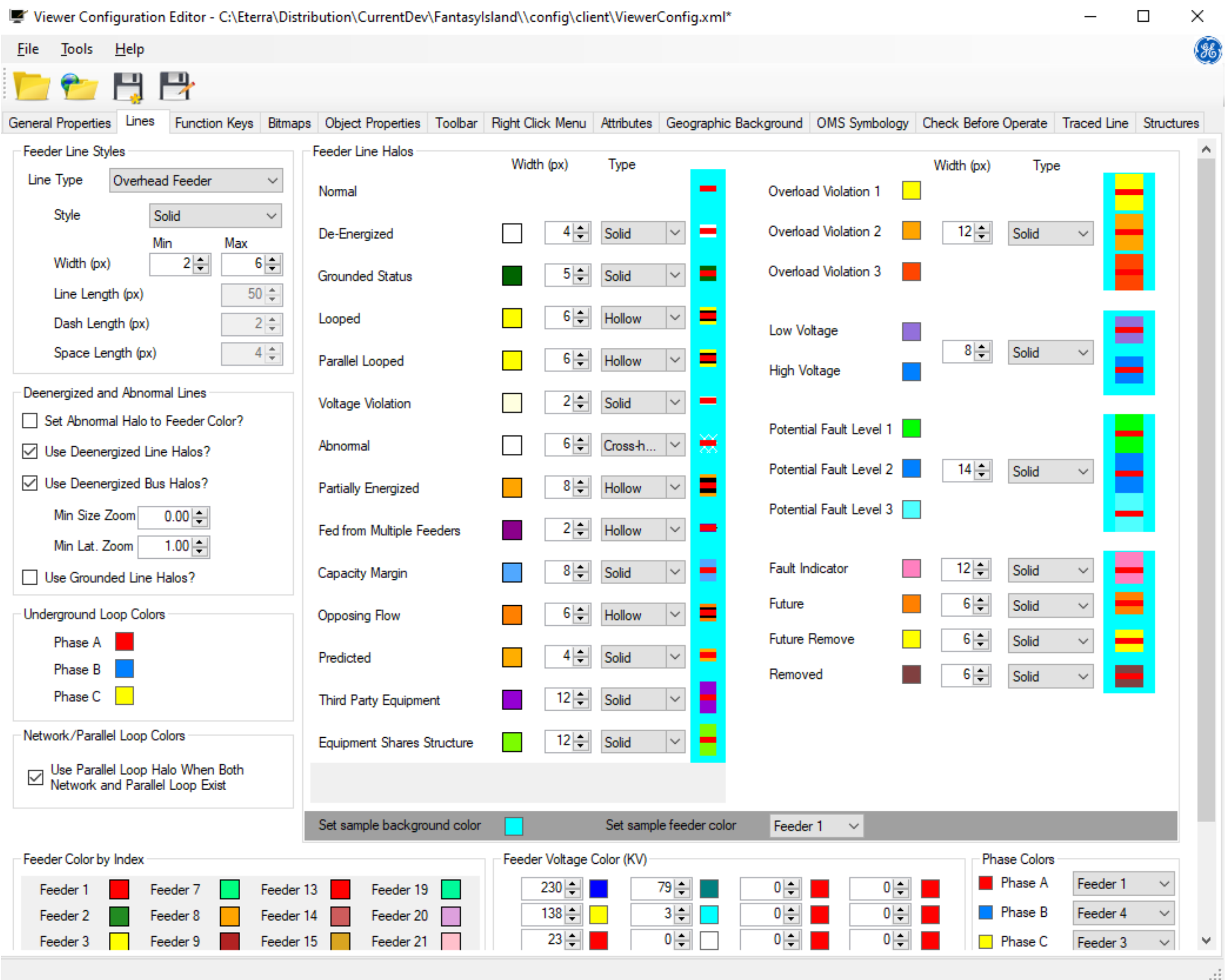


Figure 5. Lines Tab

You can change the appearance of all line types (overhead feeder, underground feeder and so on), and you can control the color and width of the halos that are associated with various violations and abnormal conditions.

4.1. Feeder Line Halos

The Feeder Line Halos group on the Lines tab allows you to see how the selected line style, color, and width look on the geographic displays, and configure the color, width, and style settings. The preview is updated automatically when the settings are modified.

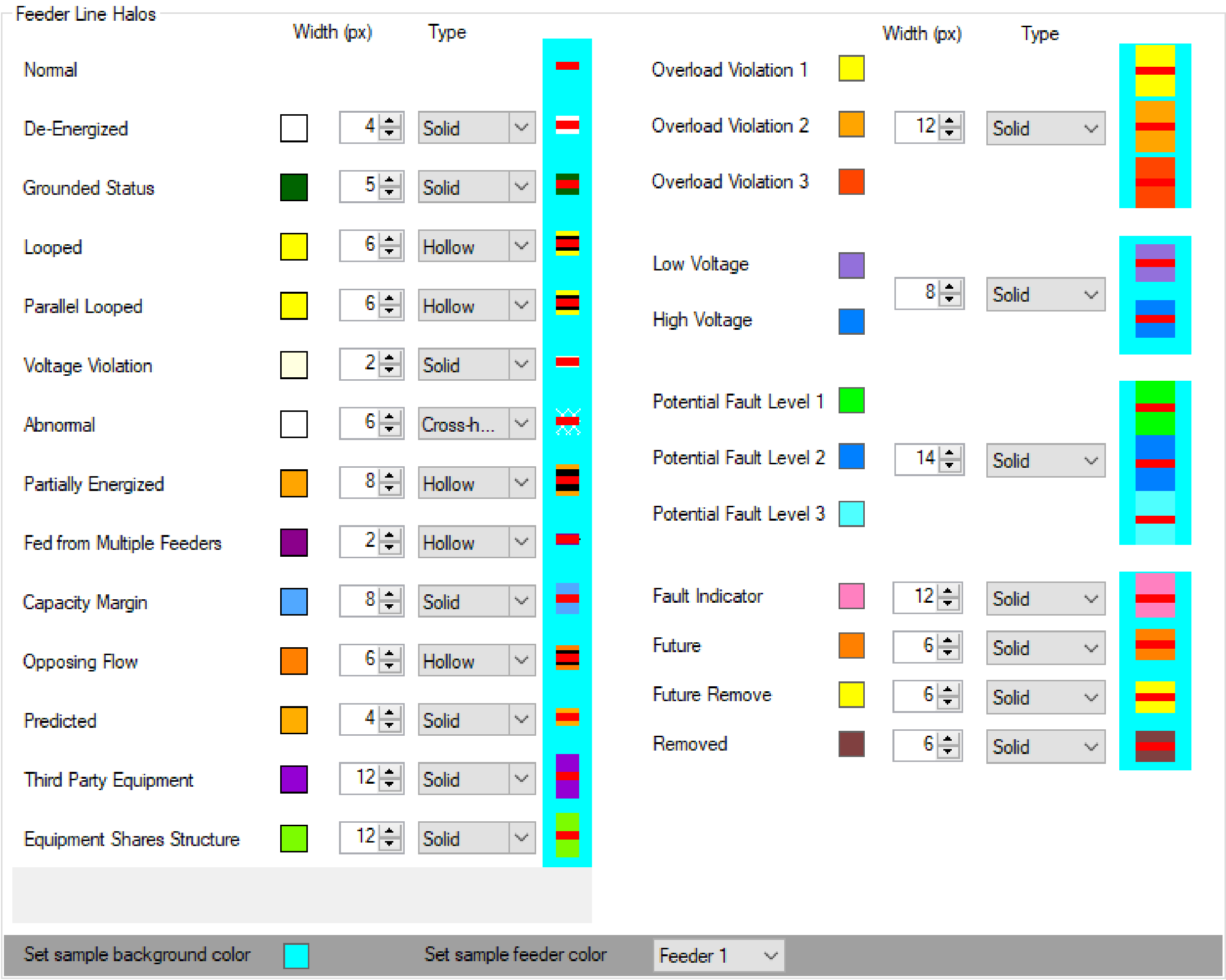


Figure 6. Feeder Line Halos Configuration Panel

Different feeder and line conditions are displayed in different colors, as described below:

- **Normal:** Nominal feeder color.
- **De-Energized:** Network line with all phases de-energized.
- **Grounded Status:** Network line with a temporary ground.
- **Looped:** When a loop with a single energization source exists (network loop), this halo is used to highlight the actual loop path.
- **Parallel Looped:** When a loop energized from more than one energization source (reference bus interface) exists, this halo is used to highlight the actual loop path between the energization points.
- **Fed from Multiple Feeders:** When a loop (either parallel or network) exists between two or more feeders, this halo is used to highlight lines. In addition, lines that are not part of the actual loop path are shown with the abnormally energized halo.
- **Voltage Violation:** Network lines that contain components with high or low voltage limit violations.

- **Abnormal:** An abnormally energized line is presented with the energizing circuit's color and abnormal halo.
- **Partially Energized:** Network lines with both energized and de-energized phases.
- **Capacity Margin:** Network lines that contain components with capacity margin violation.
- **Opposing Flow:** Any network line that has one phase fed from a direction that is opposite to the other phases on the same line.
- **Predicted:** Network equipment that is predicted to be de-energized as a result of receiving customer calls and AMI Out signals.
- **Third Party Equipment:** Network equipment that is owned by a third-party company.
- **Equipment Shares Structure:** Equipment shares a structure with equipment from a different feeder.

> **Note**
>
> The "Equipment Shares Structure" halo is only available if the "Enable Structure Information" feature was enabled in the Viewer Configuration Editor. The halos will only be shown if the model was created with a valid equipment and structure index file.

> **Note**
>
> Halo visibility is controlled using buttons on the geographic (common) toolbar.

> **Note**
>
> Higher priority halos overlap lower priority halos. Therefore, lower priority halos should be wider, so they are visible underneath higher priority halos.

The following is the halo priority list:

- TraceHalos (if configured to draw trace halo first)
- Fault Indicator Halos
- Shares Structure Halos
- Future Halos
- Future Remove Halos
- Removed Halos
- Voltage Violation Halos
- Overload Violation Halos
- Potential Fault Halos
- Dead/de-energized Halos (if de-energized is configured to be shown as halo)
- Combination of SharesStructure and Dead/de-energized Halos
- Grounded Halos

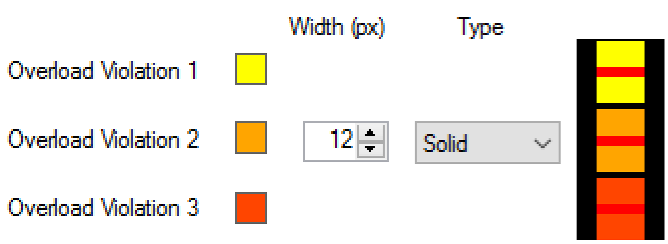
- Grounded and SharesStructure Halos
- Capacity Margin Halos
- Abnormally Energized Halos
- Predicted Out Halos
- Predicted Out and SharedStructure Halos
- Flow Halos
- Partially Energized Halos
- Looped and Parallel Looped Halos
- Powerflow Animation
- Trace Halos (if configured to be on top)

Overload and Voltage Violation Halos

Overload and voltage violation halos are used to display the results from the Distribution Power Flow (DPF) application.

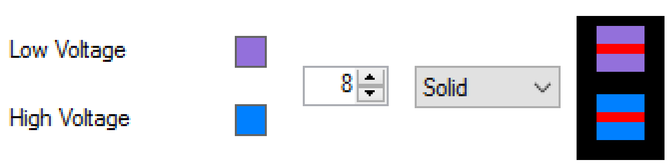
The overload violation halo is defined in three levels:

- Overload Violation 1: High
- Overload Violation 2: Very High
- Overload Violation 3: Extreme



The voltage violation halo is defined in two levels:

- Low Voltage: Under-Voltage
- High Voltage: Over-Voltage



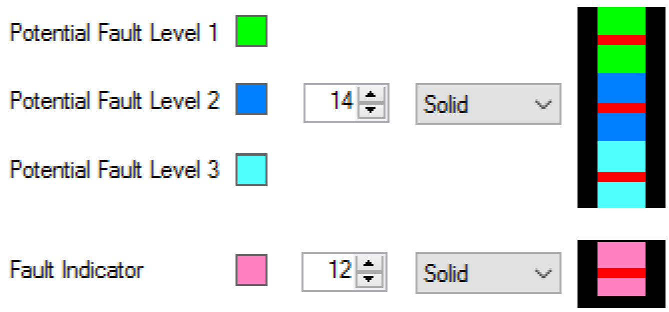
The settings for the violation levels are defined in the DNAF Configuration Editor.

Fault Indication Halos

Fault indication halos are used to show the results from fault indicator measurements and the Fault Location (FL) application.

The potential fault location halo is defined in three levels, depending on the probability that the segment of a feeder may be the location of a fault. Level1 refers to the highest fault probability. The percentage of the fault probability for each level is configured in DNAF.

The fault indicator halo highlights the segments of a feeder affected by a fault, as indicated by fault indicator measurements.



Equipment Commissioning and Decommissioning Halos

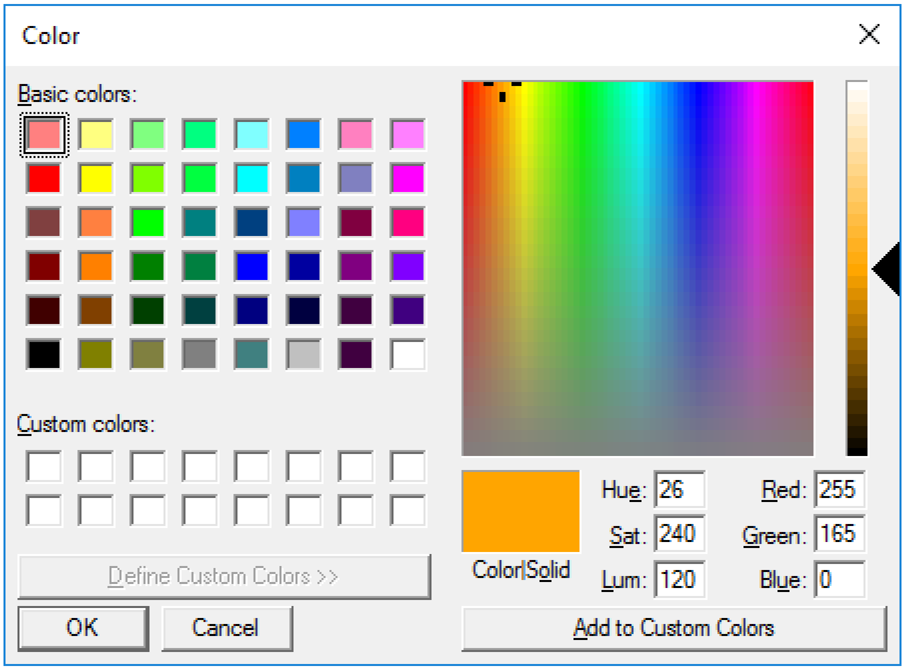
Equipment Commissioning and Decommissioning Halos are used to show components that are modeled in the distribution electrical network but not active (for example, components that are planned to be put in service, planned to be removed, or recently removed).

- **Future:** Network equipment that is constructed but not put in service.
- **Future Remove:** Network equipment that is planned to be decommissioned in future.
- **Removed:** Network equipment that was recently decommissioned.

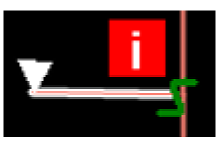
4.1.1. Configuring Line Halos

To modify line halos:

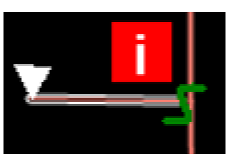
1. Click a color patch to open the color selection palette.



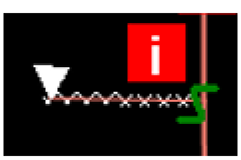
2. Select a color for each type of halo.
3. Set the halo width (in pixels) to be shown on either side of the line.
4. Select the line type:
   - **Solid:** Line with a solid halo.

   - **Hollow:** Line with a hollow halo.

   - **Cross-Hatched:** Cross-hatched line.

5. Review the changes in the halo preview window.

4.1.2. Visualizing Line Halos

To assist with the visualization of the selected feeder and halo colors, the color of the background used on the preview display can be changed. This allows the selected colors to be tested against any alternate background colors that might be used on alternate displays, such as study mode.

Figure 7. Sample Background and Feeder Colors

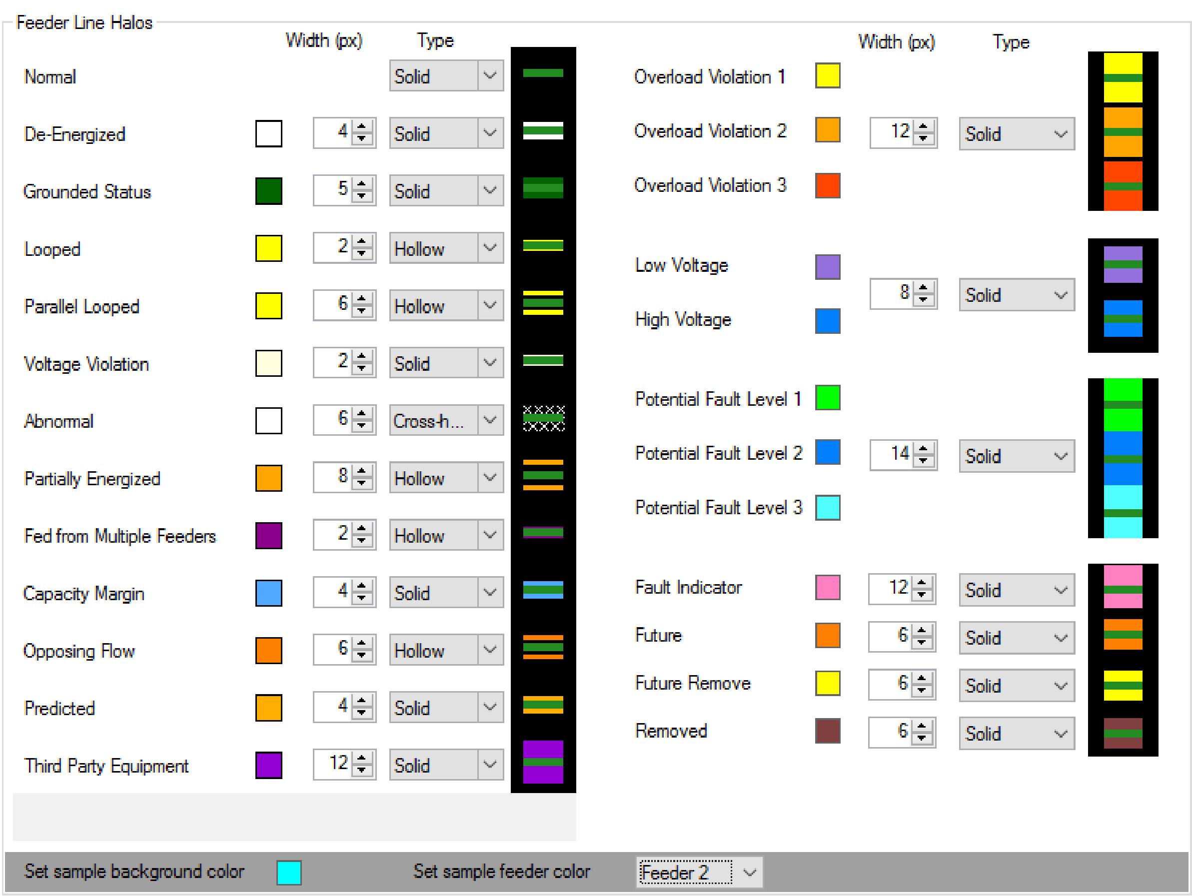
To change the sample background and feeder colors:

1. Select the feeder from the Set Sample Feeder Color drop-down list.

   The selected feeder is displayed on the color preview pane in combination with all the halos.

2. Click the Set Sample Background Color square to open the color selection palette.
3. Click the desired color on the palette.
4. Click OK to change the color.

   The background color of the preview display is updated.



Note

> This background color selection is for preview purposes only; it does not change the background color on the network view displays. That is defined in the Net Dynamics Configuration Editor, since the colored backgrounds are associated with different dynamic environments (RT, ST, and SIM).

4.2. Feeder Color by Index

The Feeder Color by Index configuration group allows you to associate a color with each feeder number. Feeder color numbers are defined within the GIS.

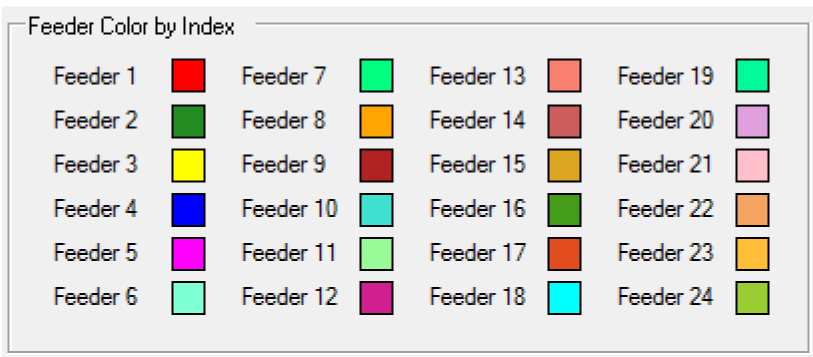


Figure 8. Feeder Color by Index Configuration Panel

To associate a color with each feeder color number:

1. Click a color patch to open the color palette dialog box.
2. Select the required color.

4.3. Feeder Voltage Color

The Feeder Voltage Color configuration group is used to configure colors of lines and where the feeder colors do not apply - primarily above feeder breakers in station internals displays. This configuration allows matching colors to be used on SCADA one-line displays.

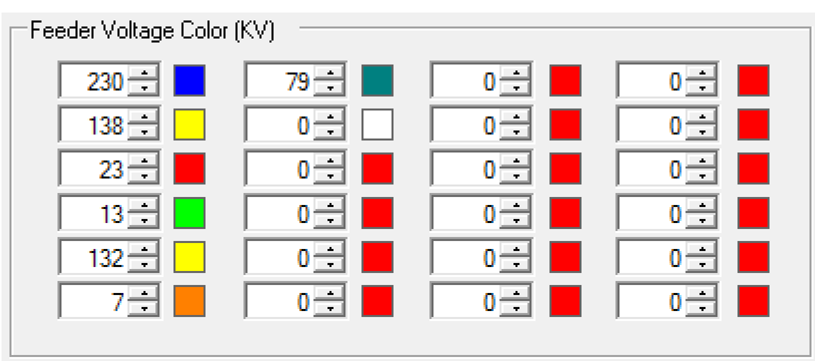
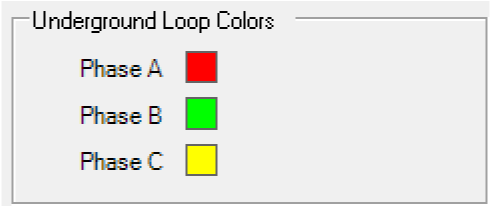


Figure 9. Feeder Voltage Color Configuration Panel

To associate a color with a feeder voltage:

1. Set the feeder voltage (in KV).
2. Click a color patch to open the color palette dialog box.
3. Select the required color.

4.4. Underground Loop Colors



The Underground Loop Colors grid is used to configure phase colors on a URD loop schematic display.

To set a phase color:

1. Click a color patch to open the color palette dialog box.
2. Select the required color.

The below image shows the colors as PhaseA=Red, PhaseB=Green, and PhaseC=Yellow.

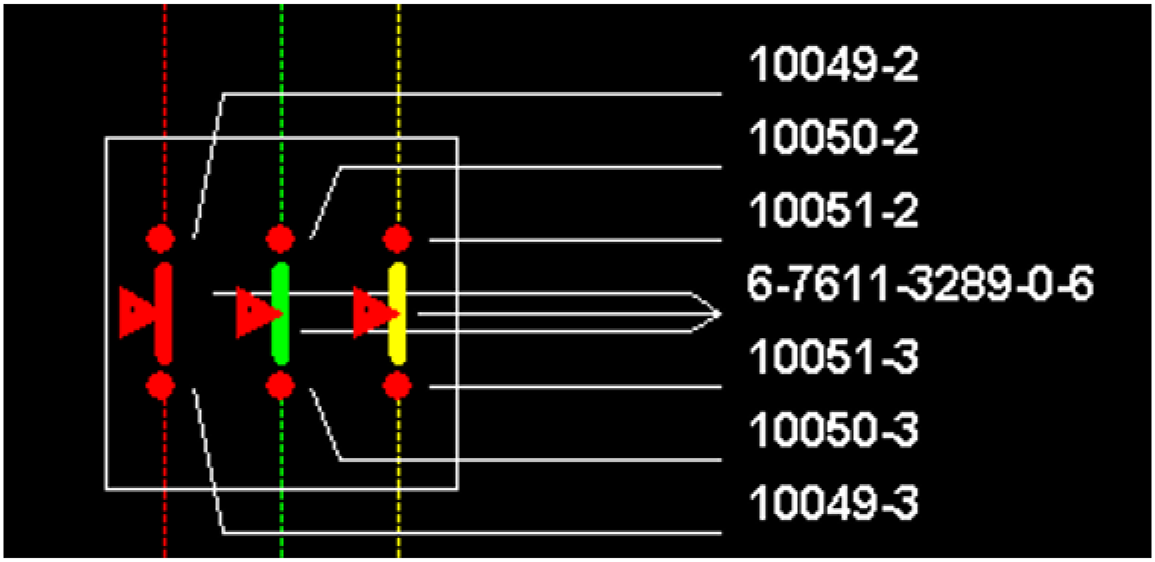


Figure 10. URD Loop Schematic

4.5. Phase Colors

The Phase Colors panel allows you to set the phase colors relative to the nominal state of the system.

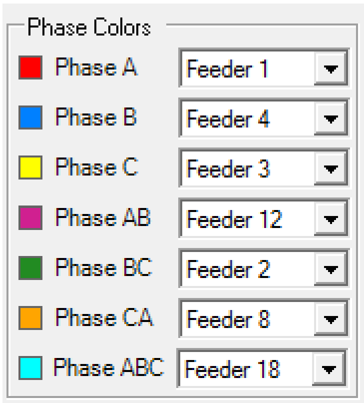


Figure 11. Phase Colors Panel

To set a phase color, select a feeder with the desired color from the drop-down list. The phase color will be the same as the feeder's color.

Note

Displaying phase colors is toggled using the Show/Hide Phase Colors button on the geographic (common) toolbar.

4.6. Feeder Line Styles

The Feeder Line Styles group (below image) allows you to modify the appearance for different types of lines.

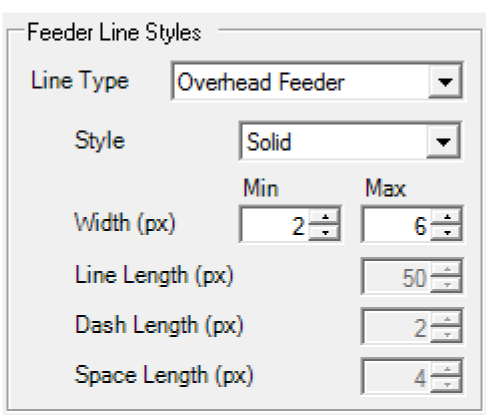


Figure 12. Feeder Line Styles Panel

To change selected line styles:

1. Select the type of line to configure from the Line Type drop-down list.
   - **Overhead Feeder:** Used for overhead three-phase backbone lines.
   - **Overhead 3 Phase:** Used for overhead three-phase lateral lines.

- **Overhead 2 Phase:** Used for overhead two-phase lines.
- **Overhead 1 Phase:** Used for overhead one-phase lines.
- **Underground Feeder:** Used for underground three-phase backbone lines.
- **Underground 3 Phase:** Used for underground three-phase lateral lines.
- **Underground 2 Phase:** Used for underground two-phase lines.
- **Underground 1 Phase:** Used for underground one-phase lines.
- **Buses:** Used for setting the zoom based width range for buses.
- **Permanent Jumper:** Used for setting the zoom based width range for permanent jumpers.

2. Select the line style to associate with the selected line type from the Style drop-down list. The possible options are:
   - Solid
   - Dashed
   - Dotted
   - Dash Dot
   - Show Phases (the number of consecutive dashes in the line style reflects the number of phases)
3. Select the width of the line (in pixels) using the Width field.

   The "Min" width is the minimum width of the line regardless of the zoom level. The "Max" width is the maximum width of the line regardless of the zoom level. In other words, this change sets the limits for the non-linear zoom.

   > **Note**
   >
   > If you increase line widths, you might need to make a comparable change to the halo widths to ensure that they are still visible on a thicker line.

4. Specify the following parameters:
   - **Line Length:** Specifies the length of a line (in pixels). Applies to a Show Phases type of line.
   - **Dash Length:** Specifies the length of a dash (in pixels). Applies to a Show Phases type of line.
   - **Space Length:** Specifies the length of a space between the lines and dashes (in pixels). Applies to a Show Phases type of line.

4.7. De-Energized and Abnormal Lines

The De-Energized and Abnormal Lines group (below image) allows you to change the appearance of a de-energized line.

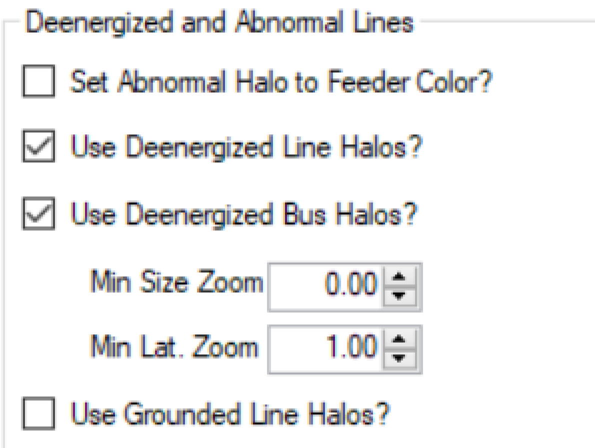
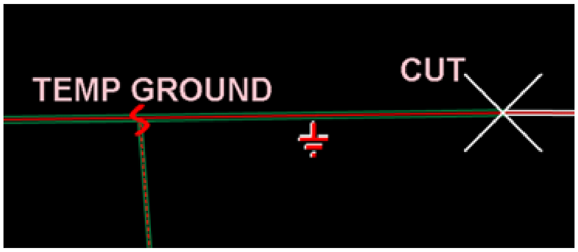


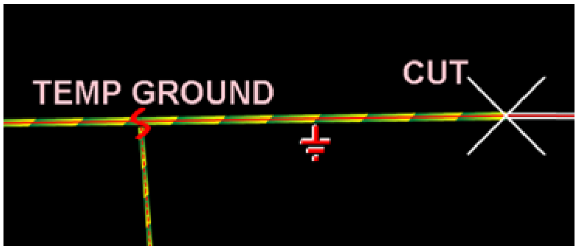
Figure 13. De-Energized and Abnormal Lines Configuration Panel

You can modify the following settings:

- **Set Abnormal Halo to Feeder Color?** Select this option to display the line with abnormal status in the feeder color.
- **Use Deenergized Line Halos?** Check/uncheck to enable/disable halos for de-energized lines.
    - **Min Size Zoom:** This is the zoom level above which the halo is visible. If a user zooms out such that the zoom level is below this value, the de-energized halos are not shown. By default, this is set to zero to ensure that de-energized lines are always visible on the displays.
    - **Min Lat. Zoom:** This item is similar to "Min Size Zoom" but it is particular to laterals. Drawing halos is computationally expensive; therefore, when a user zooms far out with the laterals turned on, performance can degrade if there is significant de-energization. It is a good idea to set this value to1 or higher (potentially the level at which laterals appear automatically).
- **Use Deenergized Bus Halos?** If selected, a deenergized bus will be shown with the Deenergized Status halo. If not selected, a deenergized bus will be shown with the deenergized conductor color (in the same manner as a line with the **Use Deenergized Line Halos** option set to false).
- **Use Grounded Line Halos?** If selected, grounded lines are shown with the Grounded Status halo. If not selected, grounded lines are shown with the Grounded Status halo with yellow dashes.



Use Grounded Line Halos selected.



Use Grounded Line Halos not selected.

5. Function Keys

The Function Keys tab allows you to configure keyboard function keys and mouse buttons to perform specific actions.

The commands that can be associated with keyboard or mouse actions are defined using the Netview URL syntax.

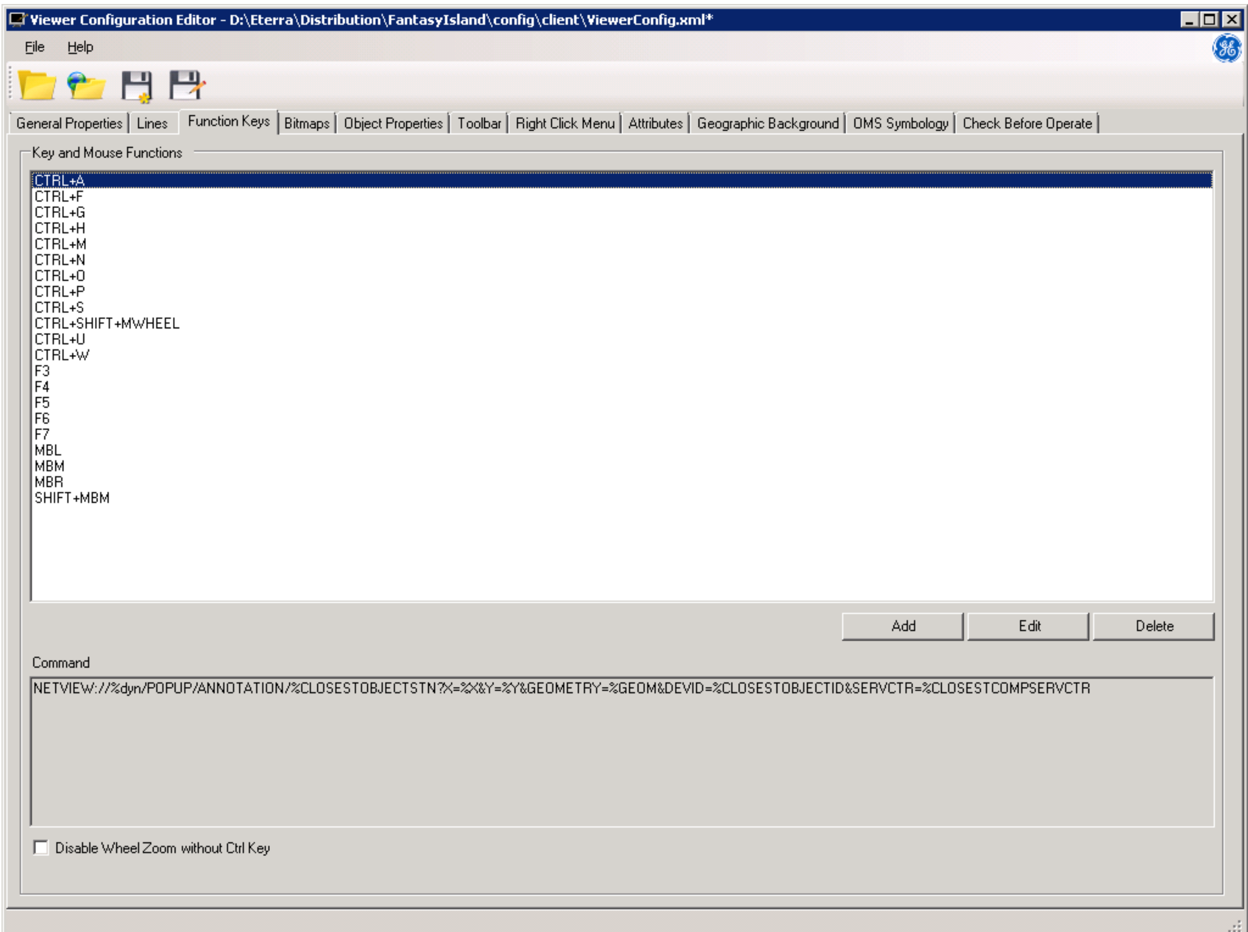


Figure 14. Function Keys Tab

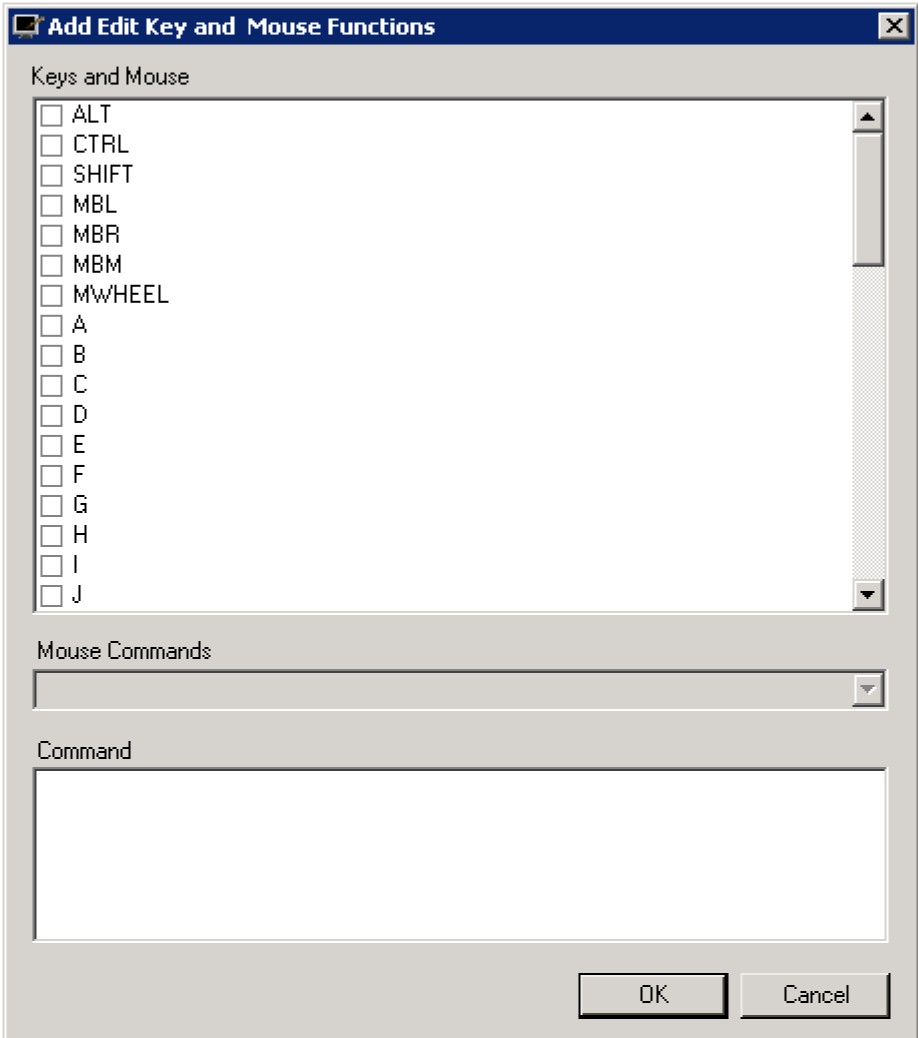
5.1. Key and Mouse Functions

The Key and Mouse Functions panel lists the already defined key and mouse combinations. Select any of the defined key (or key and mouse) sequences to see the associated command string in the Command panel. The command string can then be edited by clicking the Edit button.

To create a new keyboard function:

1. Click the Add button.

   The Add Edit Key and Mouse Functions dialog box opens.



2. In the Keys and Mouse list, select the checkboxes next to the key buttons that you want to use as a keyboard combination.

   For example, selecting the ALT and D checkboxes creates an Alt+D function key.

3. Enter the Netview URL command string to be associated with the function key in the Command field.
4. Click the OK button to add the function key to the Key and Mouse Functions list.

To define a new mouse function:

1. Click the Add button.

   The Add Edit Key and Mouse Functions dialog box opens.

2. In the Key and Mouse list, select the checkboxes next to the key and mouse buttons that you want to use as a mouse combination.

   For example, selecting the ALT and MBL checkboxes creates an "Alt+Left Mouse Button" function key.

3. In the Mouse Commands list, select a standard mouse command from the drop-down menu.
4. If an alternate command other than those on the pull-down menu is required, select "Specify Action" from the menu, and enter the Netview URL command string in the Command field.
5. Click the OK button to add the function key to the Key and Mouse Functions list.

5.2. Changing the Scroll Feature

Selecting "Disable Wheel Zoom without Ctrl Key" requires the user to press and hold the Ctrl key while scrolling the mouse wheel in order to zoom a display. The standard Microsoft user interface behavior is to have Ctrl+Scroll configured to zoom in and zoom out. However, to allow the distribution dispatcher to move the mouse with one hand and radio with the other, the default behavior (departing from the Microsoft standard) is to zoom with just the mouse wheel.

Unlike a web page or a Microsoft Office document, scrolling does not make much sense on a geographic display, so trading the scroll feature for zooming is preferred by distribution dispatchers.

6. Bitmaps

The symbols used to display network objects are derived from two sources:

- An internal set of vector images, which are used by default.

  For common objects, such as manual switches and distribution transformers, using vector images improves the drawing performance of the client.

- A set of bitmap images that can be customized to suit the utilities' needs.

The Bitmaps tab (below image) is used to define the display parameters for bitmap symbols.

> **Note**
>
> The bitmap visibility conditions are also configured on the Object Properties tab (see Object/Labels Visibility Grid).

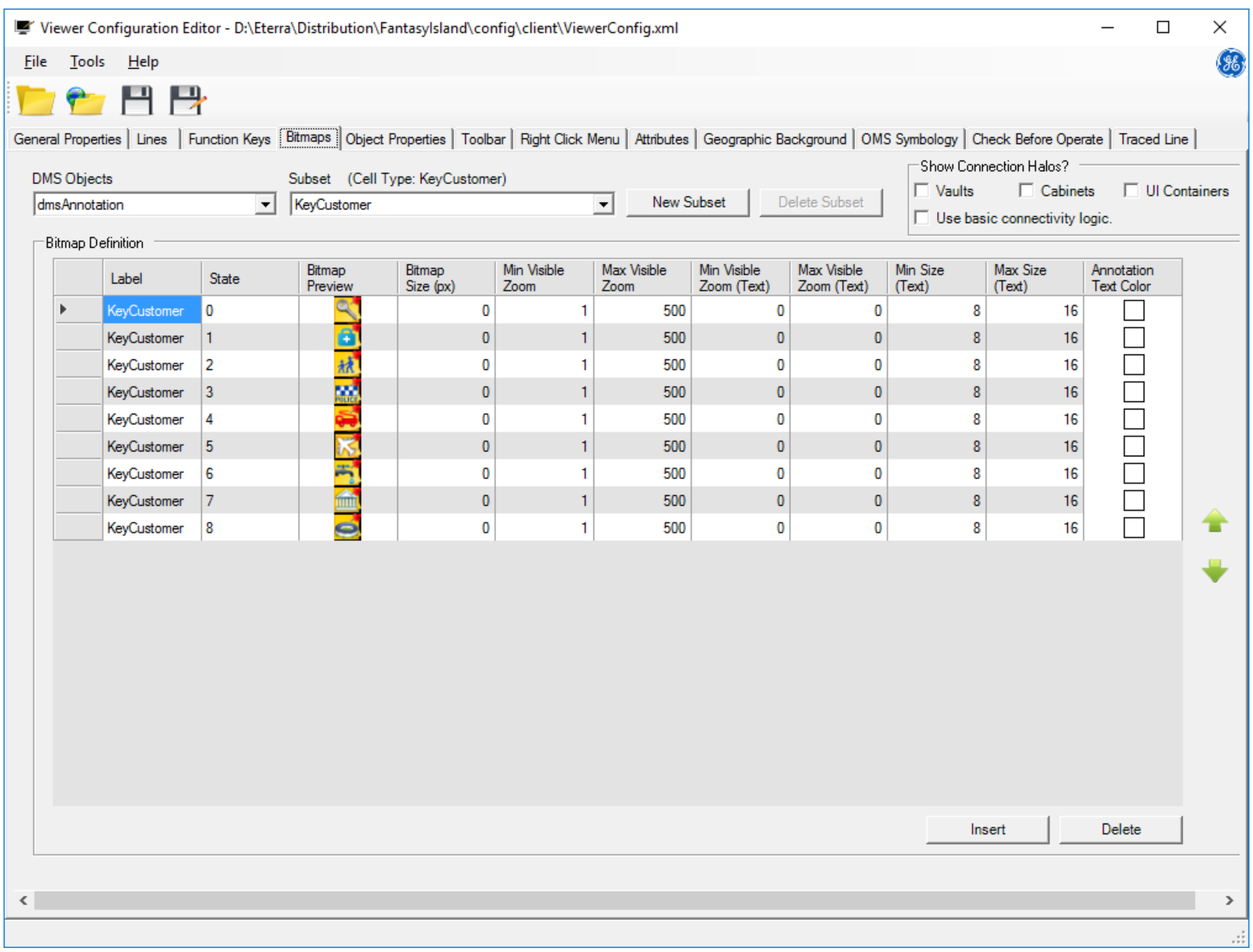


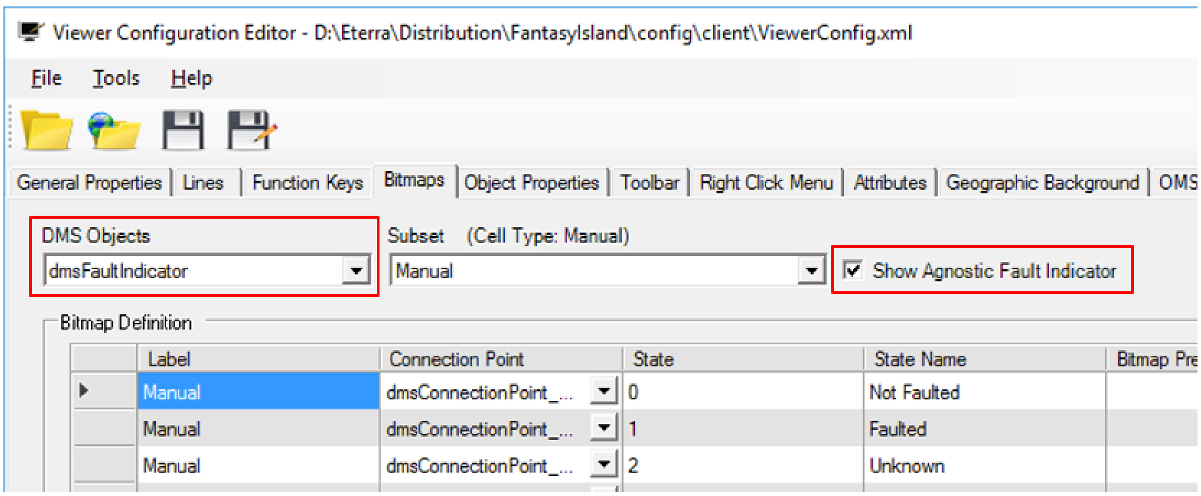
Figure 15. Bitmaps Tab

The following fields and controls are shown on the Bitmaps tab:

-

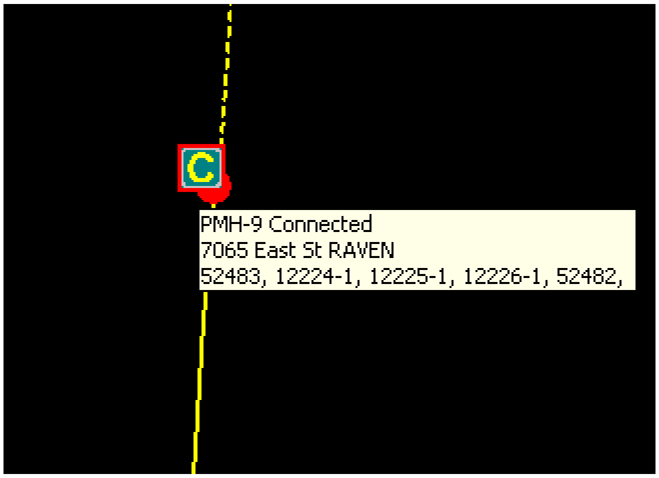
DMS Objects: A drop-down list of objects from DNOM. Select an object to modify its visual representation.

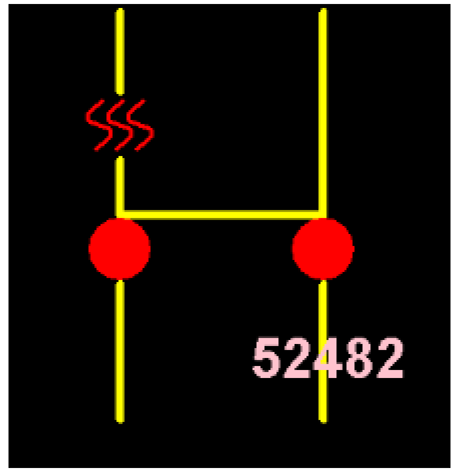
- **Subset:** This is a list of subtypes for DMS objects with more than one type. For switching objects, vaults, and cabinets, this list is derived from the Enums_Table in the `global.csv` file (see Read Global File in After Starting the Application).
- **New Subset:** Adds a new subset with a defined number of states.
- **Delete Subset:** Deletes the selected subset.
- **Show Agnostic Fault Indicator Subset:** Specifies whether phase-agnostic fault indicators are shown with the "A" symbol above the device on the network view. This option only appears when dmsFaultIndicator is selected in the DMS Objects field.



- **Show Connection Halos:** This group includes options that control the display of colored borders on the container objects to indicate the overall connection status of the switches inside the container.
  - **Vaults:** Controls whether border halos for vaults are shown.
  - **Cabinets:** Controls whether border halos for switch cabinets are shown.
  - **UI Containers:** Controls whether border halos for UI containers are shown.
  - **Use Basic Connectivity Logic:** When this option is selected, Network Topology Processing calculations are ignored and basic logic is used to determine the abnormal state of the vault, cabinet, and/or UI container. The basic logic also determines the open and connected states for recloser switch cabinets.

As shown in the image below, a solid red connection halo indicates that the switches in the cabinet are closed and that this is the normal state.



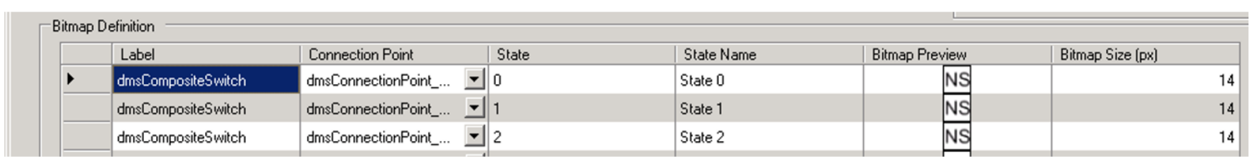


Conversely, if one of these switches is opened, the connection halo shows a broken green line. Additionally, a yellow border represents a normal partially connected setup, and a green border represents a normal open setup.

6.1. Bitmap Definition

The Bitmap Definition frame displays an auto-sized table of the bitmaps associated with the individual states of a particular object.

Some objects from DNOM represent multiple physical objects, and each physical object can be associated with various bitmaps depending on its physical state (that is, opened, closed and so on).



Bitmap Definition

Label	Connection Point	State	State Name	Bitmap Preview	Bitmap Size (px)
dmsCompositeSwitch	dmsConnectionPoint_...	0	State 0	NS	14
dmsCompositeSwitch	dmsConnectionPoint_...	1	State 1	NS	14
dmsCompositeSwitch	dmsConnectionPoint_...	2	State 2	NS	14

Figure 16. Bitmap Definition Table

The following fields are listed in the Bitmap Definition table:

- **Label:** This entry box field shows the label of a cell. This is mostly useful for objects that have different cell types, and it is only used within this editor to make things clearer for the administrator.
- **Connection Point:** This drop-down box shows the orientation of the connection point on the bitmap or vector image. There are nine possible orientations (NW, N, NE, E, SE, S, SW, W, and Center). Most bitmaps use a center connection point. Capacitors are an example of a bitmap that can be shown with a non-center connection point.

  To show a capacitor as a shunt device relative to the line with which it connects, a south connection point is used.
- **State:** This field displays a consequent numeric value for every list item of a specific cell type. The State list is a zero-based numeric list containing entries equal to the value entered in the

#States field (when adding a new subset). If the #States field contains 3, the State List shows 0, 1, and 2.

You can add a new state by clicking the Insert button at the bottom-right of the Bitmap Definition section. A new line with a label name and a consecutive state number appears at the end of the list. To delete the new line, click the Delete button at the bottom-right of the Bitmap Definition section.

> **Note**
>
> New states cannot be added to some DNOM objects.

- **State Name:** This field displays a state name for every list item of a specific cell type (for example, On/Off or Open/Close).
- **Bitmap Preview:** This field shows the bitmap image associated with the bitmap state of a specific cell type that belongs to an object. This is only useful for objects that are associated with bitmaps, not for vector images. Updating this column updates the bitmap picture.
- **Bitmap Name:** This field shows the name of the bitmap file. Updating this column updates the bitmap picture in the Bitmap Preview column.
- **Bitmap Size:** Shows the dimensions of the bitmap image in pixels.

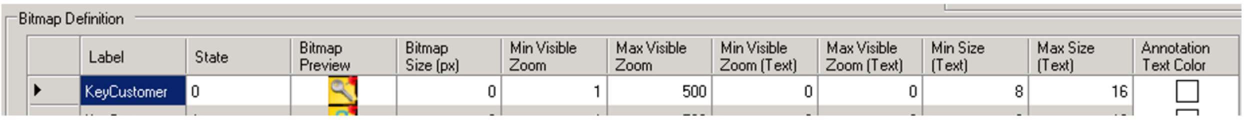
> **Note**
>
> The Bitmap Size feature has not yet been implemented; leave it at zero. You can set the bitmap size via the Object Properties tab of the Viewer Configuration Editor. For instructions, see Object Properties.

Additional parameters that appear only for specific types of objects are listed in the following sections.

The following controls are available in the Bitmap Definition table:

- **Insert:** Adds a new row to the Bitmap Definition grid. Optionally, you can right-click the desired location and select Insert from the context menu.
- **Delete:** Deletes the selected row in the Bitmap Definition grid. Optionally, you can right-click a record and select Delete from the context menu.
- **Move Up and Down Arrows:** Use these buttons to re-order the selected rows on the Bitmap Definition grid. Optionally, you can right-click a record and select MoveUp or MoveDown from the context menu. Arrows are enabled only for those subsets and states that are allowed to be arranged.

6.1.1. Annotations


Bitmap Definition

	Label	State	Bitmap Preview	Bitmap Size (px)	Min Visible Zoom	Max Visible Zoom	Min Visible Zoom (Text)	Max Visible Zoom (Text)	Min Size (Text)	Max Size (Text)	Annotation Text Color
▶	KeyCustomer	0		0	1	500	0	0	8	16	☐

Figure 17. Annotations Bitmap Definition Table

The following fields are listed in the Annotations bitmap definition table:

- **Label:** Allows you to define the name of an annotation type.

  > **Note**
  >
  > The annotation name and type must also be defined in the Net Dynamics Configuration Editor tool. For detailed instructions, refer to the *Annotation Importer/Exporter User Guide*.

- **State:** Allows you to define unique bitmaps for different states or subtypes for a particular annotation type. The State list is a zero-based numeric list containing entries equal to the value entered in the #States field. If the #States field contains 3, the State list shows 0, 1, and 2.
- **Bitmap Preview:** Shows a preview of the bitmap picture associated with the annotation.
- **Bitmap Size:** This feature has not yet been implemented; leave it at zero.
- **Min/Max Visible Zoom:** Symbols are shown when they are within the minimum and maximum zoom range. When the zoom range is above or below these settings, the symbols are hidden from view.
- **Min/Max Visible Zoom (Text):** Text is shown when it is within the minimum and maximum zoom range. When the zoom range is above or below these settings, the text is hidden from view.
- **Min/Max Size (Text):** The minimum/maximum size of the text on the display, regardless of zoom level.
- **Annotation Text Color:** The color for annotation text on the display.

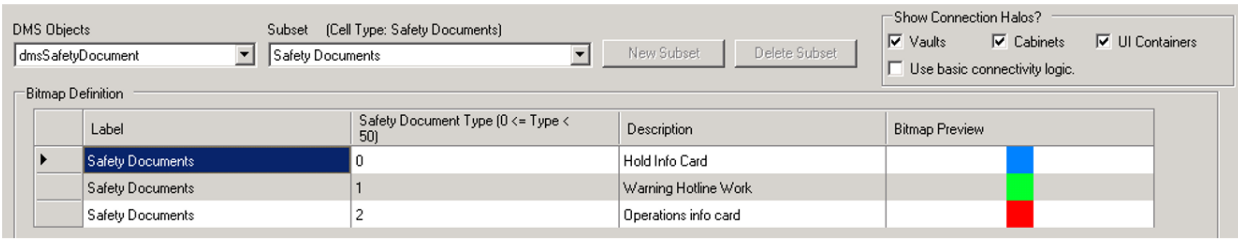
> **Important**
>
> The bitmap name assigned to an annotation state via the Viewer Configuration Editor serves as the state description. If the bitmap is renamed for an annotation index, the original state description for that index is removed and the description of a "new/different" state is added.

6.1.2. Safety Documents

A safety document is a report that confirms that it is safe to perform work on a device in the field (for example, in the case of a line outage).

For more information about safety documents, refer to the *Safety Documents User Guide* in the SCADA documentation.

To associate each safety document type with a bitmap, select the Safety Documents DMS object. You can configure all other parameters of the Safety Documents icon, such as size and zooming levels, using the Object Properties tab.



The following fields are available for the Safety Documents DMS object:

- **Label:** Name of the safety document type, as defined in the safety document application.
- **Safety Document Type:** Type of the safety document, as defined in the safety document application.
- **Description:** Description of the safety document type.
- **Bitmap Preview:** Bitmap image associated with the safety document type.

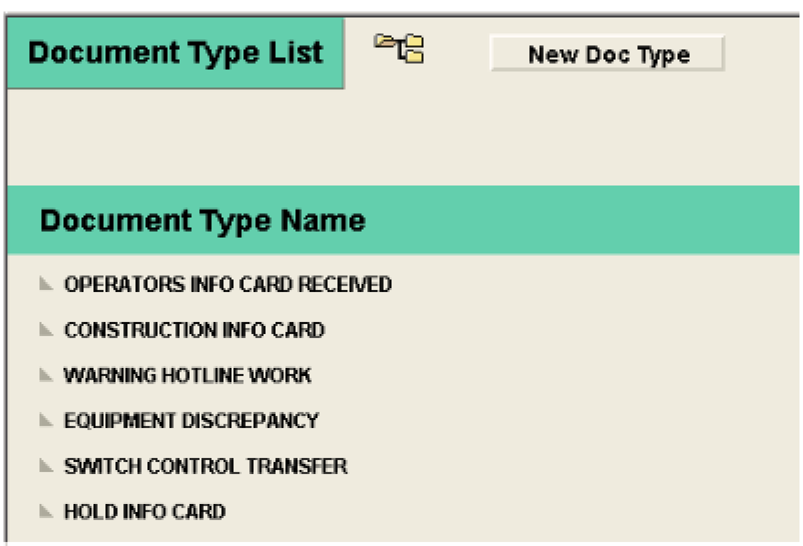
To create a safety document:

1. From the geographic network view, right-click a free space and, from the pop-up menu, select the Safety Document option.
2. Associate the safety document with a certain document type, a tag, and a device.

To assign a bitmap to a safety document icon type:

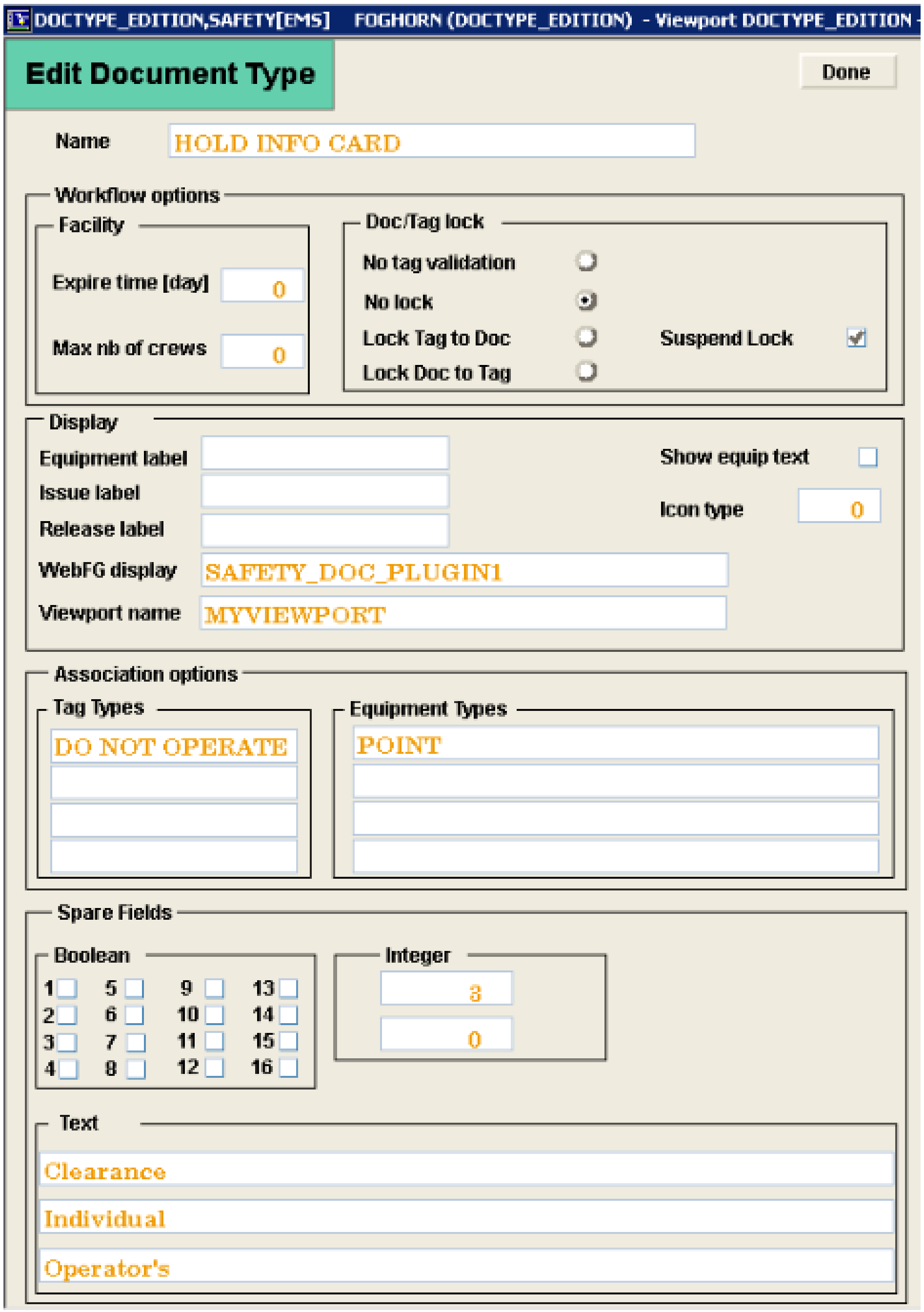
1. From the EMP Applications menu, point to Distribution Application, and select Safety List to open the Document Type List display from the geographic network view.

   The Document Type List display of the safety document application appears with the list of existing safety documents types, as shown below.

2. Select a safety document type that you need to map with a bitmap.

   The Edit Document Type pop-up display appears, as shown below.



3. Open the Bitmap tab of the Viewer Configuration Editor.
4. Select the Safety Document item from the DMS Objects list.
5. Locate a blank entry with an asterisk in the far left column of the Bitmap Definition table and enter the following information:
   - **Name:** Safety Document type name as it is defined in the Name field of the Edit Document

Type pop-up display.

- **Type:** Numerical value, as defined in the Icon Type field of the Edit Document Type pop-up display.

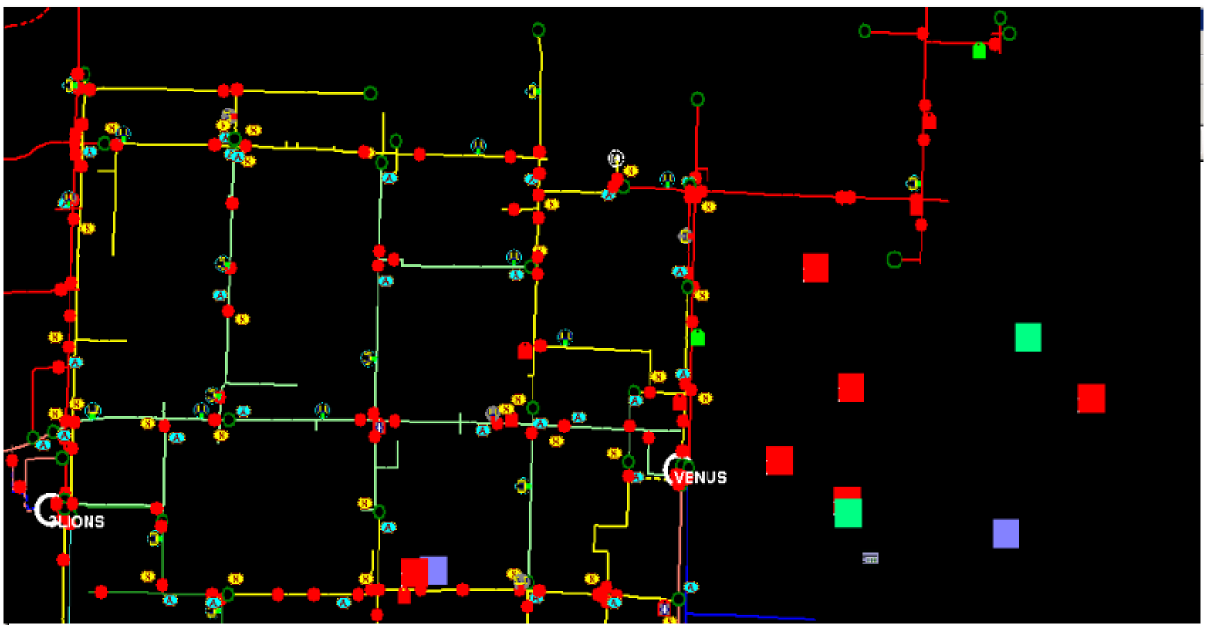
6. Select the Bitmap field.

   The Open dialog box appears with the list of bitmaps.

7. Select the desired bitmap file and click Open.

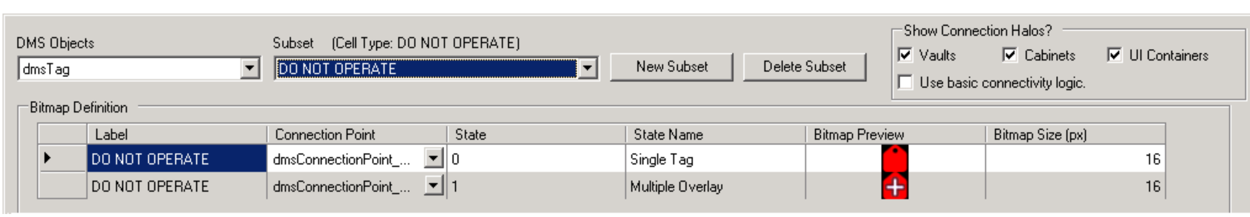
   The Bitmap field shows the preview of the new bitmap.

8. To save the changes to the `viewerconfig.xml` file, from the File menu, click Save.
9. To update the icons of the existing safety documents, click the Refresh button on the geographic network view, as shown below.



6.1.3. Tags

To associate each tag type with a bitmap, select the dmsTag object, and then use the Subset drop-down menu to select a tag type.



To reduce possible confusion in the future, clear the labels on any unused tag types.

Note

The tags in SCADAMOM under the TAGTYP records may not be defined in order of priority.

For example, when tag types are derived via the /ID_TAGTYP(*) command and copied into Viewer Configuration Editor, the tag list may be defined as follows:

TAGTYP(1) ---- PRIOR1_TAGTYP = TRUE

TAGTYP(2) ---- PRIOR3_TAGTYP = TRUE

TAGTYP(3) ---- PRIOR5_TAGTYP = TRUE

TAGTYP(4) ---- PRIOR4_TAGTYP = TRUE

TAGTYP(5) ---- PRIOR2_TAGTYP = TRUE

TAGTYP(6) ---- PRIOR6_TAGTYP = TRUE

TAGTYP(7) ---- PRIOR7_TAGTYP = TRUE

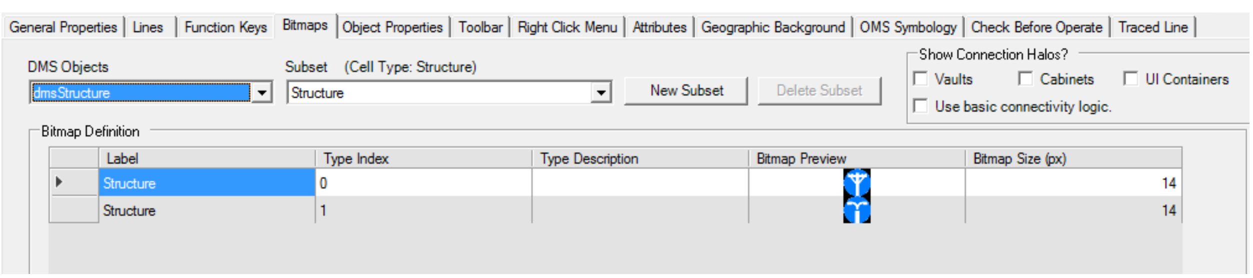
Note that tag types are not defined in order of priority. In this case, the Viewer Configuration Editor definitions need to be re-ordered to match.

6.1.4. Structures

A structure is a non-electrical asset that can be shown in the geographic display. Typically, Structures include poles, streetlights, stop lights or any other item that the utility might need to reference by geographical location. Structures can have properties that can be queried from the client, and identifiers that can be used for location purposes.

For more information about structures, refer to the *Geographic Background Compiler User Guide*.

To associate each structure type index with a bitmap, select the Structure DMS object. You can configure the zoom levels of structure bitmaps using the Object Properties tab. To set a fixed bitmap size for a specific structure type, adjust the Bitmap Size on Object Properties tab. If a size of 0 is specified, the zoom levels will be determined by the settings on the Object Properties tab.



The following fields are available for the Safety Documents DMS object:

- **Label:** Name of the structure type. (Read Only)
- **Type Index:** Index of the structure type. Corresponds to the **typeIndex** field in the structure data files. This index is also used in the Structures toolbar to control the visibility of the bitmaps

on the geographic display.

- **Type Description:** Description of the structure type.
- **Bitmap Preview:** Bitmap image associated with the safety document type.

New structure bitmaps and types can be added by using the "Insert" button at the bottom of the page.

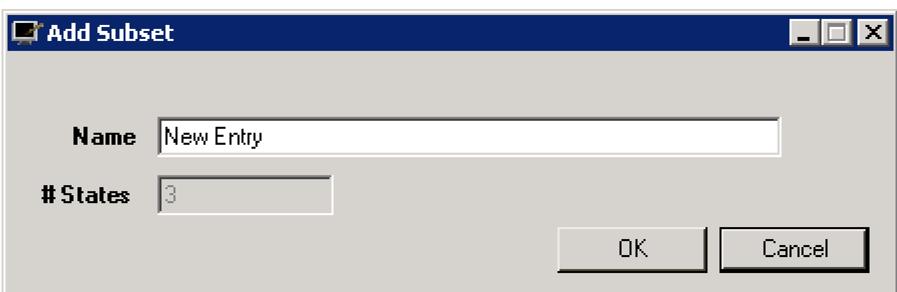
6.2. Adding New Object Types to the Bitmap Tab

New object types can be added for annotations, safety documents, tags, cuts, trouble tickets, and vehicle objects using the procedure below.

> **Note**
>
> Some objects, such as annotations, require initial definition in a separate configuration tool prior to adding them to the Viewer Configuration Editor. For detailed instructions about how to define a new annotation type, refer to the *Annotation Importer/Exporter User Guide*.

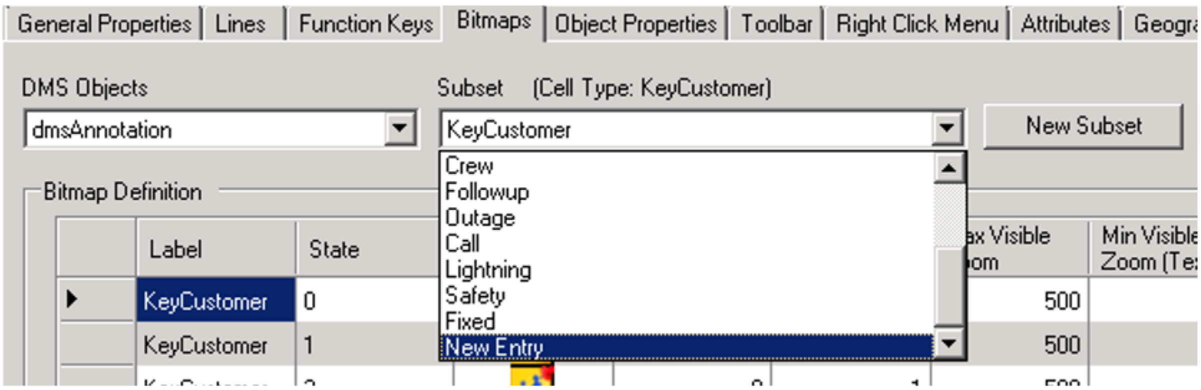
1. In the Viewer Configuration Editor, select the Bitmaps tab.
2. Select a network object in the DMS Object section.
3. Click the New Subset button to the right of the Subset section.

   The Add Subset pop-up appears.

4. Enter text in the Name field (for annotations, the name must be identical to the name defined in the Net Dynamics Configuration Editor), enter the numeric value in the #States field, and then click OK.

   A new object type entry is created at the bottom of the existing list.



The State list of a new object type is a zero-based numeric list containing entries equal to the value entered in the #States field. If the #States field contains 3, the State List shows 0, 1, and 2.

5. Enter values in the empty fields in the Bitmap Definition pane for the new object type.

   For a new annotation type, set the bitmap image, enter values in the Size, Min Visible Zoom, Max Visible Zoom, Min Visible Zoom (Text), Max Visible Zoom (Text), Min Size (Text), and Max Size (Text) fields, and set the annotation text color as necessary.

   - Zero is a reasonable value that can be used for some of the fields.
   - The maximum font size is generally 72.

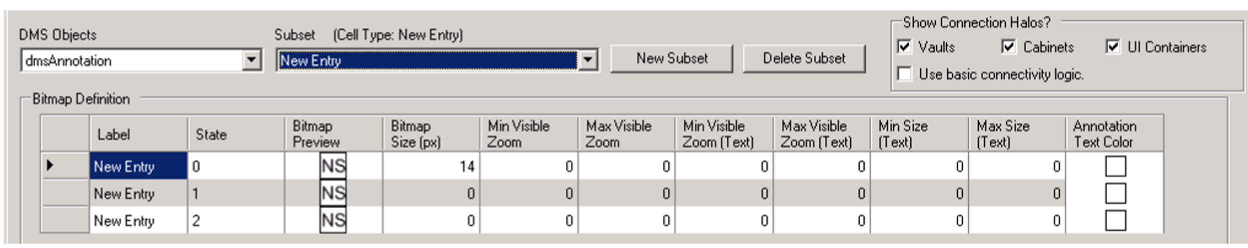


Figure 18. Adding a New Annotation Type to the Viewer Configuration Editor

6.3. Mapping New Object Types with Bitmaps

To assign a new bitmap to an object type:

1. In the Viewer Configuration Editor, select the Bitmaps tab.
2. Select a network object in the DMS Object section.

   A list of the selected DNOM object subsets appears in the Subset section.

3. Select a subset to modify.

   A list of bitmap definitions for the selected subset appears in the Bitmap Definition section.

4. Select the Bitmap Preview field for the selected state of the object type.

   An Open dialog box appears with the list of bitmaps.

5. Select the desired bitmap file name, and click Open.

> **Note**
>
> Only *.bmp and *.png images are supported.

The Bitmap Preview field shows the new bitmap.

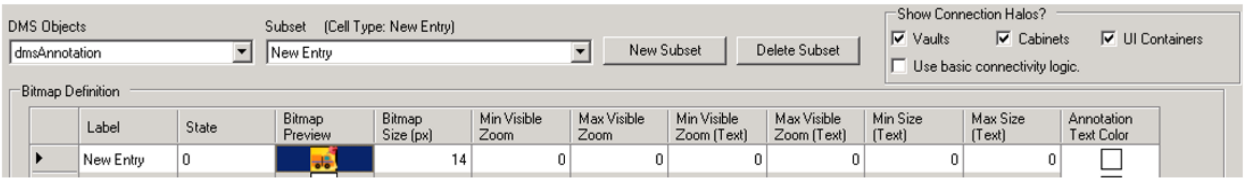


Figure 19. Mapping a New Object Type with a Bitmap

6.4. Bitmap Folder Location

The location of the bitmaps folder can be defined in the Viewer Configuration Editor startup parameter "/bpath<*bitmap path*>", where *bitmap path* is the path to the folder.

The default command defined in the Viewer Configuration Editor’s shortcut is:

`/BPATH "%IDMS_ROOT%\bitmaps"`

This provides the ability to use the exact same set of bitmaps in the client’s folder, if desired.

6.4.1. Changing the Default Bitmap Folder Location

To change the default bitmap folder location for the Viewer Configuration Editor:

1. Open the `NetDynamicsConfigEditor.exe`.

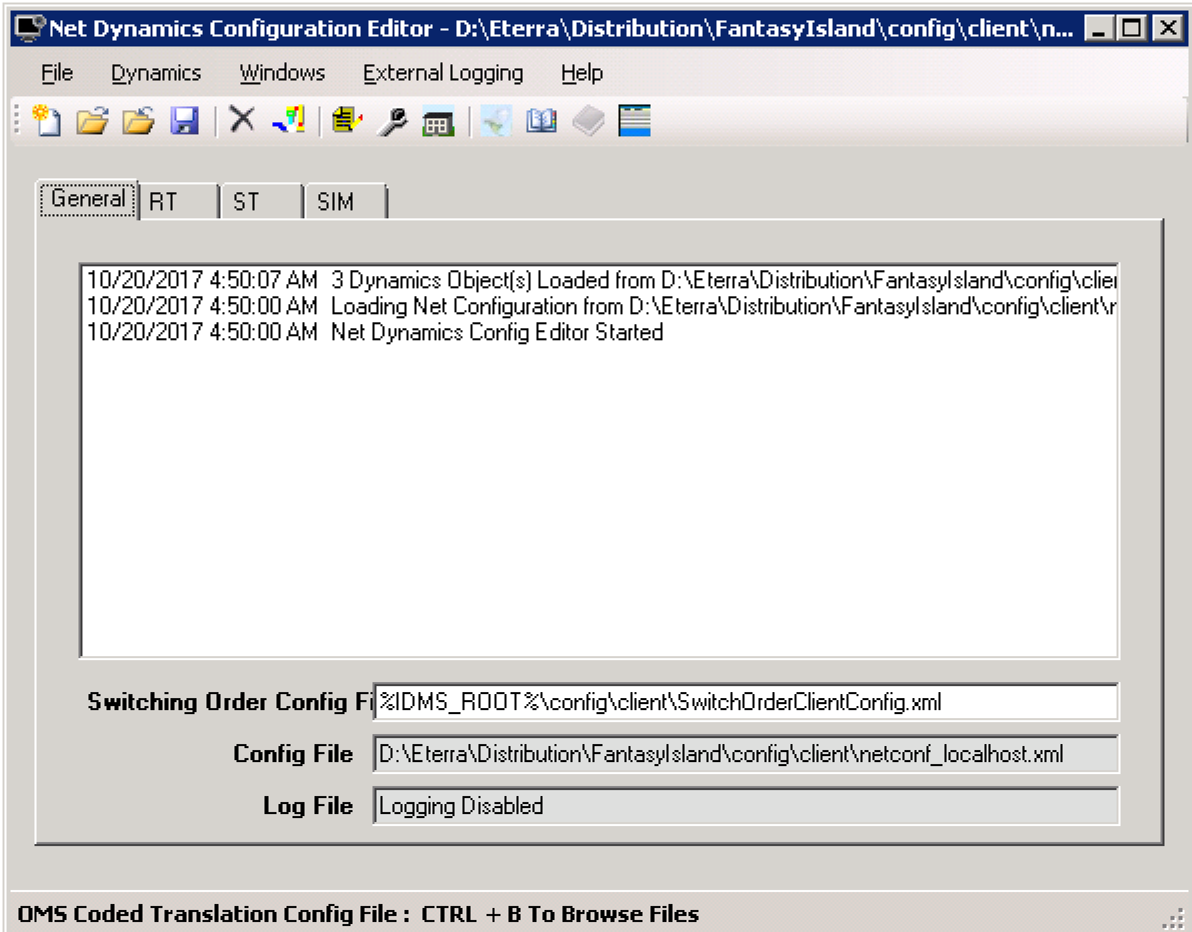
   The default installation path is:
   `D:\Eterra\Distribution\Tools\bin\NetDynamicsConfigEditor.exe`

   The Net Dynamics Configuration Editor application starts.

2. Select File > Open, and select the `netconf_localhost.xml` file from the list on the Open pop-up display, where *localhost* is a server name.

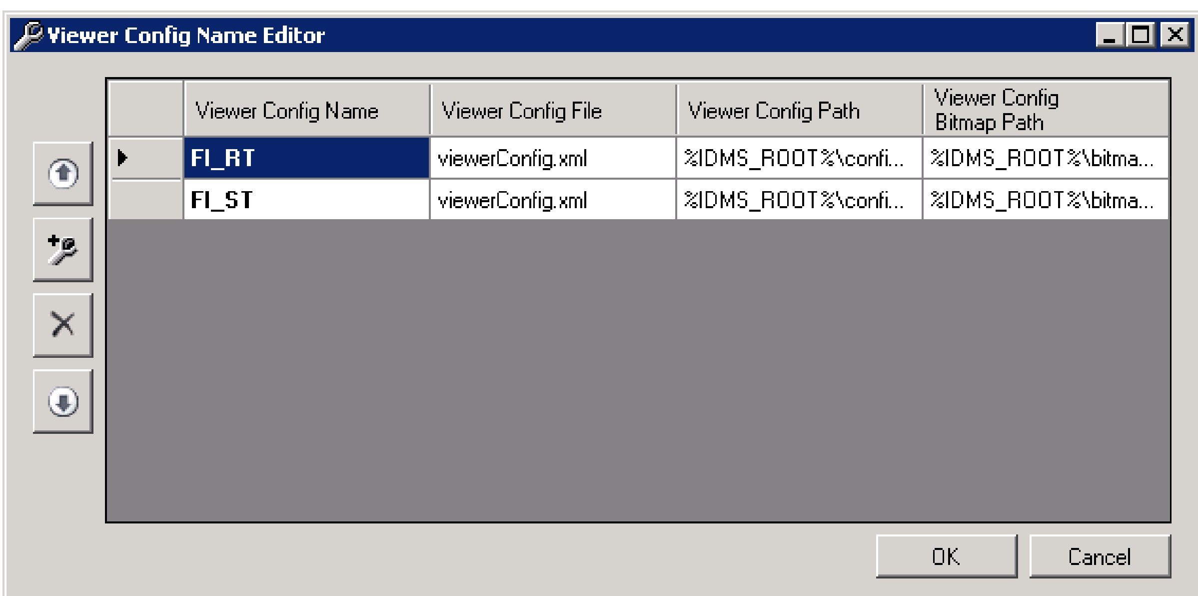
   The default configuration file’s location is:
   `D:\Eterra\Distribution\FantasyIsland\config\client`

   The network configuration is uploaded, and three new tabs named RT(real-time), ST (study), and SIM (training simulator) appear, as shown below.



3. Click the Viewer Configuration Editor icon on the top toolbar.

   The Viewer Configuration Name Editor pop-up display appears, as shown below:



4. Enter a new bitmap path or modify the existing path in the Viewer Config Bitmap Path field and click OK.
5. Save the changes via the File> Save menu.
6. Locate the Viewer Configuration Editor’s shortcut.
7. Right-click the application icon, and select the Properties option from the right-click menu.

   The Viewer Configuration Editor Properties pop-up window appears.

8. Select the Shortcut tab.
9. In the Target field, locate the "/bpath <*bitmap path*>" command, where *bitmappath* is the full path to the folder, and enter the new path exactly as defined in the Net Dynamics Configuration Editor.
10. Click the Apply button, and then click OK.

    The Viewer Configuration Editor looks for the bitmaps folder using the new path.

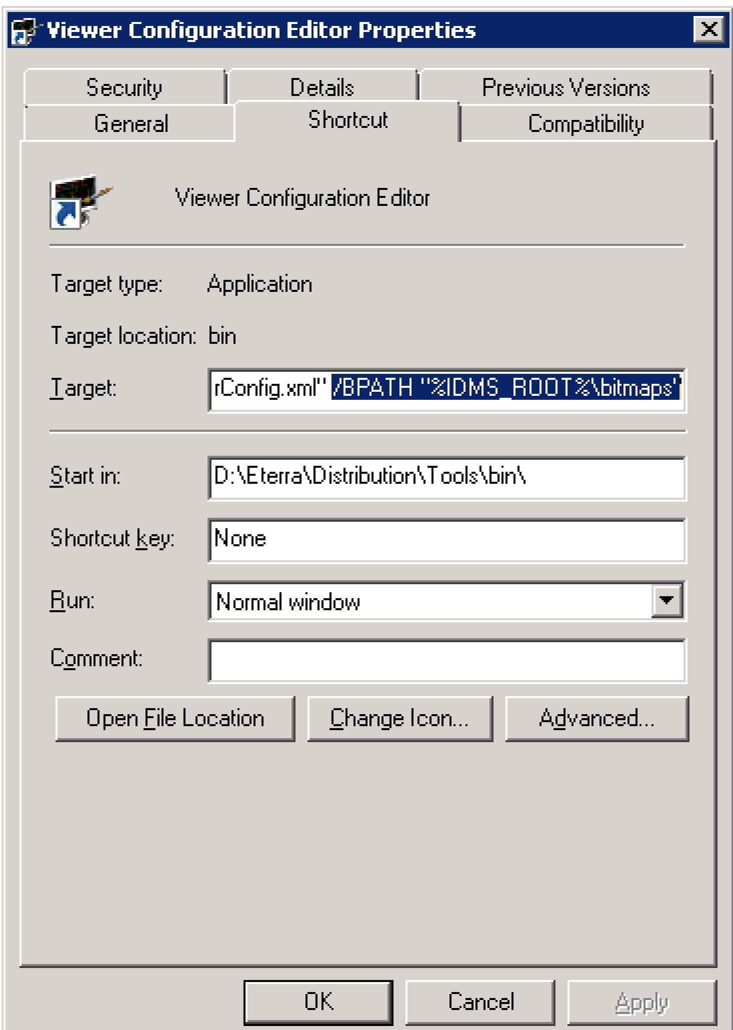


Figure 20. Viewer Configuration Editor Properties Display

7. Object Properties

The Object Properties tab (below image) is used to configure how specific device types are displayed on the geographic network view and in other displays such as internal worlds.

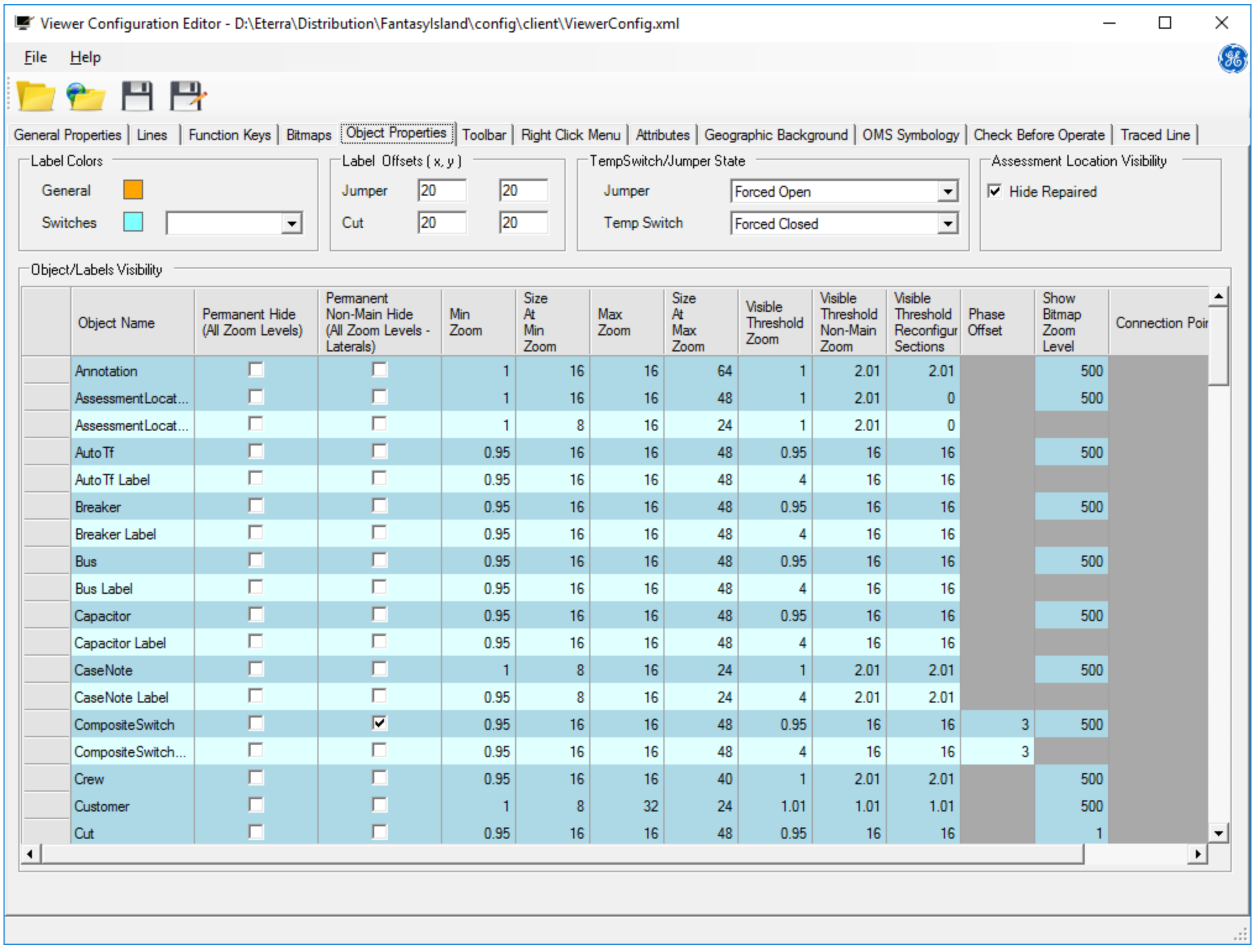


Figure 21. Object Properties Tab

7.1. Label Colors

The Label Colors group allows you to define label colors for switching devices.

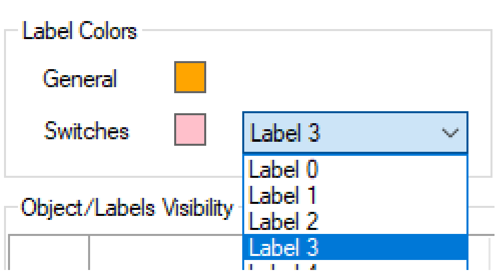


Figure 22. Setting Label Colors

The Label Colors group includes the following options:

- **General:** Deprecated. The label color for general objects (substations, transformers, capacitors and so on) is white and cannot be configured on the Label Colors group.
- **Switches:** Allows you to define switch label colors for different switch voltage levels. The switch voltage level is defined by the connectivity model, which is calculated by the Network Topology Processor (NTP).

  The label index in the configuration matches the voltage level of the switch.

  - Label 0 matches switches with the line-to-ground (LG) voltage level 0 (over 25 kV LG).
  - Label 1 matches switches with the line-to-ground voltage level 1 (0 – 5 kV LG).
  - Label 2 matches switches with the line-to-ground voltage level 2 (5 – 10 kV LG).
  - Label 3 matches switches with the line-to-ground voltage level 3 (10 – 15 kV LG).
  - Label 4 matches switches with the line-to-ground voltage level 4 (15 – 25 kV LG).

To change the label color, select a label index from the drop-down list and click a color patch to open the color selection palette.

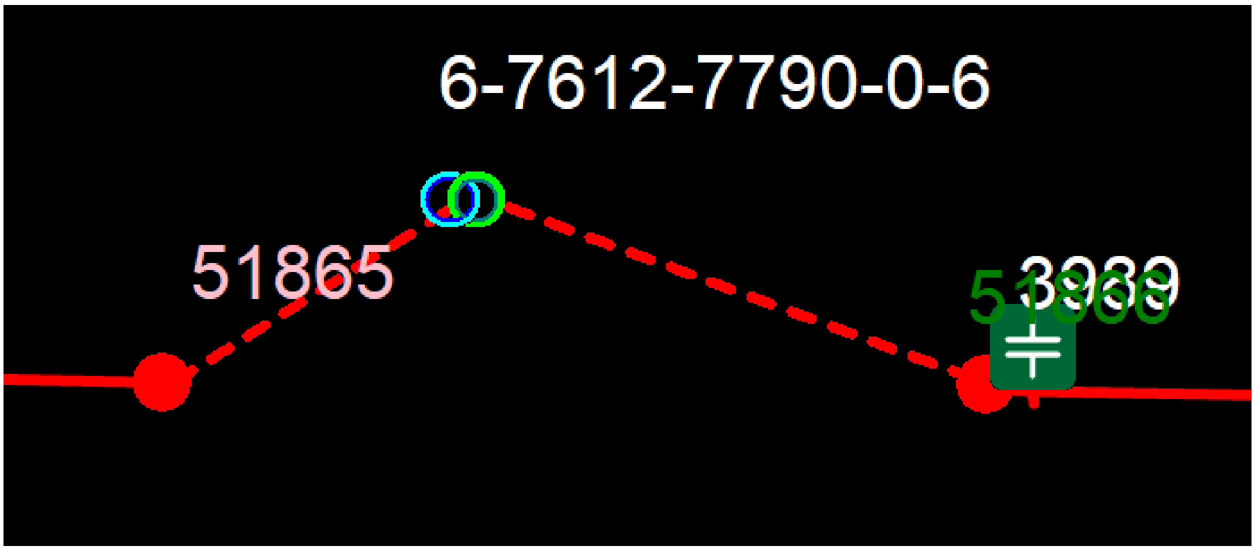
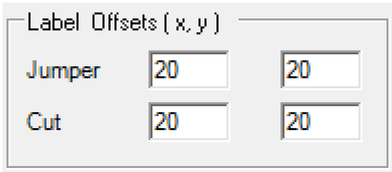


Figure 23. Switches and a Transformer with Labels Displayed

The switch label color is selected based on the switch voltage level. For example, the above image shows labels for a 7.62 kV LG switch (Label 2, green), a 13.28 kV LG switch (Label 3, pink), and an auto-transformer and capacitor (General Label, white).

7.2. Label Offsets and Sizing

The Label Offsets configuration panel allows you to define the offsets for cut and jumper labels when these objects are created on the geographic network view (in pixels).



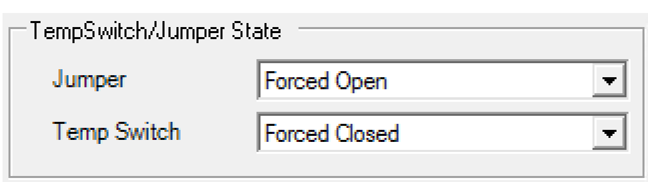
The Label Sizing Configuration supports individual label placement sizing configurations. By default, label sizing is a non-linear zoom based range specified per Object type in the Object/Labels Visibility table.

The following options are available:

- **Ignore Model Size:** Default behavior. Ignores any model provided label placement sizing information and uses the zoom based non-linear range specified in the Object/Labels Visibility table.
- **Model is min Size:** Any model label placement provided sizing is used as the "Size at Min Zoom" in the zoom based non-linear range specified in the Object/Labels Visibility table for that particular label placement. Any label placements not containing sizing information will default the zoom based non-linear sizing configured in the Object/Labels Visibility table.
- **Model is max Size:** Any model label placement provided sizing is used as the "Size at Max Zoom" in the zoom based non-linear range specified in the Object/Labels Visibility table for that particular label placement. Any label placements not containing sizing information will default the zoom based non-linear sizing configured in the Object/Labels Visibility table.
- **Size is Static:** Any model label placement provided sizing is used as the static size for that particular label placement. Any label placements not containing sizing information will default the zoom based non-linear sizing configured in the Object/Labels Visibility table.

7.3. Temporary Switch and Jumper State

This group allows you to define the initial or permanent state for a newly created temporary switch or jumper on the geographic network view, as listed below.

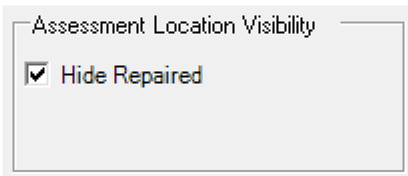


- **Blank:** The new device has no defined state.
- **Default Open:** The new device is created with the initial OPEN state. The state can be changed later using the device's toolbar.
- **Default Closed:** Indicates that new temporary switches or jumpers are created with the initial CLOSE state. The state can be changed later using the device's toolbar.
- **Forced Open:** Indicates that the new device is created with the permanent OPEN state. The user does not have an option to change the device state to CLOSE.

- **Forced Closed:** Indicates the permanent CLOSE state of the device. When you create a temporary switch or a jumper on the geographic network view, its state can only be CLOSE. The user does not have an option to change the device state to OPEN.

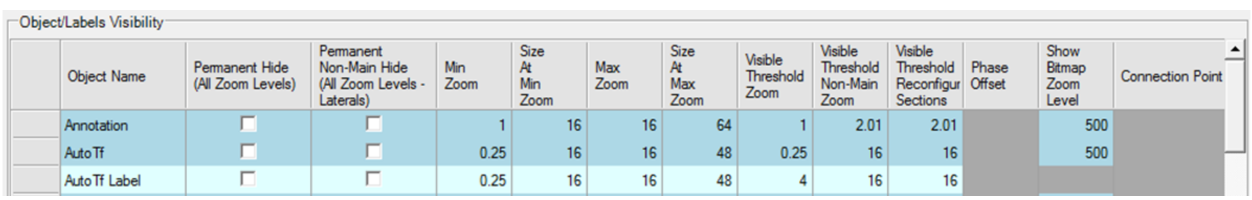
7.4. Assessment Location Visibility

This group allows you to define whether damage assessment locations are shown on the network view after being repaired.



7.5. Object/Labels Visibility Grid

The network view display for each type of device and its label can be individually configured using Object/Label Visibility panel.


Object/Labels Visibility

Object Name	Permanent Hide (All Zoom Levels)	Permanent Non-Main Hide (All Zoom Levels - Laterals)	Min Zoom	Size At Min Zoom	Max Zoom	Size At Max Zoom	Visible Threshold Zoom	Visible Threshold Non-Main Zoom	Visible Threshold Reconfigur Sections	Phase Offset	Show Bitmap Zoom Level	Connection Point
Annotation	☐	☐	1	16	16	64	1	2.01	2.01		500	
AutoTf	☐	☐	0.25	16	16	48	0.25	16	16		500	
AutoTf Label	☐	☐	0.25	16	16	48	4	16	16			

Modify the appropriate row to configure the visibility properties of a device label or image.

- **Permanent Hide (All Zoom Levels):** This option indicates whether the device or label must be permanently hidden at all zoom levels.
- **Permanent Non-Main Hide (All Zoom Levels - Laterals):** This option indicates whether the device or label must only be displayed on the Non-Feeder Main displays (that is, it only appears when the laterals are displayed or on an internal world display).
- **Min Zoom:** Sets the minimum zoom level at which the object’s size is scaled.
- **Size at Min Zoom:** Sets the minimum object’s size (in pixels) as the display is zoomed out. The object’s size does not decrease below this value.
- **Max Zoom:** Sets the maximum zoom level at which the object’s size is scaled.
- **Size at Max Zoom:** Sets the minimum object’s size (in pixels) as the display is zoomed out. The object’s size does not increase above this value.
- **Source Input Size:** This feature has not yet been implemented.
- **Visible Threshold Zoom:** This is the minimum zoom level at which the object is visible on the geographic network view.
-

Visible Threshold Non-Main Zoom: This is the minimum zoom level at which the object is visible on Non-Feeder Main displays (that is, lateral displays or internal world displays).

- **Visible Threshold Reconfigurable Sections:** This is the minimum zoom level at which the object is visible on Reconfigurable Sections displays.
- **Phase Offset:** Sets the value that is used to offset coincident objects in GIS units (typically in feet).

  If three fuses are exported with the same coordinates from the GIS (which is highly likely for A, B, and C phased fuses), this feature can be used to spread out the three symbols and their respective labels on the geographic display. This allows a dispatcher to visually determine the number of phases present where a lateral is fused off the feeder backbone.
- **Show Bitmap Zoom Level:** At this zoom level, the client switches from drawing the vector shape to drawing a configured bitmap on the geographic network view.

  Many objects, such as switches or landmarks, are represented by a default non-configurable vector-based shape. Optionally, an object can be configured to use a bitmap. Setting a certain zoom level to display a bitmap instead of a vector shape for an object improves the system performance.

  A level of 1 is equal to 100% zoom. A zoom level of500 is the highest supported zoom level. Setting the Show Bitmap Zoom Level to 500 causes the bitmap to appear at the highest zoom level only so that the object is typically represented with the vector shape.
- **Connection Point:** This drop-down box shows the orientation of the connectionpoint on the bitmap or vector image. There are nine possible orientations (NW, N, NE, E, SE, S, SW, W, and Center).

8. Toolbar

The Toolbar tab allows you to modify, create, or remove buttons on the selected toolbar in the ADMS client. You can also configure the toolbar size and background color.

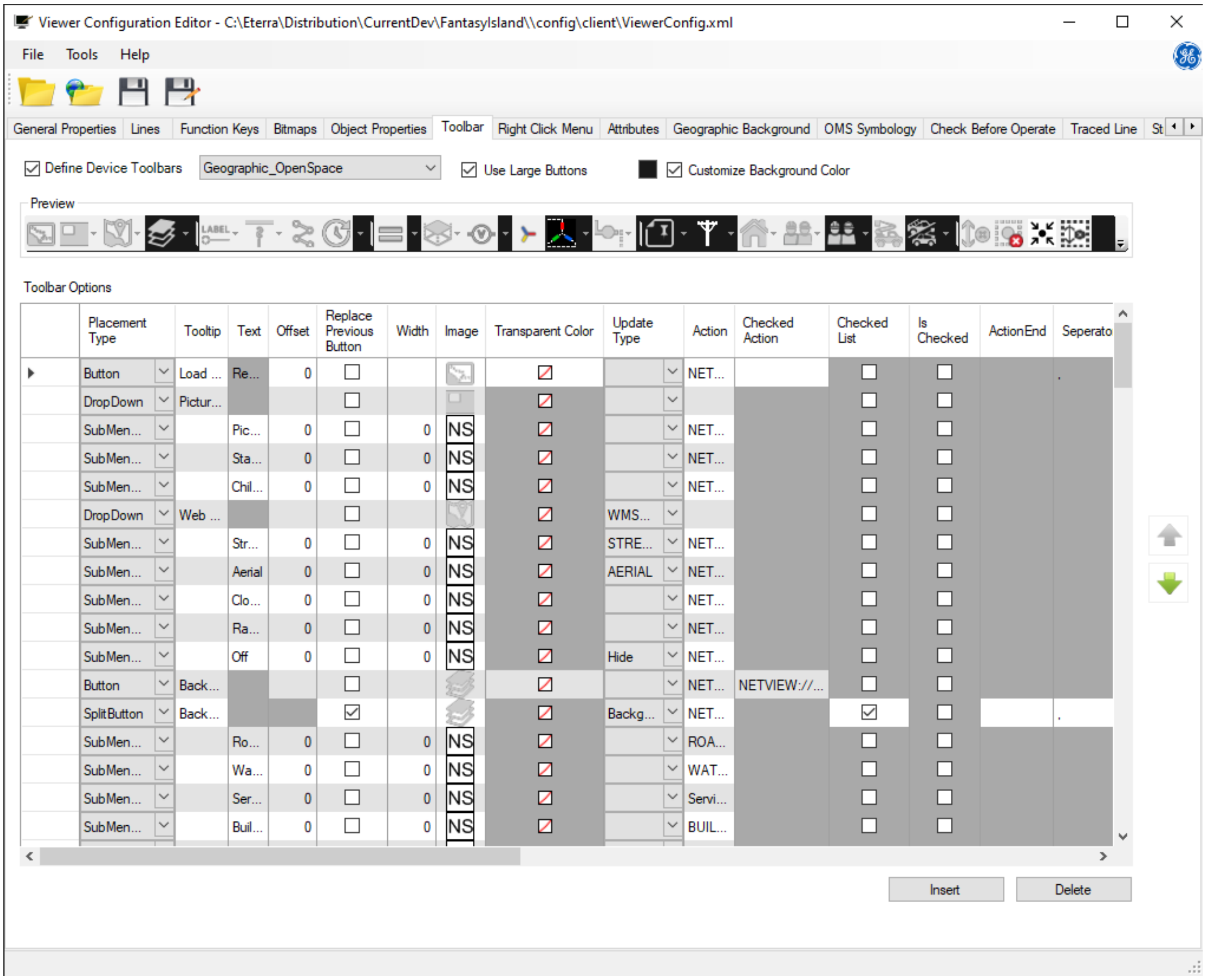


Figure 24. Toolbar Tab

The Toolbar tab includes the following fields and controls:

- **Define Device Toolbars:** This checkbox allows you to enable or disable the selection of toolbar drop-down list.
- **Toolbar Name:** Allows you to select the toolbar from a drop-down list.
- **Use Large Buttons:** If this option is selected, all toolbars are shown in the larger size. If not selected, the standard toolbar size is used.
- **Customize Background Color:** Select this option to customize the background color for all toolbars. To select the background color, click the color box at the left.

The Preview pane shows the default list of buttons that have been defined for the selected toolbar.

These buttons can be re-ordered or deleted using the Toolbar Options grid.

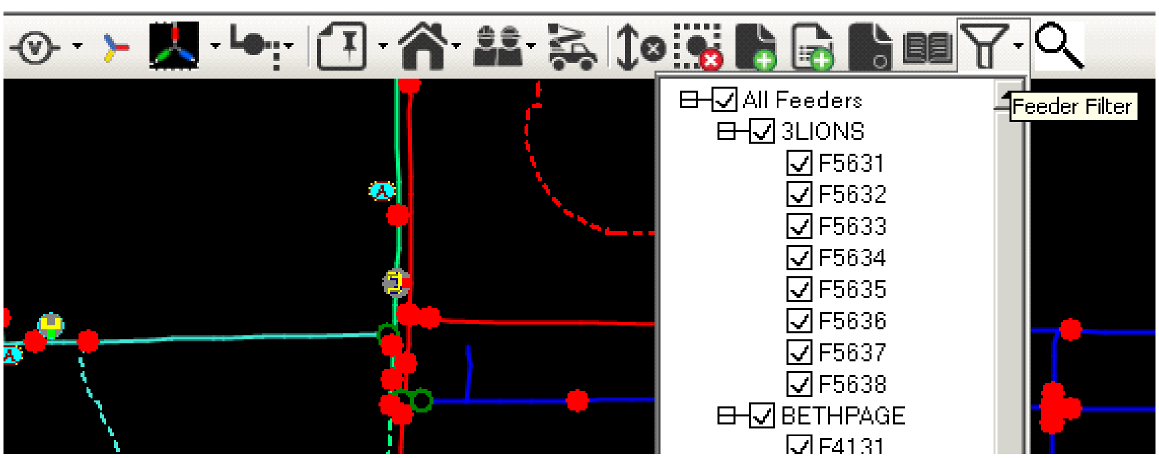
8.1. Toolbar Options

On the Toolbar Options grid, you can add, modify, and remove buttons, separators, split buttons, or pull-down menus on the selected toolbar.

The Toolbar Options grid allows you to configure the following parameters:

- **Placement Type:** Allows you to choose what type of object to add to the toolbar.
  - **Button:** Creates a button on the toolbar.
  - **Combo Box:** Creates a drop-down list of options on the toolbar. Each option has its own command.
  - **Drop Down:** Creates an icon with a drop-down arrow that opens a drop-down menu, which is populated with submenu items.
  - **Label:** Creates a non-editable text field on the toolbar.
  - **Split Button:** Creates an object with a button and a drop-down list that can be operated separately. The associated drop-down list contains checkboxes.
  - **Text Box:** Creates an enterable text box on the toolbar.
  - **Submenu Item:** Adds a submenu item for the corresponding drop-down menu, combo box, or split button.
  - **Separator:** Adds a separator bar on the toolbar.

    A separator bar can be used to separate menu sections. It has no properties except for visibility, to allow hiding separators when the whole section needs to be hidden.
  - **FeederFilter:** Adds the Feeder Filter button to the geographic (common) toolbar.

  - **Show Font Resize:** Adds the font zooming list to the geographic (common) toolbar.
- **Tooltip:** The pop-up text that appears when the user hovers the mouse pointer over a toolbar object.

- **Text:** This text appears in combo boxes, submenu items, labels and text boxes by default.
- **Offset:** Increases the spacing between icons on the toolbar (in pixels).

  The offset is relative to the end of the previous toolbar object placement. Setting the offset to a positive value in pixels moves the object placement to the right and creates empty space between it and the previous toolbar object placement. Setting the offset to a negative value moves the toolbar object placement to the left, on top of the previous object placement (that is, a button).

  This feature, in conjunction with the definition of the visibility feature, allows for objects to be conditionally displayed on the toolbar.
- **Replace Previous Button:** When selected, this replaces the previous visible placement on the toolbar. This is an alternative to setting a negative value in the Offset column.
- **Width:** Sets the width of an object (that is, a button) on the toolbar. Any associated image is centered in the object and does not get stretched or compressed to fit the space. To default the width of an object to the width of the associated image, set the Null or Zero value in the Width field.
- **Image:** Associates the image with the toolbar object.
- **Transparent Color:** Sets a color on an icon to be transparent via the Color dialog box.
- **Update Type:** Specifies which type of viewer update is used to update this toolbar item. The available update types include:
  - **ActiveCustomers:** Active customer visibility changes.
  - **Aerial:** Aerial layer visibility changes.
  - **AllCrews:** All crew visibility changes.
  - **AllCustomers:** All customer visibility changes.
  - **AnnoTypesVisible:** Annotation layer visibility changes.
  - **AvailableCrews:** Available crew visibility changes.
  - **BackgroundLayerName:** Background layer visibility changes.
  - **CallingCustomers:** Calling customer visibility changes.
  - **CapacityMarginHalosVisible:** Capacity margin halo layer visibility changes.
  - **CombinedHalos:** Combined fault indicator and fault location halo visibility changes.
  - **ConnectedCustomers:** Connected customer visibility changes.
  - **CrewLayer:** Crew layer visibility changes.
  - **CriticalCustomers:** Critical customer visibility changes.
  - **CustomerLayer:** Customer layer visibility changes.
  - **FaultIndicatorHalos:** Fault indicator halo visibility changes.
  - **FilterFeeder:** Filtered feeder changes.
  - **FutureVisible:** Future layer visibility changes.

- **GeographicPhaseColorsVisible:** Geographic phase color visibility changes.
- **HideCrews:** Crew visibility changes.
- **HideUnavailableCrews:** Unavailable crew visibility changes.
- **LabelVisible:** Label visibility changes.
- **LinesVisibility:** Line visibility changes.
- **ManualHalos:** Manual halo visibility changes.
- **NoCustomers:** No customer visibility changes.
- **NoHalos:** No halo visibility changes.
- **NonMainVisible:** Non-main layer visibility changes.
- **NonReconfigVisible:** Non-reconfigurable layer visibility changes.
- **OverheadAndUnderground:** Overhead and underground line visibility changes.
- **OverheadLines:** Overhead line visibility changes.
- **PhaseNotVisible:** Phase layer visibility changes.
- **PriorityCallingCustomers:** Customer calls of any priority visibility changes.
- **ShowFaultIndicatorHalos:** Fault indicator halos layer visibility changes.
- **ShowOverloadHalos:** Overload halos layer visibility changes.
- **ShowPotentialFaultHalos:** Potential fault halos layer visibility changes.
- **ShowFutureLayer:** Future equipment halos layer visibility changes.
- **ShowThirdPartyHalos:** Third-party owned equipment halos layer visibility changes.
- **ShowFutureRemoveLayer:** Future remove equipment halos layer visibility changes.
- **ShowRemovedLayer:** Removed equipment halos layer visibility changes.
- **ShowVoltageHalos:** Voltage halos layer visibility changes.
- **Street:** Street layer visibility changes.
- **StructTypesVisible:** Structure layer visibility changes.
- **TelemeteredHalos:** Telemetered halo visibility changes.
- **UnconnectedCustomers:** Not connected customer visibility changes.
- **UndergroundLines:** Underground line visibility changes.
- **VehiclesVisible:** Vehicle layers visibility changes.
- **VisibleCrewTypes:** Crew type layers visibility changes.
- **VisibleVehicleTypes:** Vehicle type layers visibility changes.
- **VoltLevelsNotVisible:** Voltage levels layers visibility changes.
- **WMSLayers:** Workforce Management System layers visibility changes.

For example, a background split button might use the background layer name update type so that the appropriate checkboxes are selected in the drop-down list.

- **Action:** Defines the Netview command to run when an object on the toolbar is clicked.

  For more details about Netview commands and to see a partial list of the commands and parameters, refer to Command String Definition.

- **Checked Action:** Defines an alternate action for check-on/check-off style operations.
- **Checked List:** Indicates that the drop-down list items associated with a split button use checkboxes.

  > **Note**
  >
  > When using a split button, this "Checked List" feature is almost always used with it.

- **Is Checked:** Indicates the default state of checkboxes associated with a split button.
- **ActionEnd:** String used to complete actions for all items associated with a split button.
- **Separator:** Separator used in the command syntax to list checked items associated with a split button.
- **Use Unchecked Items:** Indicates that the action associated with a split button includes a list of unchecked items.
- **Coded Translation:** Specifies the coded translation item, which defines the button's drop-down list. Applies only to split buttons with the Checked List option selected.
- **Font Size:** When the field is empty, the default text size is 8. Enter a smaller or larger number to decrease or increase the text size on the toolbar.
- **Text Color:** When the field is empty, the default text color is black. Type a color (for example, green, blue, etc.) to change the text color on the toolbar.
- **Text Alignment:** If text is displayed on the toolbar, you can choose to align it to the left, center, or right.
- **Visibility:** Defines the condition of the current toolbar item to be displayed on the toolbar.

  The visibility of a toolbar object is defined by the Visibility, Argument1, and Argument2 columns. The Visibility column specifies the type of comparison to be applied between Argument1 and Argument 2 (=, >, >=, <, < =, !=, True, False). If the comparison is satisfied, the toolbar item is displayed. The True and False values are compared only with the Argument 1 value and do not require Argument 2 to be defined.

  Argument 1 and Argument 2 support both regular text and field values.

- **Argument 1:** First argument used by visibility.
- **Argument 2:** Second argument used by visibility when two arguments are required.
- **Data Fields:** List of fields required by a toolbar item.

  The Data Fields column contains a list of all fields referenced in a given row (Tool Tip, Text, Action, Argument 1, and Argument 2). A data field only needs to be listed once, and it should not include the leading "%". Data fields are separated with a comma. "DYN" is a reserved keyword and should not be listed here

For example, for the Cut custom toolbar, the default tooltip value of the Label Placement type is %ADDRESS.NOAPPSTR, and the text field value is %TOPOSUBNAME %NAME. No other columns in this row contain the object field values. The Data Field column for this Placement type contains the following:
ADDRESS.NOAPPSTR,TOPOSUBNAME,NAME

- **Excluded Roles:** Lists client roles (Dispatcher, Operator, etc.), for which a toolbar item is hidden. To specify a client role, right-click the Excluded Roles field and select a role from the context menu, which is populated from the Client Roles list defined on the General Properties tab.
- **Included Roles:** Lists client roles, for which a toolbar item is shown.

> **Note**
>
> On the Line toolbar, in the definition of the Draw a Multiloop Schematic button, the LINEISURDLOOP flag in the Argument1 column should be specified to filter toolbar buttons for which lines allow the URDLoop Schematic button to be displayed. For the URD Loop Schematic button, visibility must be set to True.

The Toolbar Options grid includes the following buttons:

- **Insert:** Click to add a new toolbar button. Optionally, you can right-click the desired location and select Insert from the context menu.
- **Delete:** To delete a toolbar button, select the entire row and click the Delete button or right-click the desired record and select Delete from the right-click menu.
- **Move Up and Down Arrows:** Use these buttons to re-order the selected button on the toolbar. Optionally, you can right-click the desired record and select MoveUp or MoveDown from the context menu.

  > **Note**
  >
  > You can copy, paste, insert, delete, and move toolbar items via the right-click menu.

8.2. Adding a New Object to the Geographic_Openspace (Common) Toolbar

To add a new object type to the geographic (common) toolbar:

1. Select the Toolbar tab of the Viewer Configuration Editor.
2. Select the Define Device Toolbars option, and select the Geographic_Openspace toolbar from the drop-down list next to the checkbox.
3. Click the Insert button or right-click where you need to add a new record and select Insert from the context menu.

   A new blank line is added to the Toolbar Options grid.

4. Enter the new object information.
5. Select the new entry row and move it to the desired location on the toolbar using the Up or Down arrow buttons.

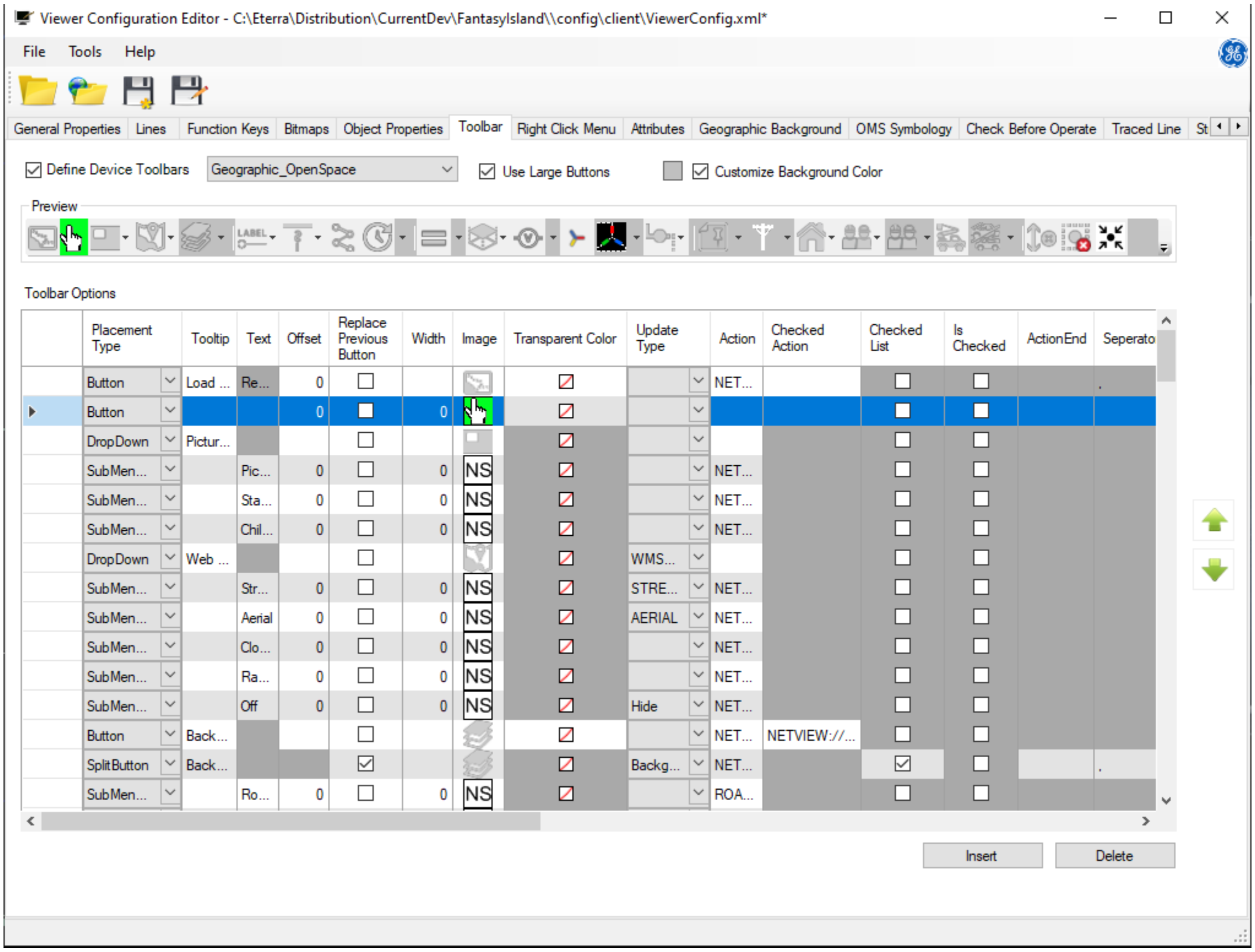


Figure 25. Adding a New Object to the Geographic Toolbar

9. Right-Click Menu

On the Right-Click Menu tab, you can configure the right-click (context) menu. You can add or delete right-click menu items as well as modify the bitmap image, menu text, action command, and other properties for menu items.

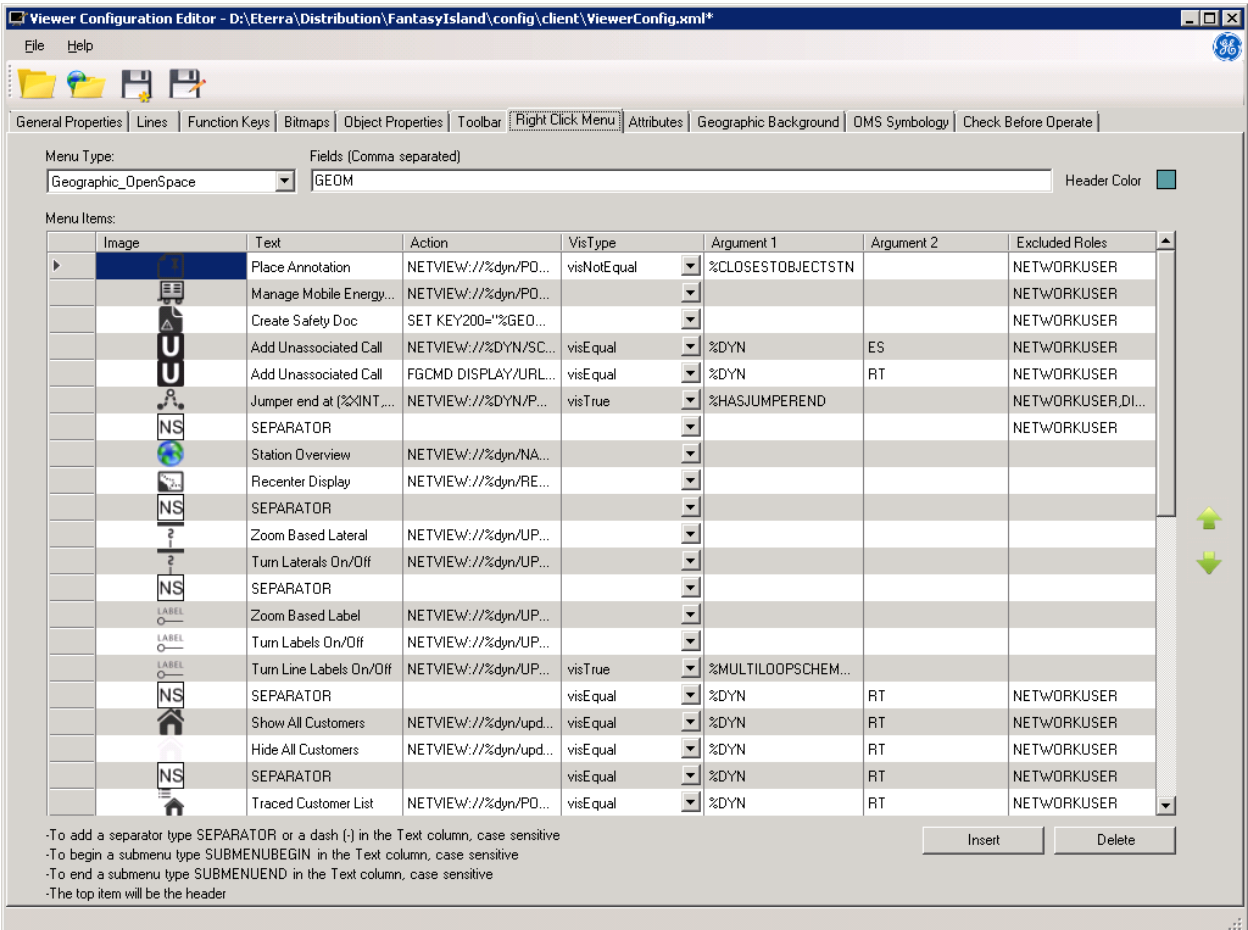


Figure 26. Right-Click Menu Tab

The Right-Click Menu tab includes the following fields and controls:

- **Menu Type:** Allows selecting the device menu to display from the drop-down list. The items that are displayed on this menu are then listed under Menu Items.
- **Fields:** A comma-separated list of fields that are queried for defining text, actions, and arguments in visibility tests. Usually, such fields are preceded by a percent sign ("%"). Fields must be typed in this text box so that the menu items can use these fields for data population and visibility conditions.

  For example, in the Action field you enable the Operator Attributes call up action for a switch by entering the following command:

  ```
  NETVIEW://%DYN/ATTRIBUTES/OPERATORATTRIBUTE/%DNOMOBJECTTYPE//%TOPOSUBNAME/%ID
  ```

For the above action, the following text must be added to the Fields parameter:

```
ID,TOPOSUBNAME
```

- **Header Color:** Specifies the color of the right-click menu header.

> **Note**
>
> The first option in the menu items list is the header of the right-click menu.

9.1. Menu Items

On the Menu Items grid, you can add, modify, and remove options, separators, or submenu options for the selected right-click menu.

The following fields are listed in the Menu Items table:

- **Image:** Specifies the image to include in the menu.
- **Text:** Sets the text to display in the menu.
- **Action:** Defines the command to run when an option is clicked.
- **Vis Type:** Type of test used to determine whether each menu item is visible. The choices are:
  - visEqual
  - visNotEqual
  - VisGreaterThan
  - VisLessThan
  - VisTrue
  - VisFalse

  For the first four types, Argument 1 is compared against Argument 2. For the VisTrue and VisFalse types, only Argument 1 is used.
- **Arguments:** Arguments used within each visibility test. For example, setting VisType to VisEqual checks to see if Argument 1 equals Argument 2 to determine if that menu item should be visible. These can be defined as static text or as fields to be resolved.
- **Excluded Roles:** Specifies client roles (Dispatcher, Operator, etc.), for which a context menu item is hidden. To specify a client role, right-click the Excluded Roles field and select a role from the context menu, which is populated from the Client Roles list defined on the General Properties tab.
- **Included Roles:** Lists client roles, for which a context menu item is shown.

> **Note**
>
> You can place a separator in the menu list by entering the word "SEPARATOR" or including a

dash ("-") in the Text or Action columns.

> **Note**
>
> You can add submenu items to a menu item. To begin a submenu, enter the word "SUBMENUBEGIN" in the Text column. To end a submenu, enter the word "SUBMENUEND" in the Text column.

The following controls are available for the Menu Items table:

- **Insert:** Adds a new row to the Menu Items grid. Optionally, you can right-click the desired location and select Insert from the context menu.
- **Delete:** Deletes the selected row in the Menu Items grid. Optionally, you can right-click a record and select Delete from the context menu.
- **Move Up and Down Arrows:** Use these buttons to re-order the selected rows on the Menu Items grid. Optionally, you can right-click a record and select MoveUp or MoveDown from the context menu.

9.2. Adding Single-Click Tag Options to the Switch Right-Click Menu

ADMS supports command syntax to initiate the creation of a specific tag type. You can create a specific type of tag directly from the Network view via the switch's right-click menu.

To add single-click tag options to the switch's right-click menu:

1. Select the Right-Click Menu tab in the Viewer Configuration Editor.
2. From the Menu Type drop-down list select dmsSwitch.
3. Click the Insert button and add a new entry with the following information:
    - **Image:** Select a Do Not Operate tag's image.
    - **Text:** Enter the "Place Do Not Operate Tag" text.
    - **Action:**

      ```
      MAIL/USER=TAGNOTES_SCADA_EMS "CRE_TAG/REC=SUBSTN/TAGTYPE=DO NOT
      OPERATE/EMSID=%TOPOEMSSUBNAME/DMSID=%TOPOSUBNAME.%FEEDERNAME.%DNOMRECORDNAME.%ID/DMSONLY=TRUE/
      AREA=%SERVCNTR.NAME/ONESTEP/USERID=DNO.%NAME/CREATOR=%DMSUSERNAME/COM1=SINGLE CLICK RED TAG
      PLACED/CREBY=%DMSUSERNAME/DMSNAME=%NAME/GEOMETRY=%OBJGEOMETRYNAME"
      ```

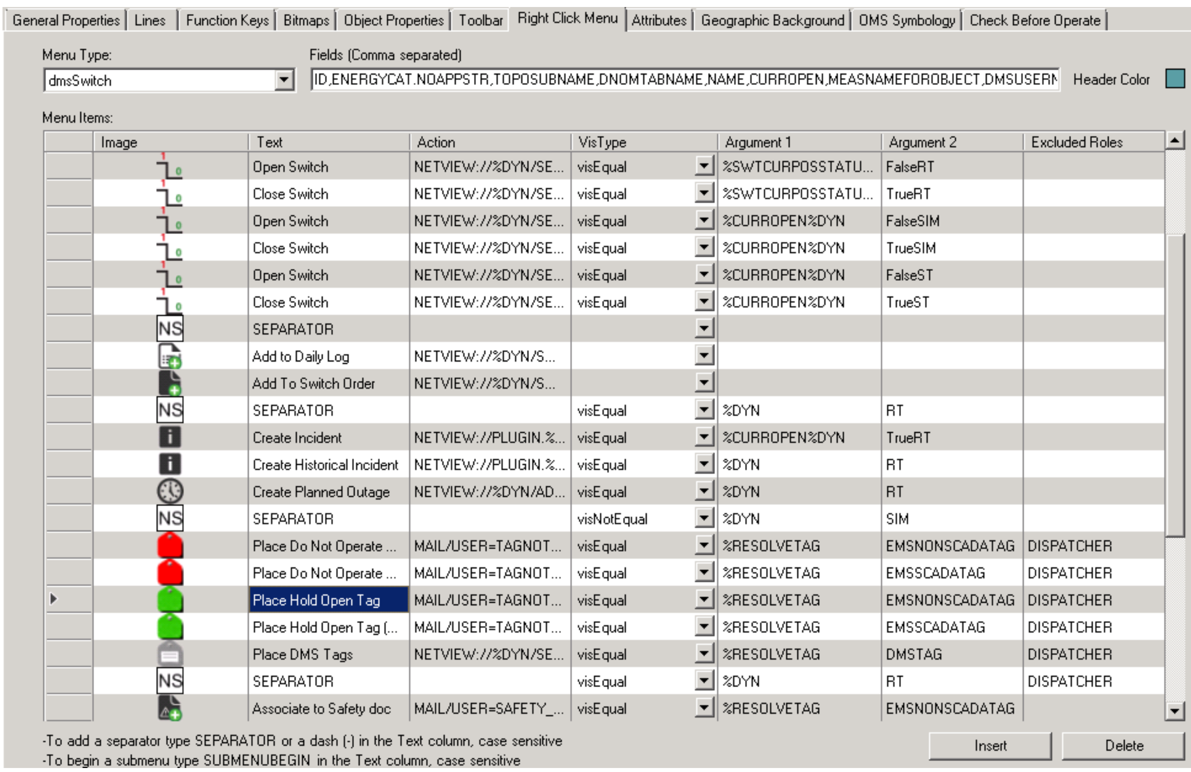
4. Add a second new item with the following information:
    - **Image:** Select a Hold Open tag's image.
    - **Text:** Enter the "Place Hold Open Tag" text.
    - **Action:**

```
MAIL/USER=TAGNOTES_SCADA_EMS "CRE_TAG/REC=SUBSTN/TAGTYPE=HOLD
OPEN/EMSID=%TOPOEMSSUBNAME/DMSID=%TOPOSUBNAME.%FEEDERNAME.%DNOMRECORDNAME.%ID/DMSONLY=TRUE/ARE
A=%SERVCNTR.NAME/ONESTEP/USERID=HO.%NAME/CREATOR=%DMSUSERNAME/COM1=SINGLE CLICK GREEN TAG
PLACED/CREBY=%DMSUSERNAME/DMSNAME=%NAME/GEOMETRY=%OBJGEOMETRYNAME"
```

5. Add the following information to the Fields text box:

```
ID,TOPOSUBNAME,NAME,DMSUSERNAME,SERVCNTR.NAME,DNOMRECORDNAME,FEEDERNAME,TOPOEMSSUBNAME,OBJGEOMETRY
NAME
```

6. Configure visibility conditions as required (for example, to show different text when adding tags from geographic or station internal views).



7. Save the file and close the Viewer Configuration Editor.

   The single-click tag options for the switch's right-click menu are now available.

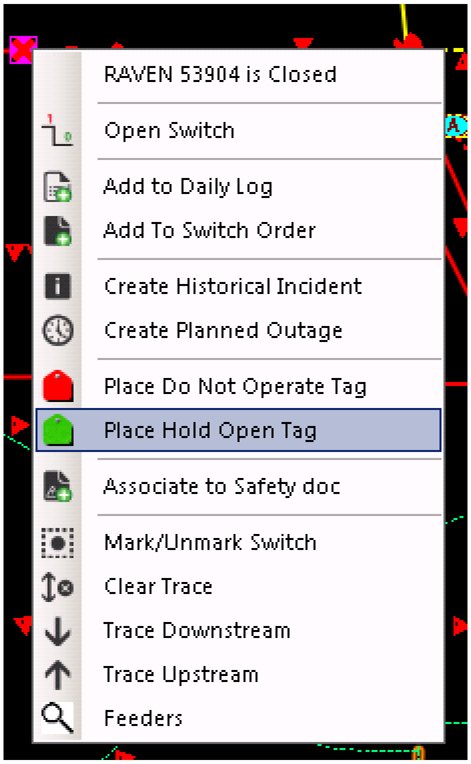


Figure 27. Single Click Tagging Options

10. Attributes

The Attributes tab is used to add and configure attribute and output parameters of various DMS objects available to analysts and operators on the geographic network view.

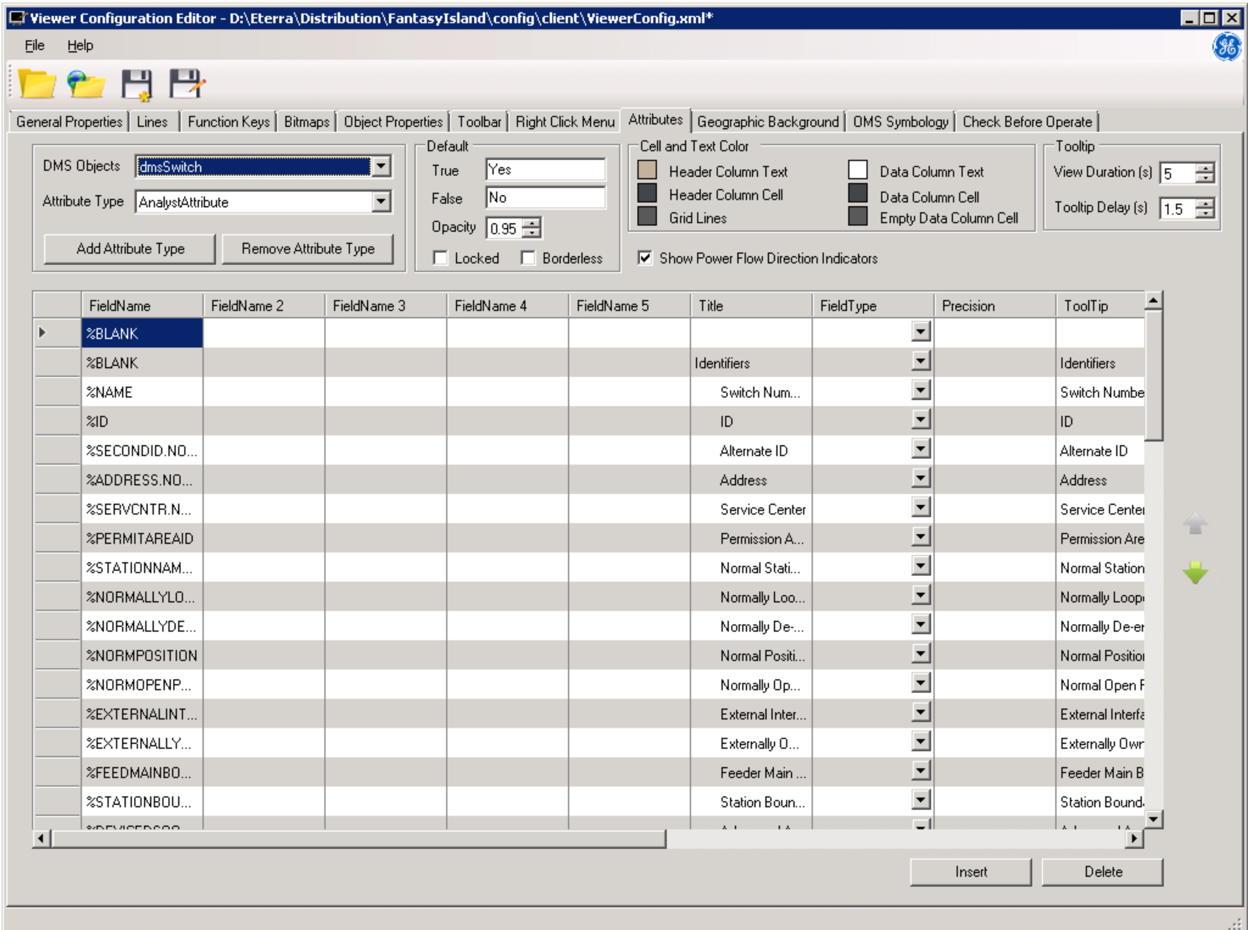


Figure 28. Attributes Tab

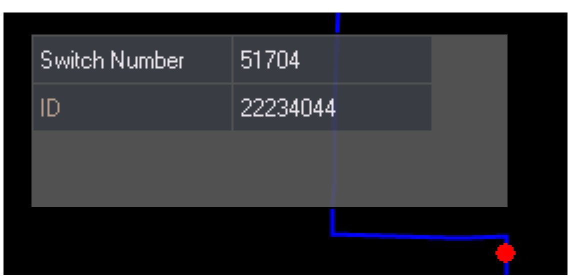
The Attributes tab includes the following fields and controls:

- **DMS Objects:** Distribution network operational model object type (for example, switch, transformer, capacitor, fuse).
- **Attribute Type:** The ADMS client supports four types of attribute panels and a tooltip panel. You can also create new attribute types by clicking the Add Attribute Type button.
    - **Analyst Attribute:** Provides detailed information for the selected device.
    - **Analyst Output:** Provides a detailed list of operating values, including the power flow output values for the selected device. The Analyst Output display also shows power flow direction indicators for the device.

      Attribute and Output displays for analysts typically provide additional information when compared to the equivalent operator displays.

- **Operator Attribute:** Provides a concise list of information that is relevant to normal operations.
- **Operator Output:** Provides a summary of the main operating values for the selected device.
- **Tooltip:** Provides key information for the selected device.

- **Add Attribute Type:** Click to create new attribute types. The Create Attribute pop-up window appears. Enter the name of the attribute and click OK.
- **Remove Attribute Type:** Click to remove the selected attribute type. The prompt dialog box appears.
- **Default Text True/False:** Enter alternate values for True and False, such as Yes and No for attribute and output pop-ups.
- **Opacity:** Specifies the opacity level of the attribute panel.
- **Locked:** If selected, an attribute panel is pinned at the left of the display.
- **Borderless:** If selected, an attribute panel is shown without borders.
- **Cell and Text Color:** Options to define the background and text colors on attribute panels.
  - **Header Column Text:** Defines the text color of the header column.
  - **Header Column Cell:** Defines the background color of the header column.
  - **Grid Lines:** Defines the color of grid lines on attribute panels.
  - **Data Column Text:** Defines the text color for data columns.
  - **Data Column Cell:** Defines the background color for data columns.
  - **Empty Data Column Cell:** Configures the color for empty cells on attribute panels.
- **Tooltip View Duration:** Specifies the duration a device tooltip is shown.
- **Tooltip Delay:** Specifies the time period before a device tooltip appears.



- **Show Power Flow Direction Indicators:** Specifies whether the Power Flow Direction Indicators pane is shown on the Analyst Output display.

10.1. Attribute Properties

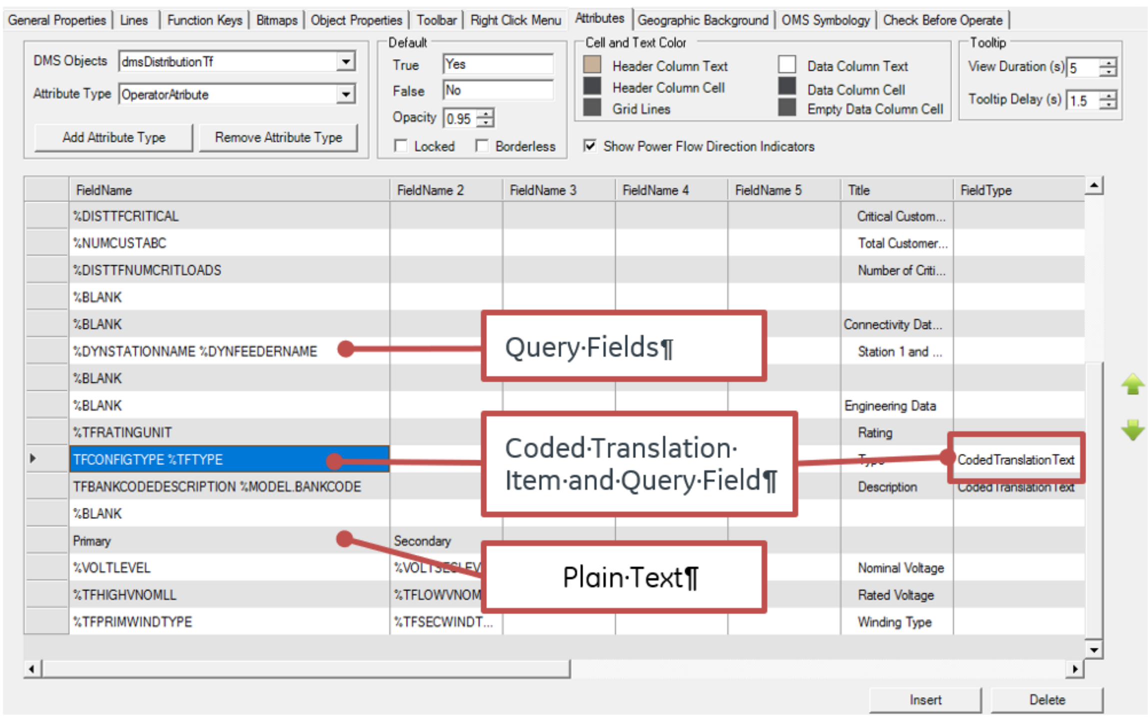
The Attribute Properties grid is populated for the selected DMS object and Attribute Type. The following fields are listed in this grid:

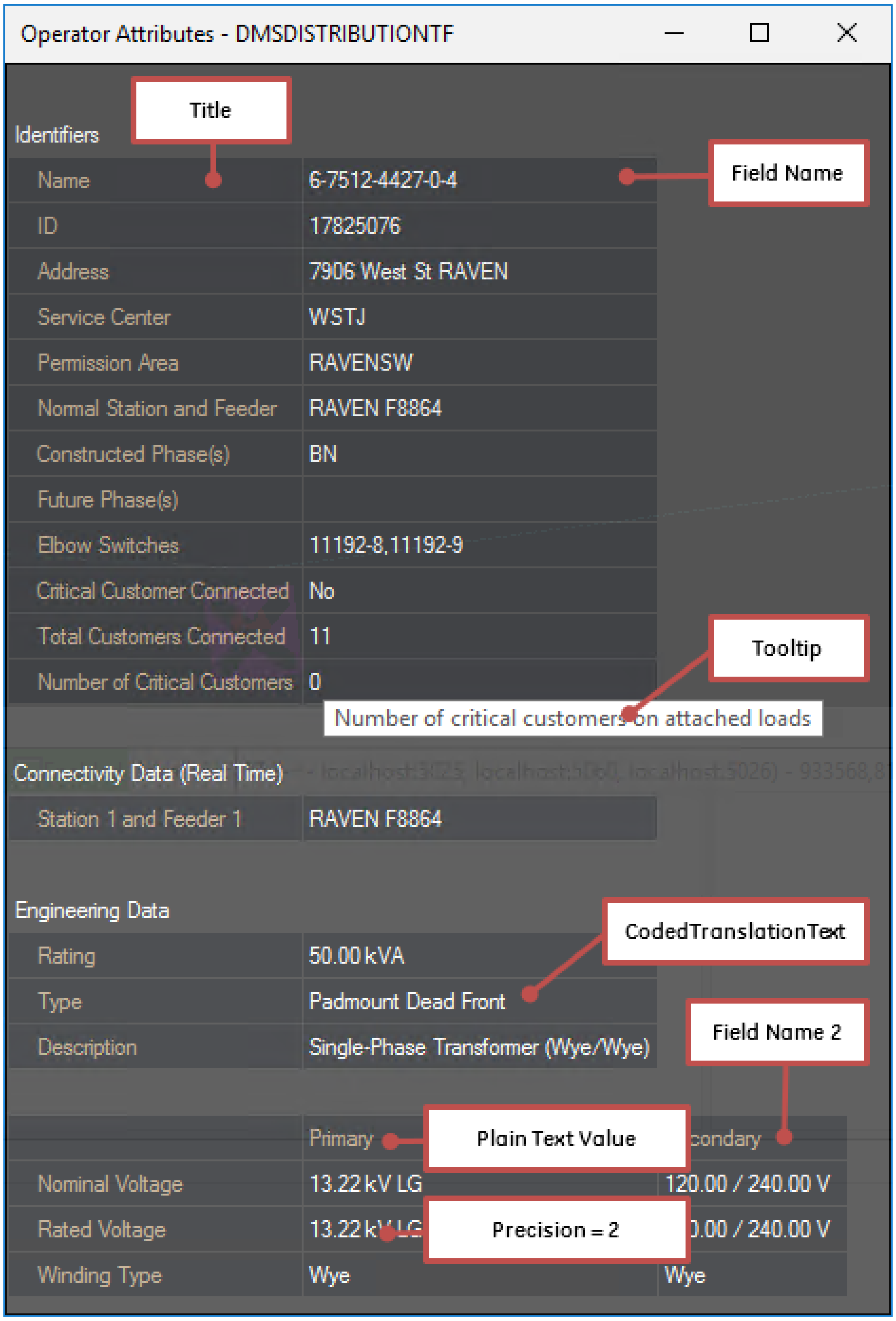
- **Field Name:** Sets the value to be displayed in the first Attribute Value column.

  The Field Name field can contain the following three types of objects:

  - **Query Field:** Use query helpers to extract object-specific data from the DNOM database. The format is %<Query Field>.
  - **Coded Translation Item:** Coded translation items are used together with query helpers to translate a numeric value returned by a query helper into human-readable text. The format is <Coded Translation Item> %<Query Field>. In addition, the "CodedTranslationText" value must be specified in the FieldType field.

    For more information about coded translation items, refer to the *Coded Translation Editor User Guide*.
  - **Plain Text:** Use a plain text value to show it for all objects of the specified type. The format is <Free form text>.





- **Field Name (2-5):** Sets the values to be displayed in 2-5 Attribute Value columns.
- **Title:** Sets the title that appears in the Attribute Name column.
- **FieldType:** Specifies the field type. Leave blank for plain text and query field values. Set to "CodedTranslationText" to indicate that the returned value is controlled by a coded translation item.

  For an example of using coded translation items in attribute displays, see the Operator Attribute pop-up of dmsDistributionTF, where the type field contains the TFCONFIGTYPE coded translation item with the %TFTYPE query field to return the correct transformer type.

- **Precision:** Sets the decimal precision to be displayed for the object's value.
- **ToolTip:** The pop-up text that appears when the user hovers the mouse pointer over the object on the overview display.
- **Vis Type:** Type of test used to determine whether each menu item is visible. The choices are:
    - visEqual
    - visNotEqual
    - VisGreaterThan
    - VisLessThan
    - VisTrue
    - VisFalse

  For the first four types, Argument 1 is compared against Argument 2. For the VisTrue and VisFalse types, only Argument 1 is used.

- **Arguments:** Arguments used within each visibility test. For example, setting VisType to VisEqual checks to see if Argument 1 equals Argument 2 to determine if that menu item should be visible. These can be defined as static text or as fields to be resolved.
- **Excluded Roles:** Specifies client roles (Dispatcher, Operator, so on), for which an attribute panel field is hidden. To specify a client role, right-click the Excluded Roles field and select a role from the context menu, which is populated from the Client Roles list defined on the General Properties tab.
- **Included Roles:** Lists client roles, for which an attribute panel field is shown.
- **Cell:** Cell color on the attribute panel.
- **Text:** Text color on the attribute panel.
- **Header Cell:** Header cell color on the attribute panel.
- **Header Text:** Header text color on the attribute panel.

The following controls are available for the Attribute Properties table:

- **Insert:** Adds a new row to the Attribute Properties grid. Optionally, you can right-click the desired location and select Insert from the context menu.
- **Delete:** Deletes the selected row in the Attribute Properties grid. Optionally, you can right-click a record and select Delete from the context menu.
- **Move Up and Down Arrows:** Use these buttons to re-order the selected rows on the Attribute Properties grid. Optionally, you can right-click a record and select MoveUp or MoveDown from the context menu.

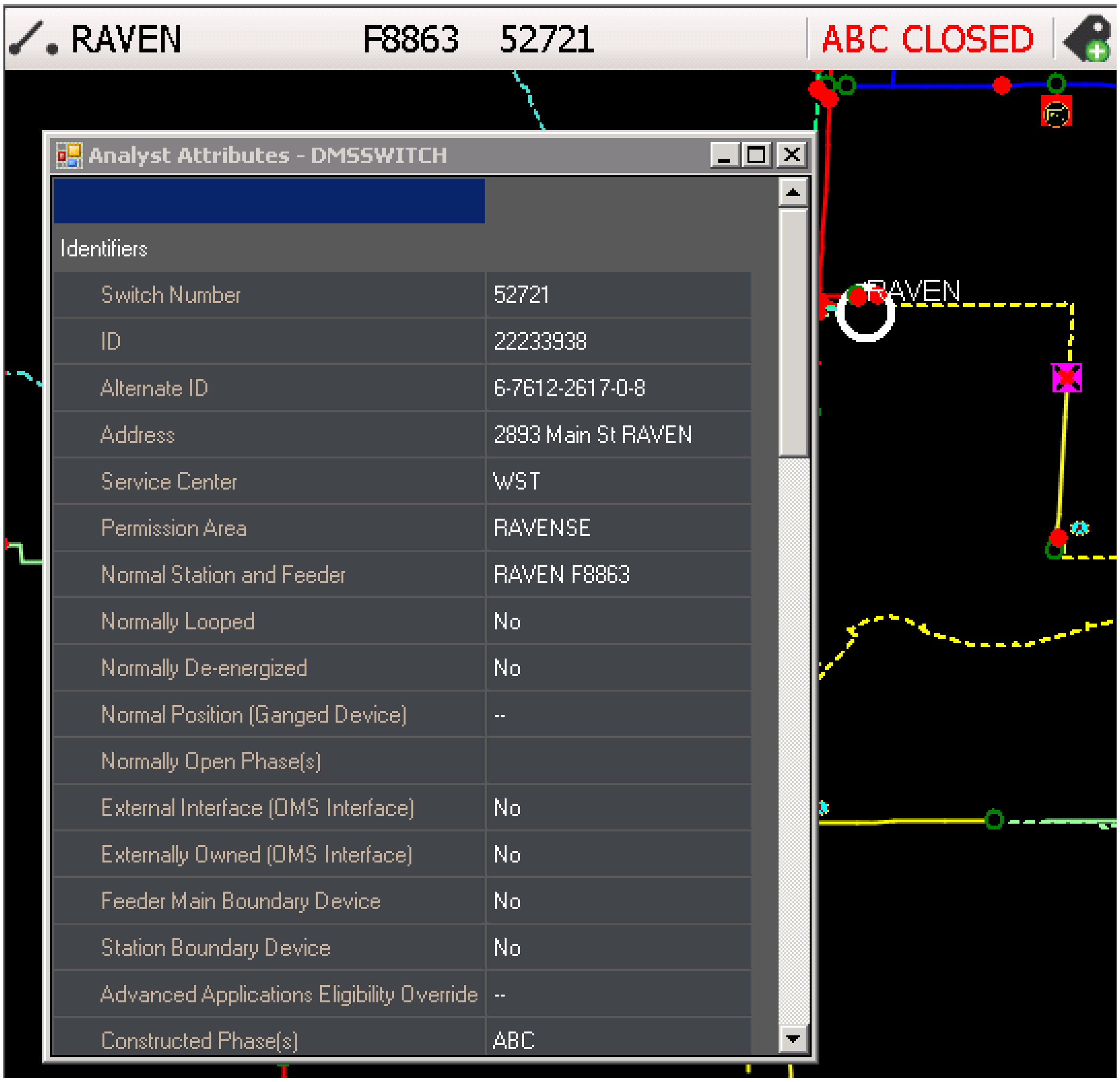


Figure 29. Analyst Attributes Pop-Up Example

11. Geographic Background

The Geographic Background tab is used to configure the colors and zoom levels of the geographic land-based objects, such as roads and building footprints.

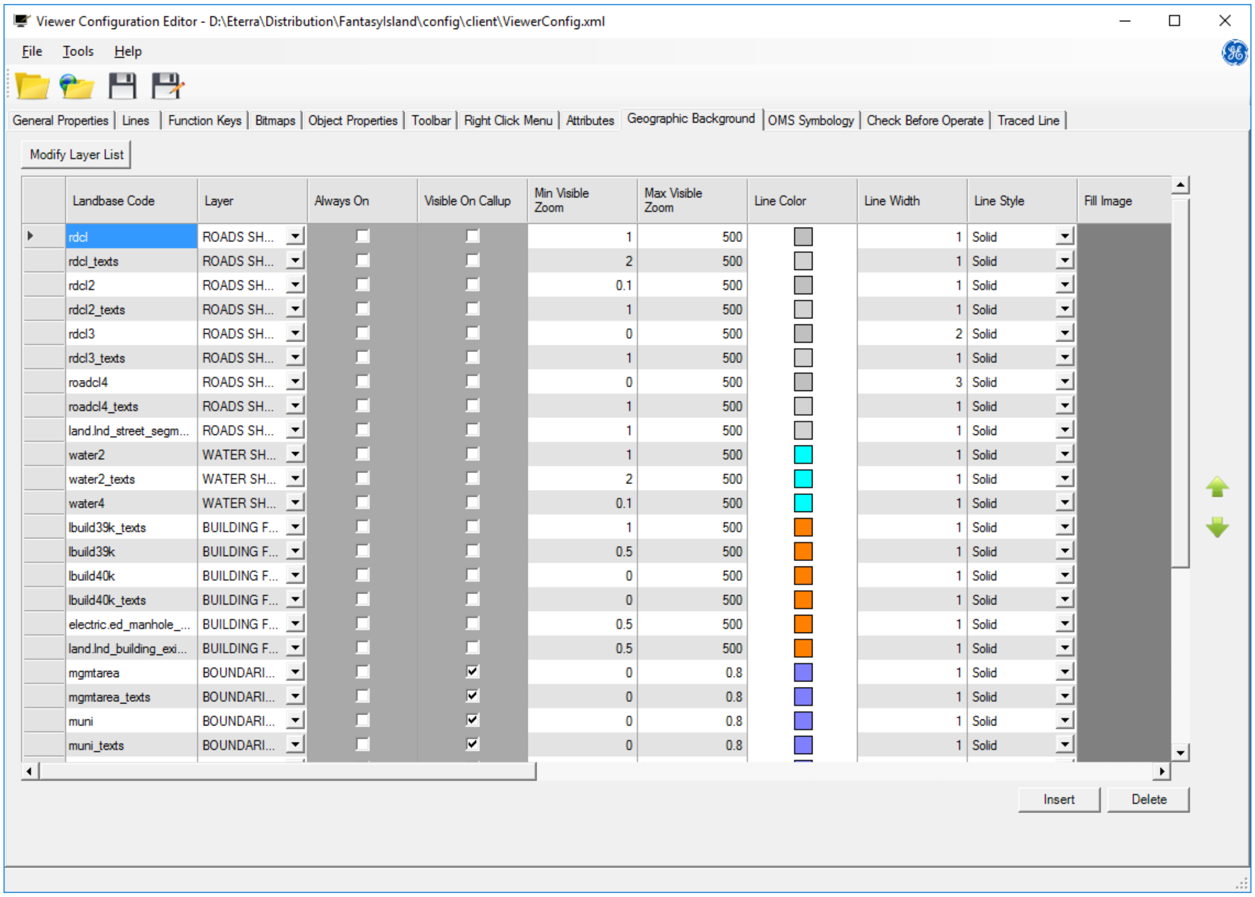


Figure 30. Geographic Background Tab

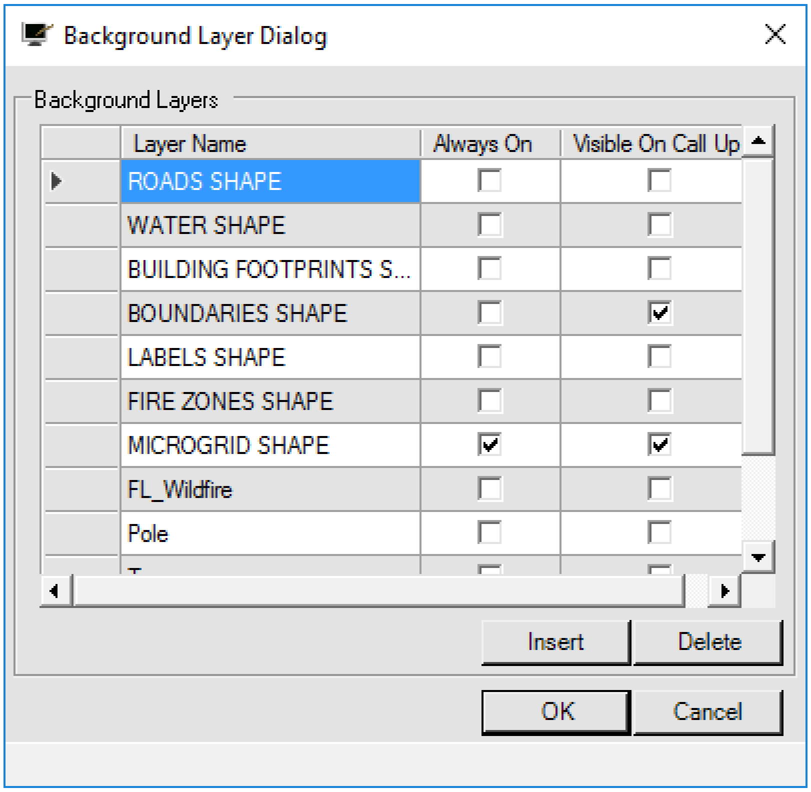
Geographic land-based objects are defined as object types and subtypes. For example, the object type "Roads" has subtypes of "Major Roads", "Large Roads", "Intermediate Roads", and "Minor Roads".

The object types and subtypes themselves are not configurable. However, for each subtype, it is possible to define the display style, color, and visibility at specific zoom levels.

To configure the properties for each geographic land-based object, select the object type and subtype from the two lists on the left side of the panel. The current settings are displayed. Each of the properties can be edited as required.

- **Modify Layer List:** Click the Modify Layer List button to display the Background Layer dialog box, as shown below.

  This allows you to choose which layers are Always On or Visible On Call Up by selecting the related options, or to add and configure a new layer.



The Always On and Visible On Call Up fields are updated on the Geographic Background tab after making any changes in the Background Layer dialog box.

The Insert button allows you to add a new line, and the Delete button removes the selected line.

- **Landbase Code:** The name of the geographic background landbase feature. The Landbase code is mapped with a Type code in the Geographic Background Compiler tool.
- **Layer:** Allows you to select a background layer for the object.
- **Always On:** When selected, this layer always appears on the display. It cannot be hidden from view.
- **Visible On Call-Up:** When selected, this layer appears on the display when initially called up. The user can later choose to hide the layer.
- **Min/Max Visible Zoom:** When the object is within the minimum and maximum zoom range, it is shown. When the zoom range is above or below these settings, the object is hidden from view.
- **Line Color:** This is the color in which the object is displayed. To change the color, click the color patch to display the color selection palette, and select a new color for the object.
- **Line Width:** Defines the width of the line in pixels.
- **Line Style:** This drop-down menu is used to select the line style for the display of the object. The following line styles are available:
  - Solid
  -

Dashed

> **Note**
>
> The Dashed line style option is enabled only when the Width parameter is set to 1. If the Width parameter is set to a value greater than 1, the system ignores the selected Dashed line style and uses the Solid line style instead.

  - Dotted
  - Dash Dotted
  - Dash Dot Dot

- **Fill Image:** Displays an image that is used to fill an object.
- **Fill Color:** This is the color in which the object is displayed. To change the color, click the color patch to display the color selection palette, and select a new color for the object.
- **Fill Style:** Sets the pattern in which the fill color is shown.
- **Fill Transparency Percent:** Specifies how transparent or opaque the fill color is.
- **Text Color:** This is the display color for any text or labels that are associated with the object, such as road names. To change the color, click a color patch to display the color selection palette, and select a new color for the object's text.
- **Min Size Zoom:** This is the minimum zoom level at which the object is visible. To make the object visible at all zoom levels, enter a value of 0.
- **Max Size Zoom:** This is the maximum zoom level at which the object can be displayed.
- **Minimum Text Size / Maximum Text Size:** These two values are used to set the zoom scaling factors for the text associated with the land-based objects, such as street names. The default value for both Minimum Text Size and Maximum Text Size is 1, which means that the text does not change size as the object is zoomed through its visible range.
- **Text Alignment:** Specifies how the text is aligned relative to the object.
- **Image Size at Min/Max Zoom:** These fields specify how small or large the image appears on the display when zoomed to the minimum or maximum value.
- **Image:** Allows defining an image to be displayed near the object.

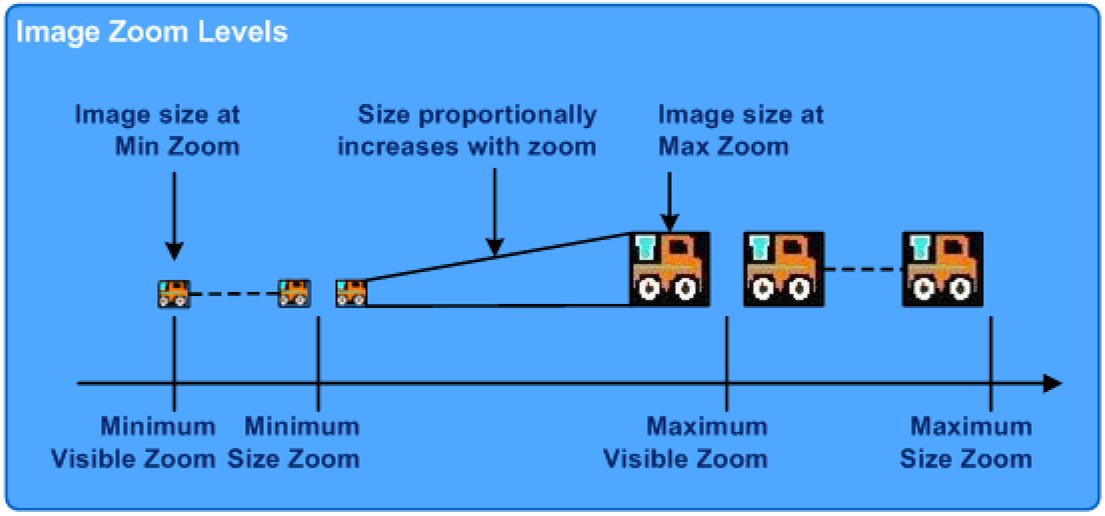


Figure 31. Image Zoom Levels

11.1. Adding a New Background Layer to the Background Layers List

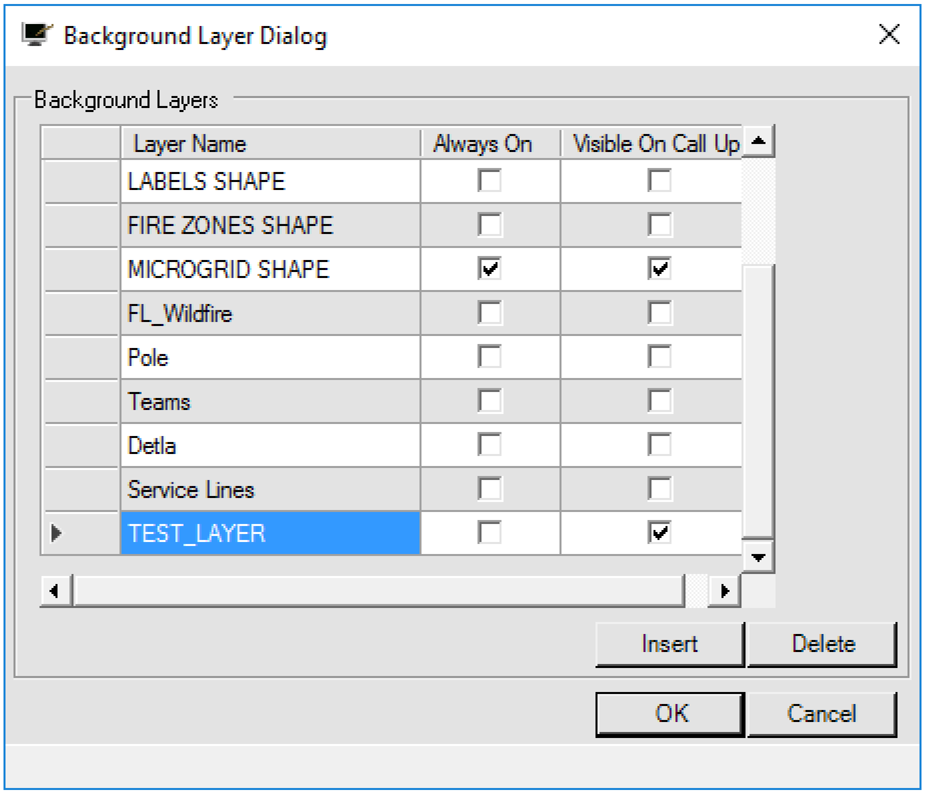
The Background Layer dialog box allows you to add and configure a new layer that has been defined in the Geographic Background Compiler.

To add a new layer:

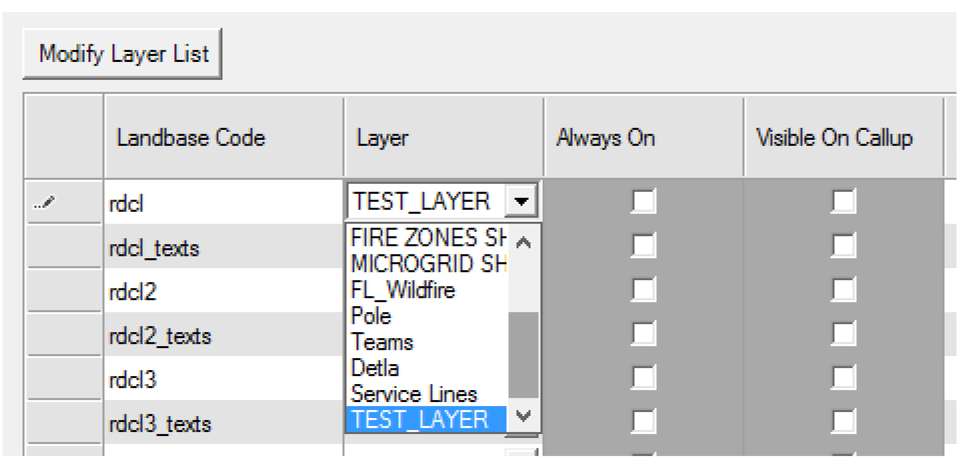
1. Select the Geographic Background tab of the Viewer Configuration Editor.
2. Click the Modify Layer List button at the top of the display.

   The Background Layer dialog box appears.

3. Click the Insert button to add a new entry.
4. Enter the name of the new background layer.



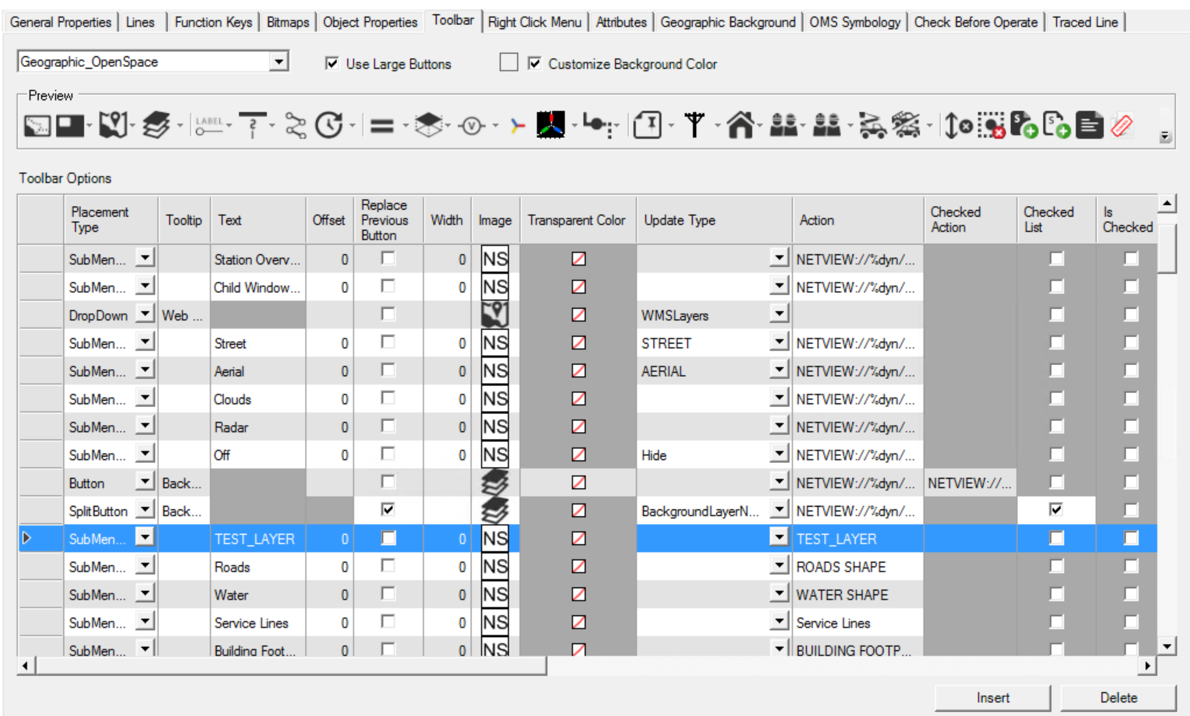
5. Select the following checkboxes, and then click OK:
    - **Always On:** When selected, this layer always appears on the display. It cannot be hidden from view.
    - **Visible On Call-Up:** When selected, this layer appears on the display when initially called up. You can later choose to turn the layer off.
6. On the Geographic Background tab, select the Layers drop-down list to verify that it contains the new entry, as shown below.



11.2. Adding a New Background Layer Label to the Geographic_Openspace (Common) Toolbar

To add a label for a geographic background layer to the Geographic_Openspace (common) toolbar:

1. Select Geographic_Openspace on the Toolbar tab of the Viewer Configuration Editor.
2. Click the Insert button to add a new entry.
3. Enter the following information for Background Layers (leave the remaining fields blank):
    - Offset = 0
    - Width = 0
    - Image = Bitmap file name for the layer
    - Text = Background layer name
    - Action = Background layer name
    - Is Checked = False
4. Use the Up and Down arrow buttons at the right of the Toolbar Options grid to move the new entry to the location of the Background Layers button group.



5. To save the changes to the `viewerconfig.xml` file, from the File menu, click Save.

For instructions on how to define a new geographic background layer, refer to the *Geographic Background Compiler User Guide*.

12. OMS Symbology

The OMS Symbology tab of the Viewer Configuration Editor allows you to define the device state decorations and customer information colors and shapes as shown in the below image.

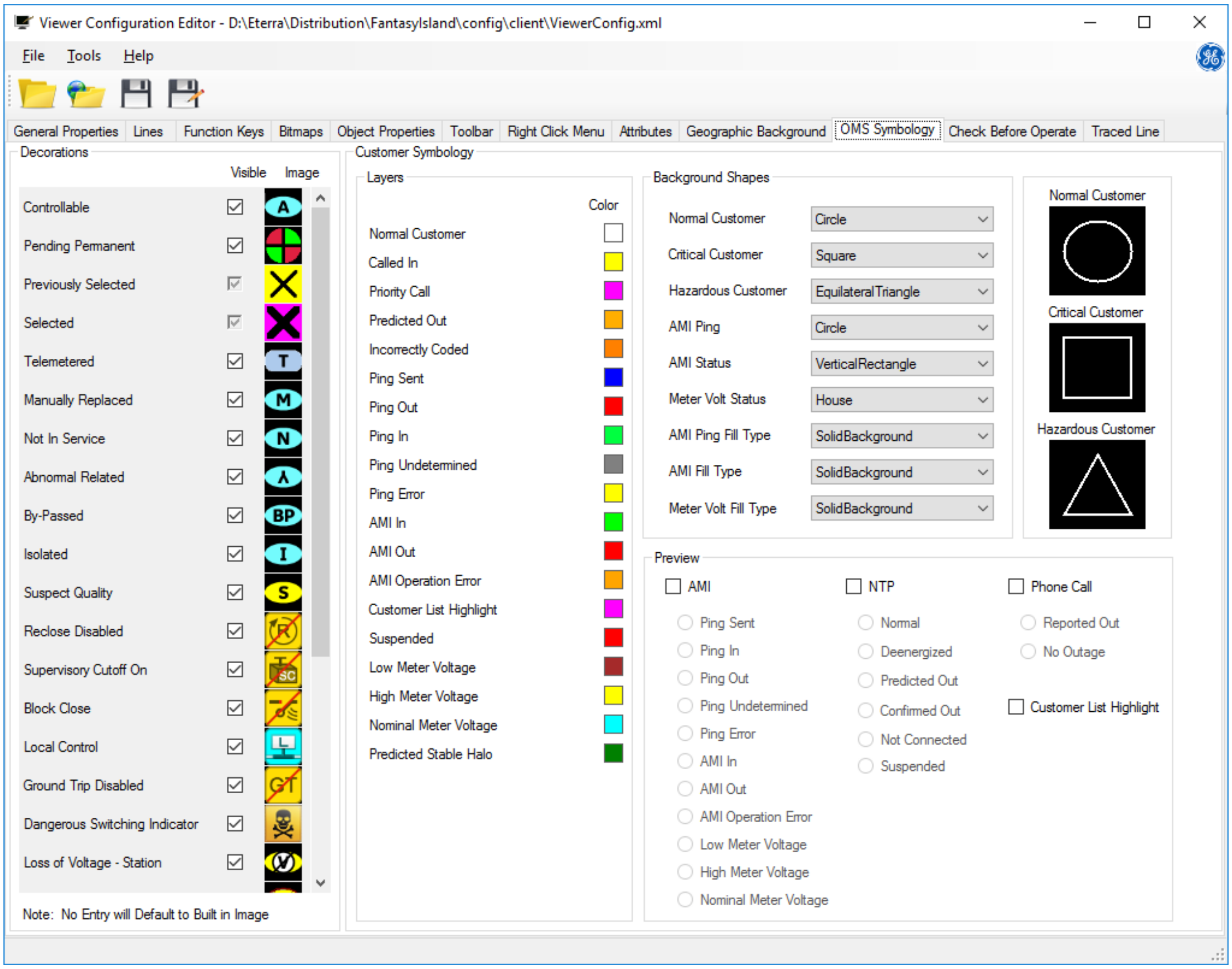
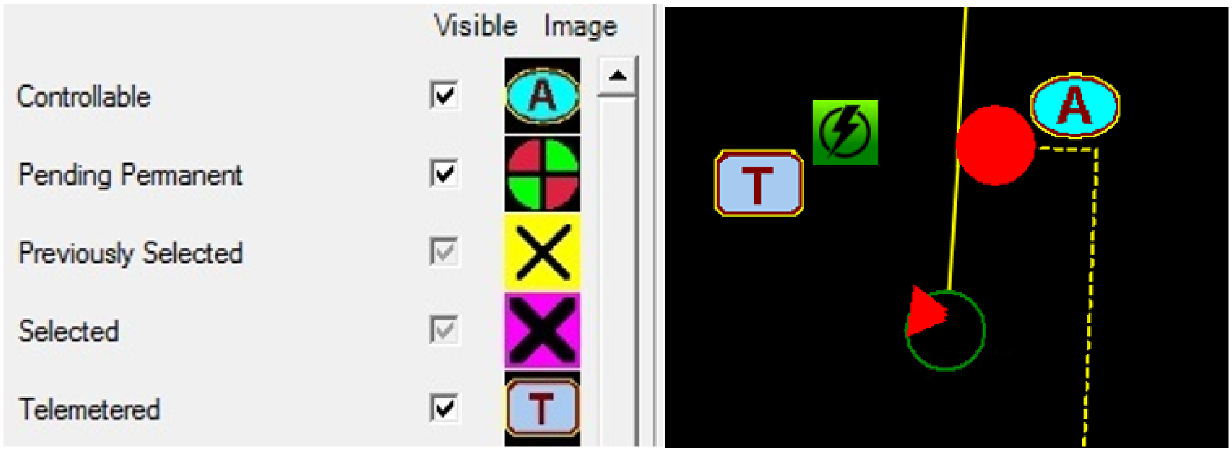


Figure 32. OMS Symbology Tab

12.1. Decorations

The Decorations group allows you to set custom bitmaps that are drawn in place of the default device state indicators.



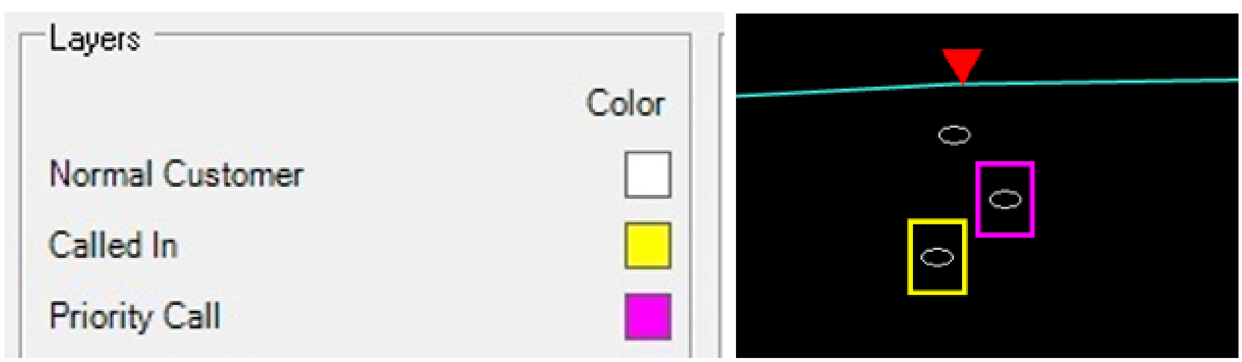
- **Visible:** Select this option to make the bitmap image visible on the network view.
- **Image:** This is a default image that indicates the device state. To select a custom image, click the bitmap preview icon.

12.2. Customer Symbology

The Customer Symbology group allows you to define background shapes, colors, and visibility of customer information.

12.2.1. Layers

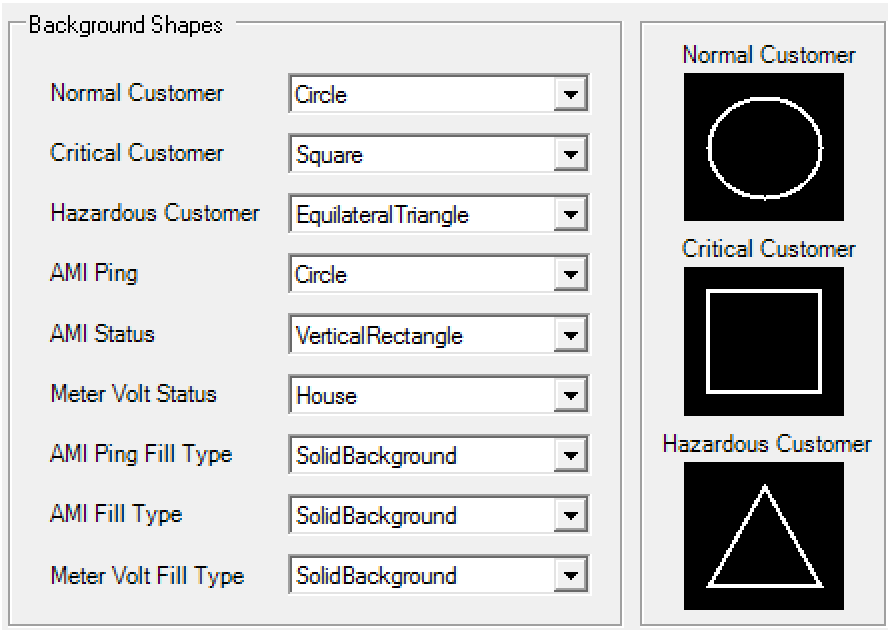
The Layers group allows you to define the colors of customer information layers.



The Color field specifies the color in which the layer appears. To change the color, click the color patch to open the color selection palette and select a new color.

12.2.2. Background Shapes

The Background Shapes panel allows you to define the icon shapes for normal customers, critical customers, and their measurement information.



The available shape types include:

- Ellipse, Circle
- Square, Horizontal Rectangle, Vertical Rectangle
- House
- Equilateral Triangle
- Diamond

12.2.3. Preview

The preview panel of the Configuration Editor allows you to preview the background shapes at different conditions for states relating to Phone Calls, Network Topology Processor (NTP) calculations, and AMI reports (standard and ping).

The options are used to select the call types, AMI types, and the possible states. The overall symbol that is shown on the geographic network view for both a normal customer and a critical customer is shown in the preview panel.

The example in the below image shows the preview of normal, critical, and hazardous customers with AMI Out and Predicted Out options selected.

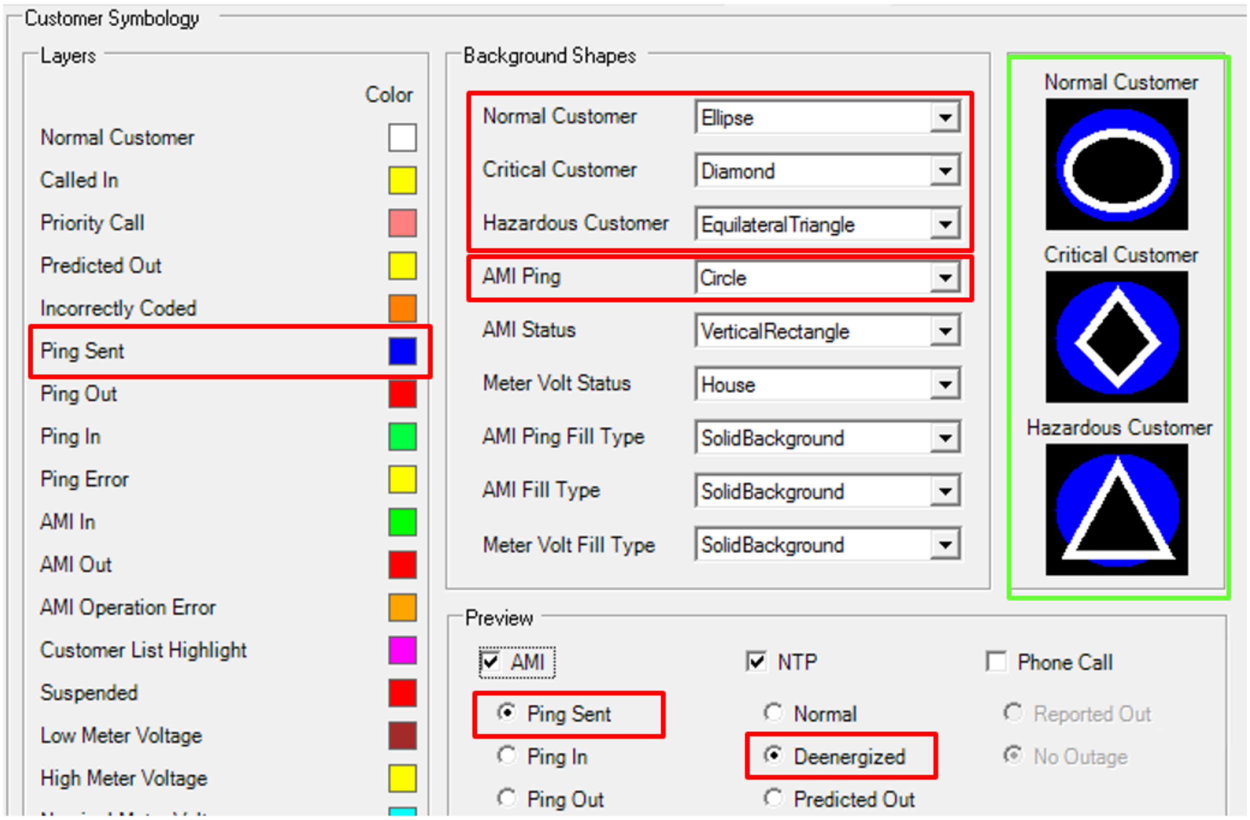


Figure 33. Customer Symbology Preview

13. Check Before Operate

The Check Before Operate tab of the Viewer Configuration Editor allows you to configure the Check Before Operate settings.

The purpose of the Check Before Operate (CBO) feature is to warn operators who are performing a switching action about certain conditions that might have an impact on customers, equipment, or crews.

CBO functionality includes a number of checks the system can automatically perform just prior to operating a switching device or placing, removing, or operating a temporary device. The CBO feature generates notifications, which the operator can consider in their decision to complete the switching action.

> **Note**
>
> Some CBO checks require the DNAF server to be running, as defined by the RequiresDNAF parameters in `ViewerConfig.xml`. These settings must not be edited.

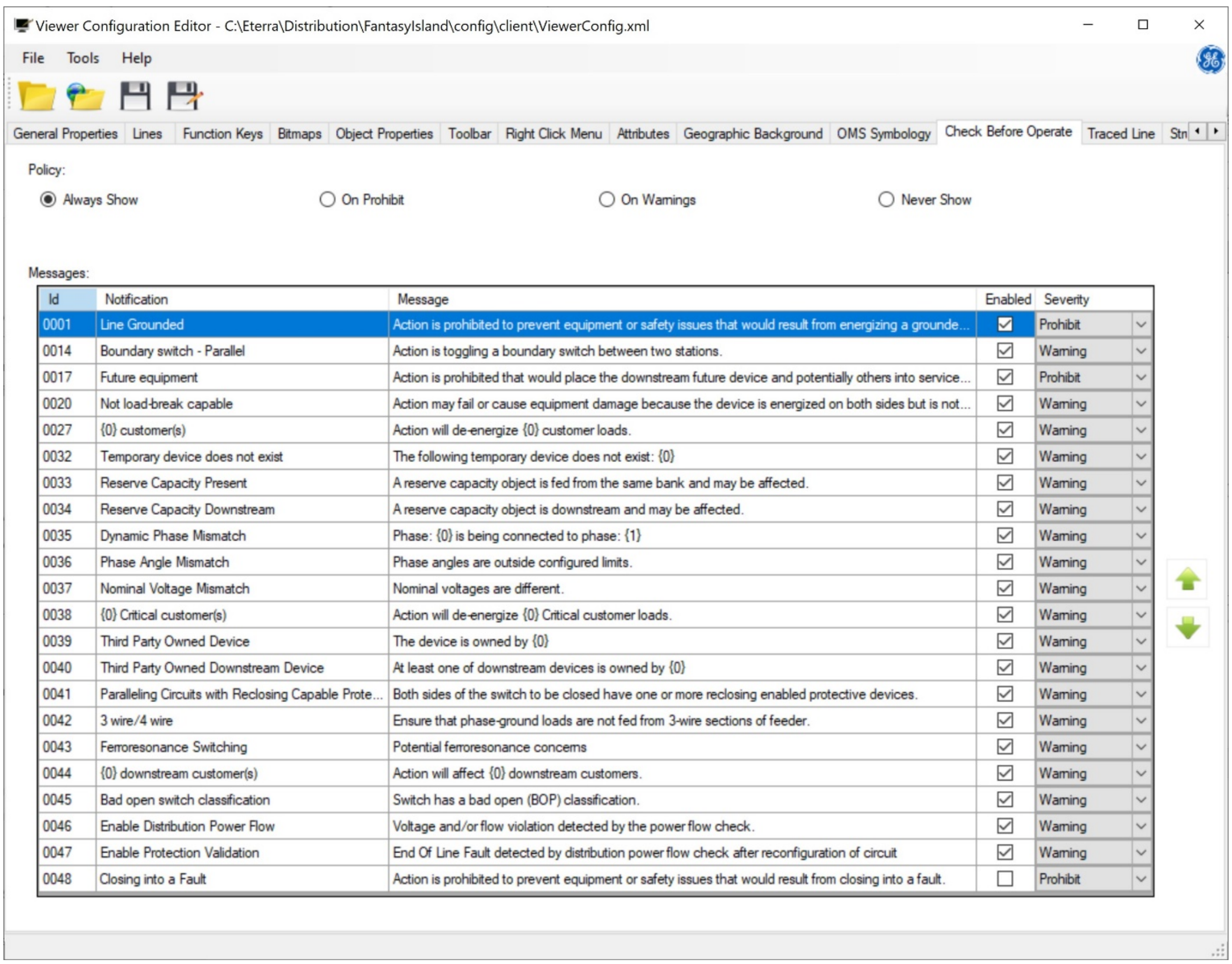


Figure 34. Check Before Operate Tab

You can choose one of the following notification issuing policies:

- **Always:** Notifications of all severity types are issued. The Check Before Operate dialog box is always presented to the user before the state of the given device can be changed.

  With this setting enabled the user cannot toggle the state of a device without being presented with a Check Before Operate dialog box.
- **On Prohibit:** Only notifications of the Prohibit severity type are issued. The user is presented with the Check Before Operate dialog box only when one or more prohibit notifications exist.
- **On Warnings:** Only notifications of the Prohibit and Warning severity types are issued. The user is presented with the Check Before Operate dialog box only when one or more prohibit or warning notifications exist.
- **Never Show:** The Check Before Operate function is disabled. No notifications are issued.

On the Messages grid, you can modify the notification header and text, and set the severity for each of the notifications. The Messages grid includes the following details:

- **ID:** ID of the message. This field cannot be edited.
- **Notification:** Notification header.
- **Message:** Notification text.
- **Enable:** Enables or disables the notification message.
- **Severity:** Notification severity (info, warning, or prohibit).

When the operator toggles a switching device or modifies a temporary device and a certain check condition is satisfied, the Check Before Operate notifications are displayed. The following table lists the currently supported checks and provides the associated operator actions.

ID	Notification	Operator Action
0001	Line grounded	Energizing a grounded line by closing a switching device.
0014	Boundary switch - parallel	Toggling a boundary switch or jumper between two stations.
0017	Future equipment	Energizing a future device by closing a switching device.
0020	No load break capable	Devices not designed to interrupt load current must not open energized circuits and should issue a warning when there is downstream load present, no warning is required when the downstream has no load. For switches with the model attribute 'LOADBRK' set to true, indicating they are load break capable, no warning is necessary even when downstream load present. Conversely, default all the cuts and jumpers, are considered non-load breakable and will trigger a warning if an attempt is made to open when downstream load present.
0027	Number of customer(s)	Opening a switching device that would result in de-energizing downstream customers.
0032	Temporary device does not exist	Modifying the Edit Cut dialog box while the cut is removed.
0033	Reserved capacity present	Closing a switching device downstream of a power transformer that has a reserved capacity object downstream. Note that closing switches above feeder breakers in station internals does not trigger this notification.

ID	Notification	Operator Action
0034	Reserved capacity downstream	Opening a switching device with a reserved capacity downstream. This message only appears if the derated limit value (reserved capacity impact) is not zero.
0035	Dynamic phase mismatch	Closing a cross-phase jumper.
0036	Phase angle mismatch	Closing a switching device that causes a phase angle mismatch.
0037	Nominal voltage mismatch	Closing a boundary switch between lines with different voltages.
0038	Critical customers	Opening a switching device that would result in de-energizing critical customers.
0039	Third Party Owned Device	Changing the status of a device that is owned by a third-party company.
0040	Third Party Owned Device Downstream	Changing the status of a device that has a downstream device owned by a third-party company.
0041	Paralleling Circuits with Reclosing Capable Protective Devices	Closing a switching device that is connected to energization sources and has reclosing-enabled protective devices at both sides. The network energization may become parallel or parallel looped and the protection functions may not operate in the expected manner.
0042	3 Wire / 4 Wire	Closing a switching device that is fed from a 3-wire section of a feeder (ABC without neutral) and has a grounded load downstream.
0043	Ferroresonance Switching	Opening one phase of a three-phase switching device at conditions that may result in a ferroresonance effect (for example, opening one phase of a switch that has a three-phase transformer with little load downstream).
0044	Downstream Customers	Opening one or more phases of a switching device when there are energized customers downstream.
0045	Bad open switch classification	Closing a switch that is in the bad open point (BOP) classification.
0046	Enable Distribution Power Flow	Operating a device results in voltage and/or flow violations based on the DPF evaluation of post-operation conditions.
0048	Closing into a Fault	Closing a switch between an energized section of the network and a de-energized section containing a fault.

Note

For the 0046 Enable Distribution Power Flow check, when the notification issuing policy is set to "On Prohibit" or "On Warnings" and no prohibit or warning severity type notifications exist for a device operation, the DPF Check Before Operate dialog does not appear. To always show the DPF check results, set the notification issuing policy to "Always Show".

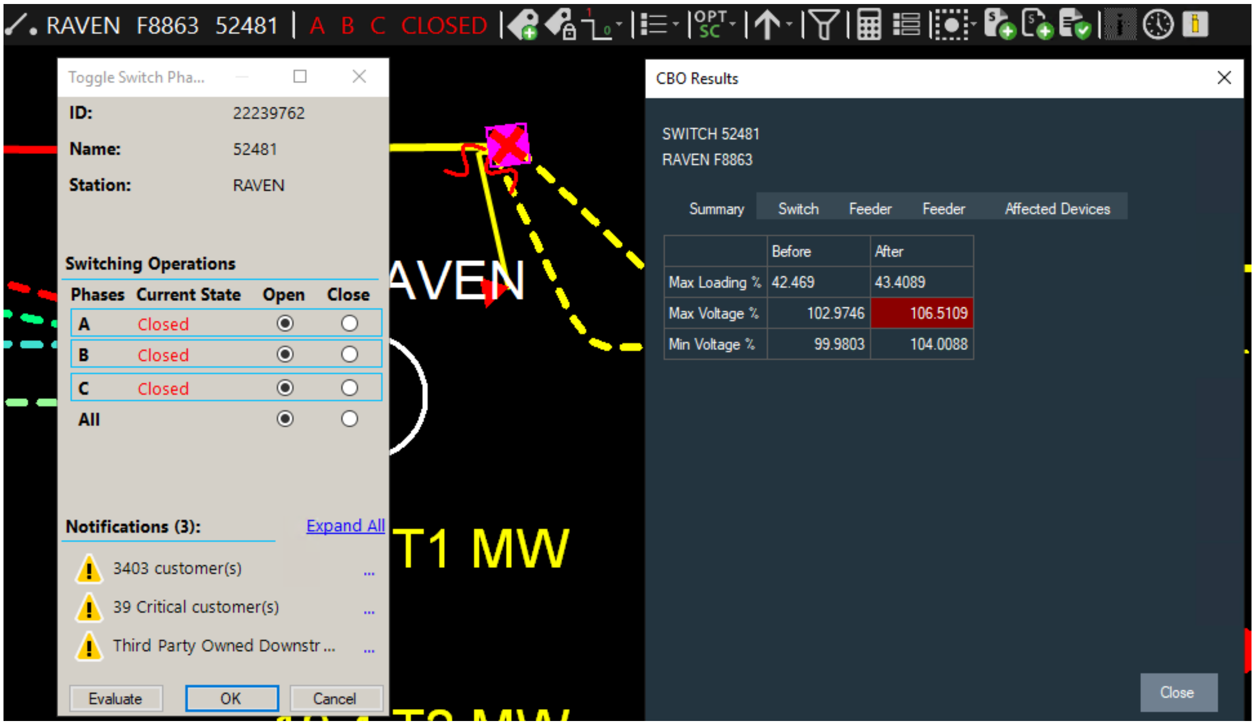


Figure 35. Check Before Operate Notifications

14. Traced Line

The Traced Line tab of the Viewer Configuration Editor allows you to define the number of trace halos that can exist simultaneously on the geographic displays, and allows you to configure the width and colors for the halos.

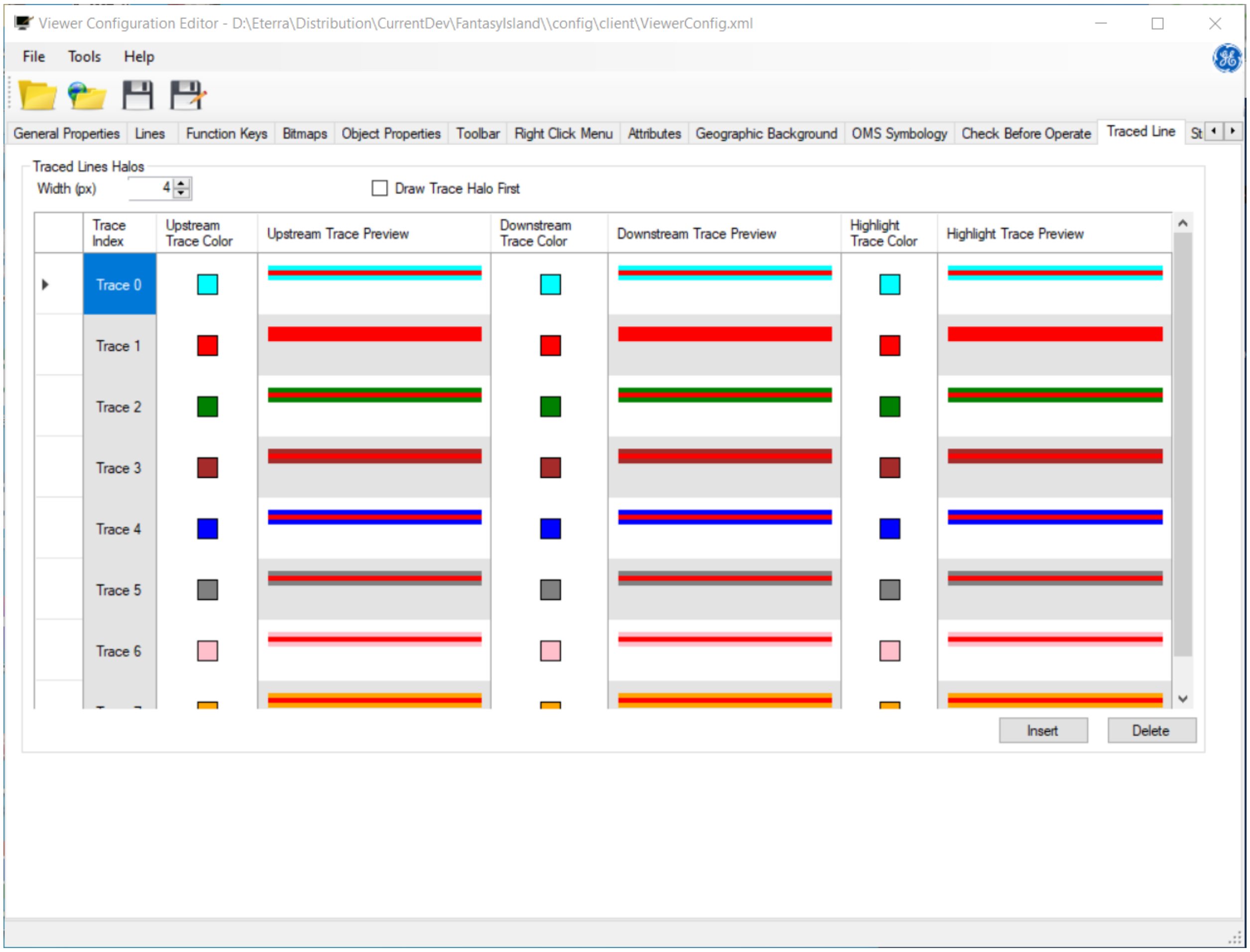


Figure 36. Traced Line Tab

The Traced Line tab includes the following fields and controls:

- **Width (px):** Defines the width for all traced line halos.
- **Draw Trace Halos First:** Results in all Trace Halos being drawn before other halos and conductor. This provides the option to see the Trace Halo with the conductor and other informational halos.
- **Trace Index:** Shows the index of the trace color configuration.

  The first traced line halo on a geographic display is shown with Trace 0 colors. The next halo will be shown with Trace 1 colors, and so on. When the maximum number of line halos on a geographic display is reached, the oldest trace is cleared and the new trace is shown with the color of the cleared trace. This logic applies until all traces are cleared using the Clear Trace option. A maximum of 8 trace colors are allowed, with support for 50 concurrent traces. After the 8th trace is generated, the 9th trace will use the first trace color, and subsequent traces will cycle through the 8 trace colors again.

- **Upstream Trace Color:** Shows the color of traced upstream line segments. To modify the line halo color, click a color patch and select another color from the selection palette.
- **Upstream Trace Preview:** Shows how the traced upstream line segments look on the geographic displays. The preview is updated automatically when the color or width is modified.
- **Downstream Trace Color:** Shows the color of traced downstream line segments. To modify the line halo color, click a color patch and select another color from the selection palette.
- **Downstream Trace Preview:** Shows how the traced downstream line segments look on the geographic displays. The preview is updated automatically when the color or width is modified.
- **Highlight Trace Color:** Shows the color of line segments traced using a trace function other than trace upstream or trace downstream (for example, trace path). To modify the line halo color, click a color patch and select another color from the selection palette.
- **Highlight Trace Preview:** Shows how line segments traced using a trace function other than trace upstream or trace downstream (for example, trace path) look on geographic displays. The preview is updated automatically when the color or width is modified.
- **Insert:** Adds a trace color configuration. Up to 8 trace color configurations can be defined.
- **Delete:** Deletes the selected trace color configuration.

15. Structures

The Structures tab of the Viewer Configuration Editor allows you define the drawing style for structures.

For more information about modeling structures, refer to the *Geographic Background Compiler User Guide*.

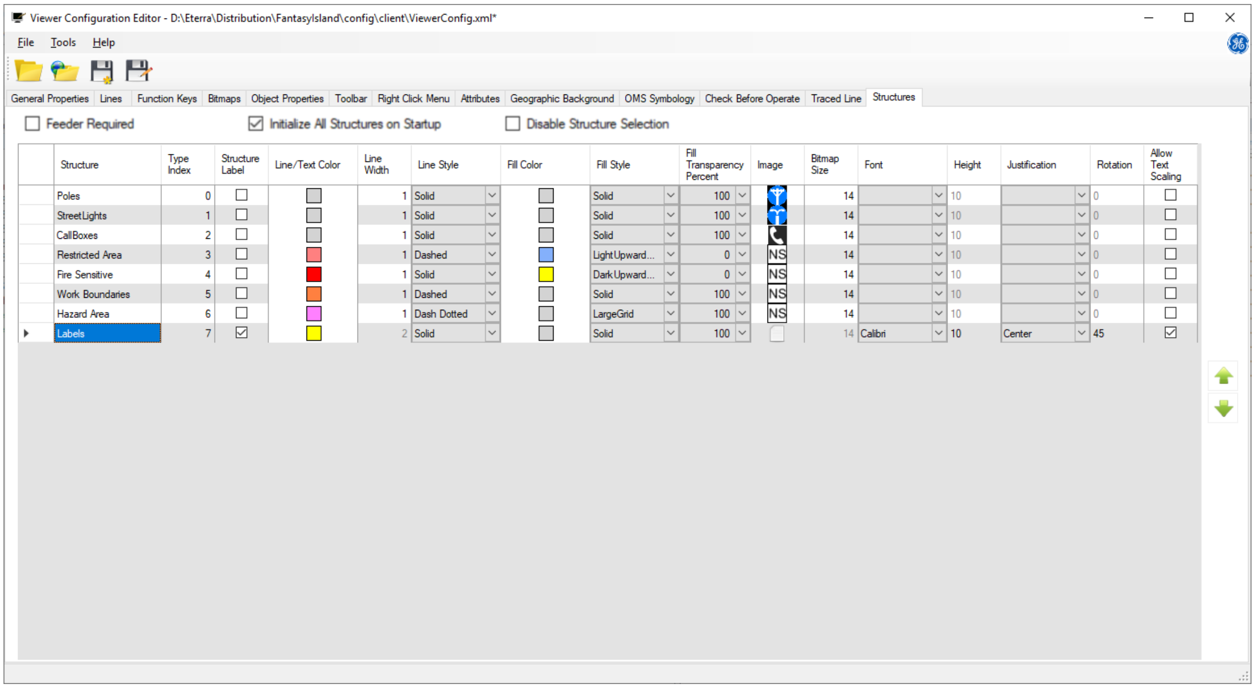


Figure 37. Structure Tab

The Structures grid includes the following fields:

- **Structure:** Structure type name.
- **Type Index:** Structure type integer value.
- **Structure Label:** Marks this structure type as a Structure Label type. This makes the system recognize a Point geometry structure as a structure label instead.
- **Line / Text Color:** This is the color in which a non-bitmap object is displayed. To change the color, click the color patch to display the color selection palette, and select a new color for the object. Only used for LineString, MultiLineString, Polygon, MultiPolygon, or Structure Label geometry structures.
- **Line Width:** Defines the width of the line or boundary in pixels, for drawn objects. Only used for LineString, MultiLineString, Polygon, or MultiPolygon geometry structures.
- **Line Style:** This drop-down menu is used to select the line style for the display of drawn objects. Only used for LineString, MultiLineString, Polygon, or MultiPolygon geometry structures. The following line styles are available:
    - Solid

- Dashed

> **Note**
>
> The Dashed line style option is enabled only when the Width parameter is set to 1. If the Width parameter is set to a value greater than 1, the system ignores the selected Dashed line style and uses the Solid line style instead.

- Dotted
- Dash Dotted
- Dash Dot Dot

- **Fill Color:** This is the color in which filled polygon objects are displayed. Only used for Polygon or MultiPolygon geometry structures.
- **Fill Style:** Sets the pattern in which the fill color is shown for filled polygon objects (if the Fill Transparency Percent is not 100). Only used for Polygon or MultiPolygon geometry structures.
- **Fill Transparency Percent:** Specifies how transparent or opaque the fill color and style are. If transparency percent is 100, the polygon is considered not filled. Otherwise, the polygon is considered filled. Only used for Polygon or MultiPolygon geometry structures.
- **Image:** Shows the assigned image. To change the image, click the image to display the browse dialog and select a new bitmap file. Only used for Point or MultiPoint geometry structures.
- **Bitmap Size:** (Information only) Shows the dimensions of the bitmap image in pixels at the 1:1 zoom. Only used for Point or MultiPoint geometry structures.
- **Font:** Shows the assigned font family. You may select any font family that is installed in the system. Be sure this font is also installed in your target client system. Only used for Structure Label geometry structures.
- **Height:** The height (in pixels) of the font at the 1:1 zoom. Only used for Structure Label geometry structures.
- **Justification:** The "anchor point" behavior of the text. Only used for Structure Label geometry structures.
    - **Left:** The left side of the text is the placement and rotation point.
    - **Right:** The right side of the text is the placement and rotation point.
    - **Center:** The center of the text is the placement and rotation point.
- **Rotation:** The text rotation (0-359 degrees). Only used for Structure Label geometry structures. The text rotation result depends on the justification. For more information, see Interaction of Structure Label Justification and Rotation.
- **Allow Text Scaling:** Defines whether the text is scaled as the display is zoomed. Only used for Structure Label geometry structures.
- **Feeder Required:** Optimizes Structure Layer drawing performance. When selected, every structure must have an associated feeder ID. This option is not selected by default.
- **Initialize All Structures on Startup:** Optimizes performance by loading all structures during

application startup. When selected, all structures are initialized at startup instead of loading on demand. This option is not selected by default.

- **Disable Structure Selection:** When checked, this option prevents structure placements from being selectable in the ADMS views. By default, the checkbox is unchecked, so structure selections remain active as before.

15.1. Interaction of Structure Label Justification and Rotation

When the left justification is applied, the label text rotates around the point at the left side of the text, as shown in the below image.

Figure 38. Left Justification

When the right justification is applied, the label text rotates around the point at the right side of the text, as shown in the below image.

Figure 39. Right Justification

When the center justification is applied, the label text rotates around the point at the center of the text, as shown in the below image.

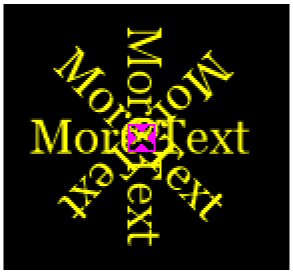


Figure 40. Center Justification

16. Command String Definition

The Action field (described in Toolbar) specifies a command string using the following syntax:

```
NETVIEW://%DYN/command1/command2/command3/command4?parameter1&parameter2
```

Where:

- **%DYN:** The name of the dynamics (for example, RT, ST, and SIM).
- **Command1:** The command (for example, VIEW, GRIDTABULAR, NAVOV, and UPDATE).
- **Command2:** Geometry name, a component type, or a display name, depending on the content of command1.
- **Command3:** A station name or a file name, depending on the content of command1.
- **Command4:** The action, component name, or component ID.
- **Parameter:** Additional data associated with each command. Parameters are separated with an "&" symbol.

For more information about Netview commands, refer to the *Grid Tabular Builder User Guide*.

16.1. Basic Commands

- **VIEW:** Calls up a geographic display centered on a specific station in the associated viewport.

  Format example: NETVIEW://%DYN/VIEW/%GEONAME/%STATION?LOCATE=%DEVICEID&ZOOM=10&FORCENONMAINON&CUSTOMERLAYERS=(SHOWACTIVECUST)

  URL example: NETVIEW://RT/VIEW/DETAIL/RAVEN

- **VIEWNEW:** Opens a new viewport and calls up a geographic display centered on a specific station.

  Format example: NETVIEW://%DYN/VIEWNEW/%GEOM/%PLANSTATIONNAME?MARK=%PLANDEVICES?CSVOUT

- **NAVOV:** Calls up the Navigation Overview display in the associated viewport.

  Format example: NETVIEW://%DYN/NAVOV

  URL example: NETVIEW://RT/NAVOV/AREA/WST

- **NAVOVNEW:** Calls up the Navigation Overview display in a new viewport.

  Format example: NETVIEW://%DYN/NAVOVNEW

- **NAVOVPIP:** Calls up the Navigation Overview display into a picture-in-picture pop-up window.

  Format example: NETVIEW://%DYN/NAVOVPIP

- **POPUPOV:** Calls up the Navigation Overview display into a pop-up window.

  Format example: NETVIEW://%DYN/POPUPOV

- **SCHEMATIC:** Calls up a geo-schematic display centered on a specific station in the associated viewport.

  Format example: NETVIEW://%DYN/SCHEMATIC/GEO/%TOPOSUBNAME

  URL example: NETVIEW://RT/SCHEMATIC/GEO/RAVEN/341027277

- **SCHEMATICNEW:** Calls up a geo-schematic display centered on a specific station in a new viewport.

  Format example: NETVIEW://%DYN/SCHEMATICNEW/MULTILOOP/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/SCHEMATICNEW/CONTAINER/WINGED FOOT/110049557?SHOWLABELS

- **SCHEMATIC/FDR:** Calls up a display with a schematic view of the feeder main and the associated devices.

  Format example: NETVIEW://%DYN/SCHEMATIC/FDR/%TOPOSUBNAME/%FEEDERNAME

  URL example: NETVIEW://RT/SCHEMATIC/FDR/3LIONS/F5634

  For testing and debug purposes, use the EXPORT parameter to export the schematic objects to C:\Temp\<Station>_<Feeder>.xml.

  URL example: NETVIEW://RT/SCHEMATIC/FDR/3LIONS/F5634?EXPORT

- **GRIDTABULAR:** Calls up a grid tabular display in the current viewport.

  Format example: NETVIEW://%DYN/GRIDTABULAR//%SOLUTIONSUBNAME/%SOLUTIONID?Frame1=AFRplanstat1_GridDisplay&Frame2=AFRplanstat2_GridDisplay

  URL example: NETVIEW://RT/GRIDTABULAR/ABNORMALFDR_GRIDDISPLAY/RAVEN/F8864

- **GRIDTABULARNEW:** Calls up a grid tabular display in a new viewport.

  Format example: NETVIEW://%DYN/GRIDTABULAR/FaultAnalysisMax_GridDisplay/%TOPOSUBNAME/%ID

- **ZOOM:** Changes the zoom level of the geographic and schematic display to the defined level. A zoom level of 1 is 100%.

  Format example: NETVIEW://%DYN/ZOOM/TO?VALUE=1

  URL example: NETVIEW://RT/ZOOM/TO?VALUE=1

- **UPDATE:** Updates the geographic view parameters.

  Format example: NETVIEW://%DYN/UPDATE?FORCENONMAINAUTO

URL example: NETVIEW://RT/UPDATE?SHOWLABELS

- **BKGD:** Updates the background layers view.

  Format example: NETVIEW://%DYN/BKGD?BACKGROUNDLAYERLIST=()

  URL example: NETVIEW://RT/BKGD?BACKGROUNDLAYERLIST=(ROADS SHAPE)

- **TRACEUP:** Draws a trace upstream from the initiating device in a geographic display in the current viewport.

  Format example: NETVIEW://%DYN/TRACEUP/SWT/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEUP/SWT/RAVEN/22233938

- **TRACEUPTOSCADA:** Draws a trace upstream from the initiating device to the next upstream SCADA controllable (not only telemetered) device. If there is no upstream SCADA device, an info message appears and the tracing is canceled.

  Format example: NETVIEW://%DYN/TRACEUPTOSCADA/TF/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEUPTOSCADA/SWT/RAVEN/13094142

- **TRACEUPTOPROT:** Draws a trace upstream from the initiating device to the next upstream protective device (fuse, recloser, sectionalizer, so on). If there is no upstream protective device, an info message appears and the tracing is canceled.

  Format example: NETVIEW://%DYN/TRACEUPTOPROT/SERIES/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEUPTOPROT/SWT/RAVEN/13094142

- **TRACEUPTOSWITCH:** Draws a trace upstream from the initiating device to the next upstream switching device, including temporary switches such as cuts and jumpers. If no device is found, an info message appears and the tracing is canceled.

  Format example: NETVIEW://%DYN/TRACEUPTOSWITCH/LN/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEUPTOSWITCH/SWT/RAVEN/13094142

- **TRACEUPTORECLOSER:** Draws a trace upstream from the initiating device to the next upstream recloser or breaker. If there is no upstream recloser or breaker, an info message appears and the tracing is canceled.

  Format example: NETVIEW://%DYN/TRACEUPTORECLOSER/LN/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEUPTORECLOSER/SWT/RAVEN/13094142

- **TRACEDOWN:** Draws a trace downstream from the initiating device in a geographic display in the currently open viewport.

  Format example:
  NETVIEW://%DYN/TRACEDOWN/LN/%TOPOSUBNAME/%ID?PHASEA=TRUE&PHASEB=FALSE&PHASEC=FALSE&PHASESSPECIFIED

  URL example: NETVIEW://RT/TRACEDOWN/SWT/RAVEN/22233938

- **TRACEDOWNTOSCADA:** Draws a trace downstream from the initiating device to the next downstream SCADA controllable (not only telemetered) device. If no device is found, an info message appears and the tracing is canceled.

  Format example: NETVIEW://%DYN/TRACEDOWNTOSCADA/TF/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEDOWNTOSCADA/SWT/RAVEN/13094142

- **TRACEDOWNTOPROT:** Draws a trace downstream from the initiating device to the next downstream protective device (fuse, recloser, sectionalizer, so on). If no device is found, an info message appears and the tracing is canceled.

  Format example: NETVIEW://%DYN/TRACEDOWNTOPROT/SERIES/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEDOWNTOPROT/SWT/RAVEN/13094142

- **TRACEDOWNTOSWITCH:** Draws a trace downstream from the initiating device to the next upstream downstream switching device, including temporary switches such as cuts and jumpers. If no device is found, an info message appears and the tracing is canceled.

  Format example: NETVIEW://%DYN/TRACEDOWNTOSWITCH/LN/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEDOWNTOSWITCH/SWT/RAVEN/13094142

- **TRACEUPANDDOWN:** Draws a trace upstream, then downstream from the initiating device in a geographic display in the currently open viewport.

  Format example: NETVIEW://%DYN/TRACEUPANDDOWN/LN/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEUPANDDOWN/SWT/RAVEN/22233938

- **TRACEPATH:** Draws a trace between two selected components in a geographic display in the currently open viewport. Only components on the direct path are highlighted.

  Format example: NETVIEW://%DYN/TRACEPATH/TF/%TOPOSUBNAME/%ID

- **TRACEBETWEEN:** Draws a trace between two selected components in a geographic display in the currently open viewport. The direct path, as well as all components connected to the path, are highlighted.

  Format example: NETVIEW://%DYN/TRACEBETWEEN/LN/%TOPOSUBNAME/%ID

- **TRACEEXTENT:** Traces the current circuit extent for the selected component in a geographic display in the currently open viewport. For a de-energized section, this traces to all open devices.

  Format example: NETVIEW://%DYN/TRACEEXTENT/SERIES/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACEEXTENT/SWT/RAVEN/22233938

- **TRACEBETWEENMARKS:** Traces outward to marked devices from the selected component in a geographic display in the currently open viewport.

  Format example: NETVIEW://%DYN/TRACEBETWEENMARKS/SHIMP/%TOPOSUBNAME/%ID

URL example: NETVIEW://RT/TRACEBETWEENMARKS/SWT/RAVEN/22233938

- **TRACESECTION:** Traces the switchable section for the initiating device in a geographic display in the currently open viewport.

  Format example: NETVIEW://%DYN/TRACESECTION/LN/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACESECTION/LN/RAVEN/22382426

- **TRACESECTION:** Traces the switchable section for the initiating device in a geographic display in the currently open viewport.

  Format example: NETVIEW://%DYN/TRACESECTION/LN/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/TRACESECTION/LN/RAVEN/22382426

- **COMBINEDCLEAR:** Clears all traces and marks in the current geographic display in the currently open viewport.

  Format example: NETVIEW://%DYN/COMBINEDCLEAR

  URL example: NETVIEW://RT/COMBINEDCLEAR

- **DISTANCETOSRC:** Calculates the network distance from the selected point or component to the source. The distance is displayed in the status bar.

  Format example:
  NETVIEW://%DYN/DISTANCETOSRC/LN/%TOPOSUBNAME/%ID?PHASEA=TRUE&PHASEB=FALSE&PHASEC=FALSE&PHASESSPECIFIED

  URL example:
  NETVIEW://RT/DISTANCETOSRC/LN/RAVEN/22382426?PHASEA=TRUE&PHASEB=FALSE&PHASEC=FALSE&PHASESSPECIFIED

- **DISTANCETOPT:** Calculates the network distance between two points or components chosen on the geographic network view. If the two points are not on the same network, no distance is calculated. The distance is shown in the status bar in geographic units.

  Format example:
  NETVIEW://%DYN/DISTANCETOPT/LN/%TOPOSUBNAME/%ID?PHASEA=TRUE&PHASEB=FALSE&PHASEC=FALSE&PHASESSPECIFIED

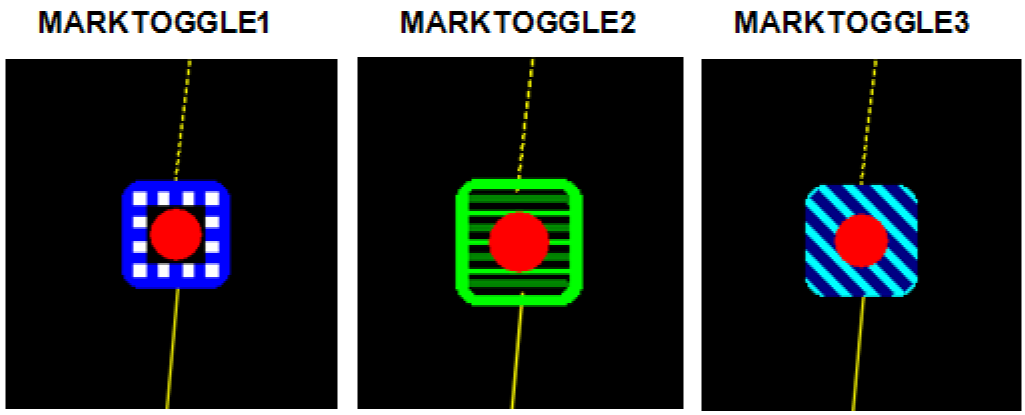
- **DISTANCESTRLINE:** Calculates the straight line distance between two points selected on the geographic network view. The distance is shown in the status bar in geographic units.

  Format example: NETVIEW://%DYN/DISTANCESTRLINE/LN/%TOPOSUBNAME/%ID

- **MARKTOGGLEX:** Places or removes a mark of the specified mark type (MARKTOGGLE1, MARKTOGGLE2, MARKTOGGLE3) in a geographic display on the specified switching device, capacitor, regulator, or distribution transformer.

  Format example: NETVIEW://%DYN/MARKTOGGLE1//%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/MARKTOGGLE2//RAVEN/22233938



- **CANCELMARK:** Clears all marks in the geographic display in the currently open viewport.

  Format example: NETVIEW://%DYN/CANCELMARK

  URL example: NETVIEW://RT/CANCELMARK

- **SHOWMARKEXTENTS:** Show all marked devices in the currently open viewport.

  Format example: NETVIEW://%DYN/SHOWMARKEXTENTS

  URL example 1: NETVIEW://RT/SHOWMARKEXTENTS

  URL example 2: NETVIEW://RT/SHOWMARKEXTENTS?MINZOOM=4&MARKINDEX=1

  URL example 3: NETVIEW://RT/VIEW/DETAIL/RAVEN?MARK=RAVEN|22233938,RAVEN|22234028,BACHELOR|22226849&SHOWEXTENTS=TRUE

- **MARKPOTENTIALX:** Marks all isolating devices that need to be tripped to isolate the selected line or switching device. Only feeder backbone (not boundary or lateral) isolating devices are considered.

  Format example: NETVIEW://%DYN/MARKPOTENTIAL3//%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/MARKPOTENTIAL2//RAVEN/22382426

- **MARKMIDPOINTX:** Marks the switch closest to the geographical midpoint of a feeder that the selected line belongs to.

  Format example: NETVIEW://%DYN/MARKMIDPOINT1//%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/MARKMIDPOINT3//RAVEN/22382426

- **MARKOPENX:** Marks all open switching devices.

  Format example: NETVIEW://%DYN/MARKOPEN1

  URL example: NETVIEW://RT/MARKOPEN1

- **MARKOFFX:** Clears a mark of the specified mark type on the specified device.

Format example: NETVIEW://%DYN/MARKOFF2//%TOPOSUBNAME/%ID

URL example: NETVIEW://RT/MARKOFF2//RAVEN/22233938

- **MARKNORMALOPEN1:** Marks all normally open switches that are connected to the selected point on the line in the current geographic display in the currently open viewport.

  Format example: NETVIEW://%DYN/MARKNORMALOPEN1//%TOPOSUBNAME/%ID

- **MARKANDTRACENORMALOPEN1:** Marks all normally open switches that are connected to the selected point on the line and draws the trace between the switches in the current geographic display in the currently open viewport.

  Format example: NETVIEW://%DYN/MARKANDTRACENORMALOPEN1//%TOPOSUBNAME/%ID

- **MARKANDTRACEPOTENTIAL1:** Marks all isolation switches and draws the trace between the switches in the current geographic display in the currently open viewport.

  Format example: NETVIEW://%DYN/MARKANDTRACEPOTENTIAL1//%TOPOSUBNAME/%ID

- **PIPON:** Enables the picture-in-picture window.

  Format example: NETVIEW://%DYN/PIPON

  URL example: NETVIEW://RT/PIPON

- **PIPOFF:** Disables the picture-in-picture window.

  Format example: NETVIEW://%DYN/PIPOFF

  URL example: NETVIEW://RT/PIPOFF

- **POPUP:** Displays a named pop-up window. The list of pop-ups is as follows: CUT, TEMPGROUND, FAULT, JUMPER, CUSTOMERCORRECT, DNAFSOLUTIONS, SAVECASE, FIND, TAG, ANNOTATION, ANNOTATIONUPDATE, MOBILESUB, SPECIALANNOTATION, SPECIALANNOTATIONUPDATE, SWITCHSTATUS, CAPACITORSTATUS, BLOCKPREDICTIONSTATUS, CURTAILPLAN, PLANNEDOUTAGE, SCHEDULEPOPUP, SWOQUALIFICATIONS, SWOEMPLOYEE, QUERY, SCHEMAQUERY, and DNAFINTERVAL.

  Format example: NetView://%DYN/POPUP/SwitchStatus/%TOPOSUBNAME/%ID

  URL example: NETVIEW://RT/POPUP/FIND

- **DESELECT:** Deselects a selected component.

  Format example: NETVIEW://%DYN/DESELECT

  URL example: NETVIEW://RT/DESELECT

- **COMMISSIONSTATEINC:** Increments the commissioning state bit for a component (Future, Active, Future Remove, or Removed).

  Format example:
  NETVIEW://%DYN/COMMISSIONSTATEINC/%DEVICETYPE/%TOPOSUBNAME/%ID

- **COMMISSIONSTATEDEC:** Decrements the commission state bit.

  Format example:
  NETVIEW://%DYN/COMMISSIONSTATEDEC/%DEVICETYPE/%TOPOSUBNAME/%ID

- **RECENTER:** Re-centers the partial model view to the specified coordinates.

  Format example: NETVIEW://%dyn/RECENTER/%GEOM?X=%CENTERX&Y=%CENTERY

  URL example: NETVIEW://RT/RECENTER/DETAIL?X=915000&Y=813000

- **MOVEMODE:** Activates the move mode on the network view. Not applicable on a loop schematic view.

  Format example: NETVIEW://%DYN/MOVEMODE?ISLOOPSCHEMATIC=%ISLOOPSCHEMATIC

  URL example: NETVIEW://RT/MOVEMODE?ISLOOPSCHEMATIC=1

- **NEWMOVEMODE:** Activates the move mode for the selected object. Applicable only to an annotation, crew, customer, incident, planned outage, safety document, mobile substation, jumper, unassociated call, damage assessment location, or vehicle.

  Format example: NETVIEW://%DYN/NEWMOVEMODE

- **REMOVE:** Removes a temporary object, such as an annotation, cut, jumper, or temporary ground.

  Format example: NETVIEW://%DYN/REMOVE/TEMPGROUND/%TOPOSUBNAME/%ID

- **MOVELABEL:** Changes the label position for a temporary device.

  Format example:
  NETVIEW://%DYN/MOVELABEL/JUMPER/%TOPOSUBNAME/%ID?LABELPOSITION=N

- **POWERFLOWANIMATIONMODE:** Enables the power flow direction animation.

  Format example: NETVIEW://%DYN/POWERFLOWANIMATIONMODE/PABC

  URL example: NETVIEW://RT/POWERFLOWANIMATIONMODE/QB

- **NORMALMODE:** Disables the power flow direction animation.

  Format example: NETVIEW://%DYN/NORMALMODE

  URL example: NETVIEW://RT/NORMALMODE

- **UPDATEVC:** Loads a new `viewerconfig.xml` configuration file without restarting the client. After executing this command, changes are available in new viewports.

  URL example: NETVIEW://RT/UPDATEVC

- **TOGGLE:** Changes the real-time status of the device to Closed or Open.

  Format example: NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%ID/TOGGLE

- **TOGGLECONFIRM:** Confirms the device status change and changes the real-time status of the device to Closed or Open.

Format example: NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%ID/TOGGLECONFIRM

- **TOGGLECONFIRMCHECK:** Confirms the device status change and changes the real-time status of the device to Closed or Open. For protective switching devices, such as fuses and reclosers, this option also invokes a pop-up display with a list of all downstream single customer and predicted incidents, including those in the predicted stable state. On the pop-up display, you can select incidents to be merged with the newly created upstream confirmed incident; the incidents that are not selected become predicted stable incidents.

  Format example: NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%ID/TOGGLECONFIRMCHECK

- **PENDPERM:** Changes the Pending/Permanent status. Places a special mark at the device that informs about changing permanent status.

  Format example: NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%ID/PENDPERM

- **CONFIRMCLOSED:** Blocks or unblocks outage predictions for a protective device.

  Format example: NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%ID/CONFIRMCLOSED

- **SEARCH:** Opens the Switching Order Search window, which is used to filter and open switching orders for edit or review.

  Format example: NETVIEW://%dyn/SWITCHORDER/SEARCH

- **Initialize Study Mode:** Opens the Study Case Setup display and initializes the study environment for the real-time or nominal state for the specified substation.

  Format example NETVIEW://ST/POPUP/SAVECASE?STATION=<Station>&ACTION=<INITNOMINAL or INITREALTIME>

  Command example:
  NETVIEW://ST/POPUP/SAVECASE?STATION=3LIONS&ACTION=INITREALTIME

16.1.1. OMS Commands

- **CREATE:** Creates an incident (from a call or for a device) upon an operator's request.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CREATE/MANUALINCIDENT?DEVICEID=%SELECTCOMPID

- **CREATEMULTIPLE:** Creates incidents from calls.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CREATEMULTIPLE/INCIDENT_FROM_CALL?CALLID=%ID

- **DELETE:** Deletes an OMS component, such as a planned outage or call.

Format example:
NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/DELETE/PLANNEDOUTAGE?ID=%ID

- **CANCEL:** Cancels a customer call or an incident.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CANCEL/CALL?ID=%SELECTCOMPI
  D

- **SETPREDICTEDSTATIC:** Creates a predicted stable incident for a protective device (such as distribution transformer, fuse, recloser, or breaker).

  Format example: NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%ID/SETPREDICTEDSTATIC

- **CREATE/MANUALINCIDENT:** Splits information about an open device or a customer from an incident into a new incident.

  Format example 1:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CREATE/MANUALINCIDENT?DEVIC
  EID=%SELECTCOMPID

  Format example 2:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CREATE/MANUALINCIDENT?CUSTO
  MERID=%SELECTCOMPID

- **CREATE/ISOLATEDINCIDENT:** Creates a non-outage isolated incident for a device.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CREATE/ISOLATEDINCIDENT?X=%X
  &Y=%Y

- **FOUNDOPEN:** Opens the device. Creates a confirmed incident with the FoundOpen flag set. Applies to non-SCADA switching devices and distribution transformers.

  Format example: NETVIEW://%DYN/SET/SW/%TOPOSUBNAME/%ID/FOUNDOPEN

- **CONFIRMIN:** Confirms that a customer is energized.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CONFIRMIN/CUSTOMER?ID=%SELE
  CTCOMPID

- **CONFIRMOUT:** Confirms that a customer is de-energized.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CONFIRMOUT/CUSTOMER?ID=%SE
  LECTCOMPID

- **AMI:** Issues a command to a customer’s AMI meter.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/AMI/DISCONNECTMETER?ID=%SEL
  ECTCOMPID

- **SUSPEND:** Marks the customer as being temporarily out of service due to a known cause (such as a house fire).

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/SUSPEND/CUSTOMER?ID=%SELECTCOMPID

- **UNSUSPEND:** Marks the customer as being in service.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/UNSUSPEND/CUSTOMER?ID=%SELECTCOMPID

- **MERGEDOWNSTREAMPREDICTEDINCIDENTS:** Merges downstream predicted and single customer incidents to a confirmed incident. Applies to downstream incidents on the same phase.

  Format example:
  NETVIEW://%DYN/OMSCOMMAND/MERGEDOWNSTREAMPREDICTEDINCIDENTS?ID=%SELECTCOMPID

- **MERGEDOWNSTREAMPREDICTEDINCIDENTSXPHASE:** Merges downstream predicted and single customer incidents to a confirmed incident. Applies to downstream incidents on all phases.

  Format example:
  NETVIEW://%DYN/OMSCOMMAND/MERGEDOWNSTREAMPREDICTEDINCIDENTSXPHASE?ID=%SELECTCOMPID

- **MERGEDOWNSTREAMPREDICTEDANDISOLATEDINCIDENTS:** Merges downstream predicted and single customer incidents, and isolated incidents to a confirmed incident. Works similarly to the MERGEDOWNSTREAMPREDICTEDINCIDENTS command.

  Format example:
  NETVIEW://%DYN/OMSCOMMAND/MERGEDOWNSTREAMPREDICTEDANDISOLATEDINCIDENTS?ID=%SELECTCOMPID

- **SETPREDICTEDSTABLE:** Sets the Predicted Stable status for a predicted incident.

  Format example:
  NETVIEW://%DYN/OMSCOMMAND/SETPREDICTEDSTABLE?ID=%SELECTCOMPID&INCIDENTNAME=

- **RESETPREDICTEDSTABLE:** Resets the Predicted Stable status for a predicted incident.

  Format example:
  NETVIEW://%DYN/OMSCOMMAND/RESETPREDICTEDSTABLE?ID=%SELECTCOMPID&INCIDENTNAME=

- **PUSHPREDICTEDSTABLE:** Rolls down a predicted incident and creates Predicted Stable incidents for downstream customer calls and predicted incidents.

Format example:
NETVIEW://%DYN/OMSCOMMAND/PUSHPREDICTEDSTABLE?ID=%SELECTCOMPID&INCIDENTNAME=

- **MOVEINCIDENTUP:** Moves the incident to the next upstream protective device and predicts the device out. This option is available for single customer and predicted incidents.

  Format example:
  NETVIEW://%DYN/OMSCOMMAND/MOVEINCIDENTUP?ID=%SELECTCOMPID&INCIDENTNAME=

- **ASSOCIATE:** Associates incidents with another incident.

  Format example:
  NETVIEW://%DYN/OMSCOMMAND/ASSOCIATE/MERGEINCIDENT?ID=%SELECTCOMPID&INCIDENTNAME=

- **ASSESSMENTLOCATION:** Deletes a damage assessment location.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/DELETE/ASSESSMENTLOCATION?ID=%SELECTCOMPID

- **ASSIGNCREW:** Assigns crews to an incident.

  Format example:
  NETVIEW://%DYN/OMSCOMMAND/ASSIGNCREW/?ID=%SELECTCOMPID&NAME=%SELECTCOMPNAME

- **ASSIGNSPECIFICCREW:** Assigns a specific crew type to an incident via WFM. In the command, a CREWTYPE value corresponds to a CrewType key from the coded translation file.

  Format example:
  NETVIEW://%DYN/OMSCOMMAND/ASSIGNSPECIFICCREW/?ID=%SELECTCOMPID&CREWTYPE=2015

- **SPLITINCIDENTUI:** Creates a historical incident for a device.

  Format example: NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/SPLITINCIDENTUI?ID=%SELECTCOMPID&PHASEA=%COMPPHASEA&PHASEB=%COMPPHASEB&PHASEC=%COMPPHASEC&MEASNAME=%MEASNAMEFOROBJECT

- **HISTORYCORRECTIONDISPLAY:** Commands for a historical incident.

  Format example:
  NETVIEW://PLUGIN.%DYN/HISTORYCORRECTIONDISPLAY/INCIDENT/UPDATE?ID=%ID&ISOUTAGE=%ISOUTAGE

- **INCIDENT:** Opens the Update Incident display for an incident.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/INCIDENT?ID=%ID&ISOUTAGE=%ISOUTA

GE&VIEWPORTNAME=%VIEWPORTNAME

- **ADDUNASSOCIATEDCALLPLUGINFORM:** Opens the Add Unassociated Call display.

  Format example: NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/ADDUNASSOCIATEDCALLPLUGINFORM?X=%X&Y=%Y&ID=%ID

- **ADDCALLPLUGINFORM:** Opens the Add Unassociated Call display.

  Format example: NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/ADDCALLPLUGINFORM?ID=%SELECTCOMPID

- **FOLLOWUPPLUGINFORM:** Opens the Follow-Up Work Order display.

  Format example: NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/FOLLOWUPPLUGINFORM?ID=%SELECTCOMPID

- **CONVERTINCIDENTPLUGINPOPUP:** Opens the Convert Incident display to convert a non-outage incident into an outage incident or vice versa.

  Format example: FGCMD DISPLAY/URL "NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/CONVERTINCIDENTPLUGINPOPUP?ID=%SELECTCOMPID"

- **DAMAGEASSESSMENTSTATUSFORM:** Opens the Add Damage Assessment Location dialog box to add or edit a damage assessment location report.

  Format example: NETVIEW://PLUGIN.%DYN/NETADDINS/DAMAGEASSESSMENTSTATUSFORM?MODE=ADD&X=%X&Y=%Y

- **CUSTOMERCALLSFORM:** Opens the Customer Calls dialog box, which shows call history for a customer.

  Format example: NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/CUSTOMERCALLSFORM?ID=%SELECTCOMPID

- **MERGEDOWNSTREAMINCIDENTS:** Opens the Merge Downstream Incidents pop-up display to merge downstream confirmed, manually confirmed, predicted, and single customer incidents to a confirmed incident. Applies to downstream incidents on all phases.

  Format example: FGCMD DISPLAY/URL "NETVIEW://PLUGIN.%DYN/NETADDINS/MERGEDOWNSTREAMINCIDENTS?TYPE=INIT&ID=%SELECTCOMPID"

- **AREANOTE:** Opens the Operating Center Area Note display to create or update an area note.

  Format example: NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/AREANOTE?ID=%ASSOCIATIONNAME

- **ACKNOWLEDGE:** Acknowledges (recognizes) the incident.

Format example:
NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/ACKNOWLEDGE/INCIDENTACK?NAME=%NAME

- **CANCELMOVE:** Cancels the previous change of a customer’s spatial location.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/COMMAND/CANCELMOVE/CUSTOMER?ID=%CUSTOMERKEY

- **CREWUPDATEFORM:** Opens the Crew Update display to create or update a temporary crew.

  Format example: NETVIEW://PLUGIN.%DYN/NETADDINS/CREWUPDATEFORM?ID=%ID

- **UPDATECREW:** Updates the crew status.

  Format example 1:
  UPDATECREW?ID=%ID&CREWSTATUSCODE=5&CURRENTINCIDENTACTION=ASSIGN

  Format example 2:
  NETVIEW://PLUGIN.RT/OMSDISPLAYPLUGIN/COMMAND/UPDATE/CREW?ID=%SELECTCOMPID&CREWSTATUSCODE=6

  Format example 3: UPDATECREW?ID=%ID&CREWAREA=NEWCREWAREA

- **SETETR:** Opens the Set ETR display, which allows you to set ETRs for the selected incidents.

  Format example: NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/SETETR?ID=%ID

- **CHANGEETR:** Opens the Change ETR display, which allows you to change ETRs for the selected incidents.

  Format example: NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/CHANGEETR?ID=%ID

- **CANCELINCIDENTS:** Opens the Cancel Incidents display, which allows you to cancel the selected incidents.

  Format example: NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/CANCELINCIDENTS?ID=%ID

- **INCIDENTCUSTOMFIELDSFORM:** Opens the Incident’s Custom Fields display, which allows you to enter, view, and update custom incident properties.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/INCIDENTCUSTOMFIELDSFORM?ID=%ID&NAME=%NAME

- **HISTORYCORRECTIONCUSTOMFIELDSFORM:** Opens the Incident’s Custom Fields display, which allows you to enter, view, and update custom incident properties for historical incidents.

  Format example:
  NETVIEW://PLUGIN.%DYN/OMSDISPLAYPLUGIN/INCIDENTCUSTOMFIELDSFORM?ID=%ID&NAME=%NAME

- **POPUPCUSTOMERLIST:** Opens a Traced Customer List (Create Planned Outage) display for downstream customers.

Format example:
NETVIEW://%DYN/POPUPCUSTOMERLIST/LN/%TOPOSUBNAME/%ID/TRACEDOWN?PHASEA=TRUE&PHASEB=FALSE&PHASEC=FALSE&PHASESSPECIFIED

URL example:
NETVIEW://RT/POPUPCUSTOMERLIST/LN/RAVEN/141631273/TRACEDOWN?PHASEA=TRUE&PHASEB=FALSE&PHASEC=FALSE&PHASESSPECIFIED

- **DEASSIGNCREWS:** De-assigns crews from the selected incident.

  Format example:
  NETVIEW://PLUGIN.%DYN/NETADDINS/MERGEDOWNSTREAMINCIDENTS?TYPE=DEASSIGNCREWS&ID=%ID&CREWS=%CREWRADIONUMBER

16.2. Basic Parameters

- **FUTURE:** Shows specified types of future equipment in a geographic display (SHOWFUTUREHALOS, SHOWFUTUREREMOVEHALOS, SHOWREMOVEDHALOS, or ALL).

  Format example:
  NETVIEW://%dyn/UPDATE?FUTURE=(SHOWFUTUREHALOS,SHOWFUTUREREMOVEHALOS,SHOWREMOVEDHALOS)

- **SHOWFUTUREHALOS:** Shows future equipment.
- **SHOWFUTUREREMOVEHALOS:** Shows future remove equipment.
- **SHOWREMOVEDHALOS:** Shows removed equipment.
- **FORCENONRECONFIGON:** Shows non-reconfigurable sections.

  Format example: NETVIEW://%DYN/UPDATE?FORCENONRECONFIGON

- **FORCENONRECONFIGOFF:** Hides non-reconfigurable sections.
- **FORCENONMAIN:** Toggles the display of laterals.
- **FORCENONMAINAUTO:** Displays laterals based on the viewer configuration zoom level.
- **FORCENONMAINON:** Shows laterals at all zoom levels.
- **FORCENONMAINOFF:** Hides laterals at all zoom levels.
- **LABELS:** Toggles the display of labels.
- **SHOWLINELABELS:** Toggles the display of line labels. Applicable only for a loop schematic view.
- **AUTOLABELS:** Displays labels based on the viewer configuration zoom level.
- **SHOWLABELS:** Shows labels at all zoom levels.
- **HIDELABELS:** Hides labels at all zoom levels.
- **SHOWVOLTAGEHALOS:** Shows voltage halos.

- **HIDEVOLTAGEHALOS:** Hides voltage halos.
- **SHOWOVERLOADHALOS:** Shows overload halos.
- **HIDEOVERLOADHALOS:** Hides overload halos.
- **SHOWCAPACITYMARGINHALOS:** Shows capacity margin halos.
- **HIDECAPACITYMARGINHALOS:** Hides capacity margin halos.
- **SHOWPOTENTIALFAULTHALOS:** Shows potential fault halos.
- **HIDEPOTENTIALFAULTHALOS:** Hides potential fault halos.
- **SHOWFAULTINDICATORHALOS:** Shows fault indicator halos.
- **HIDEFAULTINDICATORHALOS:** Hides fault indicator halos.
- **FLTINDHALOS:** Shows the specified type of fault indicator halos (NONE, MANUAL, TELEMETERED, or BOTH).

  URL example: UPDATE?FLTINDHALOS=(BOTH)

- **CUSTOMERLAYERS:** Shows the specified type of customers (ALL, NONE, SHOWCRITICALCUST, SHOWNORMALCUST, SHOWNOTCONNECTEDCUST, SHOWCALLINGCUST, SHOWPRIORITYCALLINGCUST, or SHOWACTIVECUST)

  URL example: UPDATE?CUSTOMERLAYERS=(SHOWACTIVECUST, SHOWCRITICALCUST)

- **CREWLAYERS:** Shows the specified crews by status (ALL, NONE, SHOWASSIGNEDCREWS, SHOWAVAILABLECREWS, or HIDEUNAVAILABLECREWS).

  URL example: UPDATE?CREWLAYERS=(SHOWAVAILABLECREWS)

- **CREWTYPELAYERS:** Shows the specified crew types, as specified in the Crew Type coded translation entry (for example, ALL, NONE, 0, 1, Foreman, or First Responder).

  URL example: UPDATE?CREWTYPELAYERS=(0,1)

- **VEHICLELAYERS:** Displays vehicles (ALL, NONE).

  URL example: UPDATE?VEHICLELAYERS=(NONE)

- **VEHICLETYPELAYERS:** Shows the specified vehicles types, as specified in the Vehicle Type coded translation entry (for example, NONE, 0, 1, A, B, Pickup, or Bucket Truck).

  URL example: UPDATE?VEHICLETYPELAYERS=(1,2)

- **WMSLAYERS:** Shows the specified web map layers (Aerial, Street, or null).

  URL example: UPDATE?WMSLAYERS=Aerial

  URL example: UPDATE?WMSLAYERS=

- **HIDEVOLT:** Hides the specified voltage levels.

  URL example: UPDATE?HIDEVOLT=(230,138)

- **HIDEPHASES:** Hides the specified phases.

URL example: UPDATE?HIDEPHASES=(A,B)

- **SHOWANNOTATIONS:** Shows the annotation layer.
- **HIDEANNOTATIONS:** Hides the annotation layer.
- **SHOWANNOTATIONS&SHOWANNO:** Shows specified types of annotations. Annotations are referenced by an index as defined in configuration files.

  URL example: UPDATE?SHOWANNOTATIONS&SHOWANNO=(1,2,4,6)

- **SHOWSTRUCTURESS:** Shows the structure layer.
- **HIDESTRUCTURESS:** Hides the structure layer.
- **SHOWSTRUCTURESS&SHOWSTRUCT:** Shows specified types of structures. Structures are referenced by an index as defined in configuration files.

  URL example: UPDATE?SHOWSTRUCTURESS&SHOWSTRUCT=(1,2)

- **SHOWOVERHEAD:** Shows the overhead equipment.

  URL example: UPDATE?SHOWOVERHEAD&HIDEUNDERGROUND

- **HIDEOVERHEAD:** Hides the overhead equipment.
- **SHOWUNDERGROUND:** Shows the underground equipment.

  URL example: UPDATE?HIDEOVERHEAD&SHOWUNDERGROUND

- **HIDEUNDERGROUND:** Hides the underground equipment.
- **SHOWGEOPHASECOLOR:** Shows lines with phase colors.
- **HIDEGEOPHASECOLOR:** Shows lines with colors of the feeders they belong to.
- **SHOWMEASUREMENTVL:** Shows SCADA measurement value placements.
- **HIDEMEASUREMENTVL:** Hides SCADA measurement value placements.
- **SHOWCALCULATEDVL:** Shows DNAF measurement value placements.
- **HIDECALCULATEDVL:** Hides DNAF measurement value placements.
- **SHOWINCID:** Shows incident symbols on the geographic view.
- **HIDEINCID:** Hides incident symbols on the geographic view.
- **FONTZOOM:** Sets the label font zoom level.

  URL example: UPDATE?FONTZOOM=0.8

- **BACKGROUNDLAYERLIST:** Updates the geographic view with background layers (roads shape, water shape, building footprints shape, boundaries shape, or labels).

  URL example: BKGD?BACKGROUNDLAYERLIST=(ROADS SHAPE)

- **CLEARMARKSMANUALLY:** If this parameter is added to the Add Marked devices to SWO or Add Marked Devices to Daily Log command, the marked devices remain marked after adding to a switching order or a daily log.

Format example:
NETVIEW://%DYN/SWITCHORDER/MARKEDADD?MARKINDEX=1&SILENT=FALSE&CLEARMARKSM ANUALLY

- **KEEPZOOMSETTINGS:** Keeps the existing zoom settings when opening another display or navigating to another view.
- **KEEPLAYERSETTINGS:** Keeps the existing setting for displaying customers when opening another display or navigating to another view.
- **KEEPBACKGROUNDLAYERSETTINGS:** Keeps the existing settings for displaying geographic background layers when opening another display or navigating to another view.
- **KEEPHALOSETTINGS:** Keeps the existing setting for displaying halos when opening another display or navigating to another view.
- **PIPZOOM:** Specifies the picture-in-picture view zoom level.
- **PIPMODE:** Specifies the picture-in-picture view mode.
- **PIPX:** X position of the selection in the picture-in-picture view.
- **PIPY:** Y position of the selection in the picture-in-picture view.

  Format example:
  NETVIEW://%DYN/PIPON?PIPZOOM=40&PIPMODE=2&PIPX=%XINT&PIPY=%YINT 0683224135

  URL example: NETVIEW://rt/PIPON?PIPZOOM=40&PIPMODE=2&PIPX=915987&PIPY=798487

- **NOGEOADJACENCY:** Disables the Geo Adjacent feature.

  URL example: NETVIEW:// %DYN/VIEW/DETAIL/RAVEN&NOGEOADJACENCY

- **X:** X position of the selection.
- **Y:** Y position of the selection.

  Format example:
  NETVIEW://%DYN/VIEW/%GEOM?LOCATEOMSCOMP=%ID&X=%X&Y=%Y&CUSTO MERLAYERS=(ALL)&ZOOM=4

  URL example: NETVIEW://RT/VIEW/DETAIL/RAVEN?ZOOM=1.6&X=937937&Y=811449

- **ADJACENT:** Specifies the level of adjacency to show.

  Format example:
  NETVIEW://%DYN/VIEW/%GEO/%SUB?LOCATETAG=%ID&FORCENONMAINON&ZOOM =20&ADJACENT=2

  URL example:
  NETVIEW://RT/VIEW/DETAIL/RAVEN?ZOOM=1.6&X=937937&Y=811449&ADJACENT=2

- **LOCATE:** Locates and selects a component on the network view.

  Format example:

NETVIEW://%DYN/VIEW/%GEONAME/%STATION?LOCATE=%DEVICEID&ZOOM=10&FORCENONMAINON&CUSTOMERLAYERS=(SHOWACTIVECUST)

URL example: NETVIEW://RT/VIEW/DETAIL/RAVEN?LOCATE=22233938

- **LOCATEOMSCOMP:** Locates and selects an OMS component, such as an incident, unassociated call, or planned outage.

  Format example:
  NETVIEW://%DYN/VIEW/%GEOM?LOCATEOMSCOMP=%ID&X=%X&Y=%Y&CUSTOMERLAYERS=(ALL)&ZOOM=4

- **LOCATECUSTOMER:** Locates and selects a customer on the network view.

  Format example:
  NETVIEW://%DYN/VIEW/%GEOM?LOCATECUSTOMER=%CUSTOMERKEY&ZOOM=20&X=%X&Y=%Y&CUSTOMERLAYERS=(ALL)

  URL example:
  NETVIEW://RT/VIEW/%GEOM?LOCATECUSTOMER=555190559&ZOOM=20&X=932905&Y=831654&CUSTOMERLAYERS=(ALL)

- **LOCATEMOBILESUB:** Locates a mobile substation.

  Format example:
  NETVIEW://%DYN/VIEW/%GEOM?LOCATEMOBILESUB=%StationName&ZOOM=5&X=%X&Y=%Y

- **LOCATEDEVICE:** Locates a protective device on the network view from an OMS display.

  Format example:
  LOCATEDEVICE=%PROTECTIVEDEVICEID,%ORIGINALDEVICETYPE,%PROTECTIVESTATION

- **Mark:** Marks a device.

  Format example:
  NETVIEW://%DYN/VIEWNEW/%GEOM/%SUBNAME?MARK=%SUBNAME|%OPTMOVDEVID

- **SHOWPIP:** Shows a picture-in-picture view.

  Format example:
  NETVIEW://%DYN/VIEW/%GEOM/%TOPOSUBNAME?SHOWPIP&PIPPOSITION=5&X=%XINT&Y=%YINT

- **FILTERFEEDER:** Filters the display by the specific feeder.

  Format example:
  NETVIEW://%DYN/UPDATE?FILTERFEEDER=%DYNSTATIONNAME/%DYNFEEDERNAME,
  %DYNSTATIONNAME2/%DYNFEEDERNAME2

  URL example: NETVIEW://RT/UPDATE?FILTERFEEDER=

- **SHOWHALOS:** Shows the specified halo layers (SHOWVOLTAGEHALOS, SHOWOVERLOADHALOS,

SHOWCAPACITYMARGINHALOS, SHOWPOTENTIALFAULTHALOS, SHOWFAULTINDICATORHALOS, or null).

Format example: NETVIEW://%DYN/UPDATE?SHOWHALOS=(SHOWVOLTAGEHALOS,SHOWOVERLOADHALOS)

- **LABELPOSITION:** Specifies the position of the temporary device label (N, NE, E, SE, S, SW, W, or NW).

  Format example: NETVIEW://%DYN/MOVELABEL/JUMPER/%TOPOSUBNAME/%ID?LABELPOSITION=N

- **PIPPOSITION:** Specifies the position of a picture-in-picture view display.

  Format example: NETVIEW://%DYN/VIEW/%GEOM/%TOPOSUBNAME?SHOWPIP&PIPPOSITION=5&X=%XINT&Y=%YINT

  URL example: NETVIEW://RT/PIPON?PIPPOSITION=5

- **ZOOMBY:** Increases or decreases the current zoom level by the specified value.

  Format example: NETVIEW://%dyn/UPDATE?ZOOMBY=1.25

- **EVALUATECBO:** Enables the Check Before Operate functionality for SCADA switching devices and shows the Evaluate CBO display, where you can review the notifications and proceed to the SCADA Control display.

  Format example: NETVIEW://%DYN/SET/SW/%MEASSTATIONFOROBJECT/%MEASNAMEFOROBJECT/SCADA?PICTURE=fantasyisland_internals_scada_control_popup.fgpopup&LOCATION=1&FOLLOWMONITOR=1&EVALUATECBO

- **CHECKGROUND:** Performs the temporary ground check for SCADA devices before calling the SCADA Control display.

  Format example: NETVIEW://%DYN/SET/SW/%MEASSTATIONFOROBJECT/%MEASNAMEFOROBJECT/SCADA?PICTURE=fantasyisland_internals_scada_control_popup.fgpopup&LOCATION=1&FOLLOWMONITOR=1&CHECKGROUND=1&EVALUATECBO

- **FOLLOWMONITOR:** When several workstation screens are used, this parameter enables the SCADA Control display to appear in the viewport where it was called from.
- **LOCATION:** Specifies whether the SCADA Control display opens in the same viewport (LOCATION=1) or in a new viewport (LOCATION=6).
- **SCALETOFIT:** Scales the display to show the entire network or schematic view. Note that zooming in or out is disabled in this mode.

  URL example: NETVIEW://RT/UPDATE?SCALETOFIT=TRUE

- **NOSCALETOFIT:** Disables scaling the display to show the entire network or schematic view.

URL example: NETVIEW://RT/UPDATE?NOSCALETOFIT

- **PHASESSPECIFIED:** Specifies whether a Netview command applies to specific phases.

  Format example: NETVIEW://%DYN/TRACEDOWN/LN/%TOPOSUBNAME/%ID?PHASEA=TRUE&PHASEB=FALSE&PHASEC=FALSE&PHASESSPECIFIED

  URL example: NETVIEW://RT/TRACEDOWN/LN/RAVEN/141631273?PHASEA=TRUE&PHASEB=FALSE&PHASEC=FALSE&PHASESSPECIFIED

16.3. Environment Fields

Environment fields are used with commands and parameters when it is necessary to use specific data, such as current cursor position (x, y) or the ID of the nearest switch.

For example, the command for adding an unassociated call from the geographic right-click menu is:

```
NETVIEW://%DYN/SCRIPTADDUNASSOCIATEDCALLEVT?%X\&amp;%Y
```

This command uses three environment fields (%DYN, %X, and %Y) to check for the dynamics, and determine the network position (x, y) where the unassociated call needs to be added.

Environment fields are typically used for commands that are invoked from the Geographic_OpenSpace (common) toolbar and the geographic right-click menu. The fields are prefixed with "%"to indicate that a substitution is needed. The substituted value is always a string.

The following table provides the complete list of environment fields.

Environment Field	Possible Values
%CENTERX	Invariant culture X. Represents coordinate at the center of the screen.
%CENTERY	Invariant culture Y. Represents the coordinate at the center of the screen.
%CLOSESTOBJECTID	ID of the closest device.
%CLOSESTCOMPSERVCTR	Name of the service center of the closest device.
%CLOSESTOBJECTSTN	Name of closest station.
%CLOSESTOBJECTDMSSTN	Name of closed DMS station.
%CUSTOMERSHOW	0: No customers visible 1: Only critical customers visible 2: Only connected customers visible 3: Only unconnected customers visible 4: Only customers with activity visible 5: All customers visible

Environment Field	Possible Values
%DMSOBJECTTYPE	dmsNone
%DNOMOBJECTTYPE	DNOM object type
%DYN	Names of dynamics stored in `netconf.xml`, such as. RT, ST, or SIM.
%GEOM	Current geometry name, such as DETAIL or GEOGRAPHIC. **Note** Similarly, the %GEONAME DNOM query field can be used.
%HIDECREWS	0: Crews are visible 1: Crews are hidden
%HIDEVEHICLES	0: Vehicles are visible 1: Vehicles are hidden
%ISGEOGRAPHIC	True or false expressed in current culture. Based on the geometry name marked as geographic in the converter.
%ISNAVOV	True or false expressed in current culture. True if the display is NETVIEW://.../NAVOV
%ISPIP	True or false. Based on selection made inside a picture-in-picture window.
%ISREALTIME	True or false expressed in current culture. Based on the current mode.
%ISSIMULATOR	True or false expressed in current culture. Based on the current dynamics.
%ISSTATIONINTERNAL	True or false expressed in current culture. Based on the geometry name marked as Station Internal in the Converter.
%ISLOOPSCHEMATIC	True or false expressed in current culture. Based on the network view type.
%ISOUTAGE	True or false expressed in current culture. Based on the incident type (outage or non-outage).
%ISNONOUTAGE	True or false expressed in current culture. Based on the incident type (outage or non-outage).
%CANCONVERTINCIDENT	True or false expressed in current culture. Based on whether the incident can be converted from outage to non-outage or vice-versa.
%ISRECORDING	True or false expressed in current culture. Based on whether the Switching Order recording mode is on.

Environment Field	Possible Values
%LABELSTATUS	0: Default (display based on zoom level) 1: Forced On 2: Force Off
%LATERALSOFF	0: Feeder backbone is not visible 1: Feeder backbone is visible
%LATERALSON	0: Laterals are currently not visible 1: Laterals are currently visible
%MOUSEX	Invariant culture X. Represents the mouse position in device units.
%MOUSEY	Invariant culture Y. Represents the mouse position in device units.
%PICTUREPATH	Path to pictures (tabulars) as defined in `netconf.xml.`
%RECONFIGONLY	0: All networks are displayed. 1: Only reconfigurable networks are displayed.
%SHOWFAULTINDICATORHALOS	0: The fault indicator halos are not displayed. 1: The fault indicator halos are displayed.
%SHOWFUTURE	0: The future equipment is not displayed. 1: The future equipment is displayed.
%SHOWOVERLADHALOS	0: The overload halos are not displayed. 1: The overload halos are displayed.
%SHOWPIP	0: The picture-in-picture window is not displayed. 1: The picture-in-picture window is displayed.
%SHOWPOTENTIALFAULTHALOS	0: The fault location results are not displayed. 1: The fault location results are displayed.
%SHOWVOLTAGEHALOS	0: The voltage halos are not displayed. 1: The voltage halos are displayed.
%SHOWOVERLOADHALOS	0: The overload halos are not displayed. 1: The overload halos are displayed.
%VAULTID	ID of the vault.
%VIEWPORTNAME	Name of the viewport in Browser, such as Station or NavOverview.
%X	Invariant culture X. Represents the position of the selection in world coordinate units.
%XINT	Invariant culture X (converted to integer first). Represents the position of the selection in world coordinate units.

Environment Field	Possible Values
%Y	Invariant culture Y. Represents the position of the selection in world coordinate units.
%YINT	Invariant culture Y (converted to integer first). Represents the position of the selection in world coordinate units.
%ZOOM	Invariant culture zoom.
%ANNX	Represents the X position of the annotation in world coordinate units.
%ANNY	Represents the Y position of the annotation in world coordinate units.
%PIPPOSITION	Specifies the position of a picture-in-picture view display. 0: Docked in the upper-left corner of the display. 1: In the upper-left corner of the display. 2: In the upper-right corner of the display. 3: In the lower-right corner of the display. 4: In the lower-left corner of the display. 5: Docked in the right half of the display.
%ID	ID of the device.
%NAME	Name of the device.
%STATION	Name of the station.
%SWITCHTYPE	Switch type.
%SELECTSTN	Selected station name.
%SELECTCOMPID	Selected component ID.
%SELECTCOMPNAME	Selected component name.
%SELECTQTYPE	Select query type.
%SELECTX	Invariant culture X. Represents the position of the selection in world coordinate units.
%SELECTY	Invariant culture Y. Represents the position of the selection in world coordinate units.
%SELECTDNOMTYPE	Selected component DNOM type.
%SELECTTYPE	Selected component type.
%ALTERNATEENDTIME	Alternate end time of the planned outage.
%ALTERNATESTARTTIME	Alternate start time of the planned outage.
%CREATEDTIME	Created time of the planned outage.
%CREATORCOMMENTS	Creator comments of the planned outage.
%CUSTOMERACCOUNTLIST	Customer account list of the planned outage.
%DEVICEID	Device ID of the planned outage.
%DEVICENAME	Device name of the planned outage.

Environment Field	Possible Values
%OUTAGEDEVICETYPE	Outage device type.
%OUTAGEMESSAGE	Outage message.
%PRIMARYENDTIME	Primary end time of the planned outage.
%PRIMARYSTARTTIME	Primary start time of the planned outage.
%REPRESENTATIVENAME	Representative name that is associated with the planned outage.
%REPRESENTATIVENUMBER	Representative number that is associated with the planned outage.
%SWITCHORDERID	Names of switching orders that are associated with the incident (or planned outage).
%SWITCHORDERNAME	IDs of switching orders that are associated with the incident (or planned outage).
%USEALTERNATEDATE	True or false expressed in current culture. Based on whether the alternate date is used for the planned outage.
%ISPREDICTEDSTABLE	True or false expressed in current culture. Based on whether the incident state is predicted stable.
%ISPREDICTED	True or false expressed in current culture. Based on whether the incident state is predicted.
%ISCONFIRMED	True or false expressed in current culture. Based on whether the incident type is Confirmed.
%ISTYPEPREDICTED	True or false expressed in current culture. Based on whether the incident type is Predicted.
%ISSINGLECUSTOMER	True or false expressed in current culture. Based on whether the incident type is Single Customer.
%ISDEENERGIZED	True or false expressed in current culture. Based on whether the component is de-energized.
%ISDISCONNECTED	True or false expressed in current culture. Based on whether the component is disconnected.
%AMI	True or false expressed in current culture. Based on whether the customer has an AMI meter.
%MEASNAMEFOROBJECT	Composite ID of a SCADA Point of the device (typically status, but could be analog or control).
%HASJUMPEREND	True or false expressed in current culture. Based on whether the line has the first jumper end.
%MULTILOOPSCHEMATIC	True or false expressed in current culture. Based on the network view type.
%ISUSINGEMPSCADA	True or false expressed in current culture. Based on whether the Scada component is enabled.

Environment Field	Possible Values
%RESOLVETAG	Tag type.

16.4. DNOM Query Fields

DNOM query fields are used with commands and parameters when it is necessary to use specific data, such as substation name and component ID.

For example, the command for opening an Operator Attributes panel for a component is:

```
NETVIEW://%DYN/ATTRIBUTES/OPERATORATTRIBUTE/%DNOMTABNAME//%TOPOSUBNAME/%ID
```

This command uses three DNOM query fields (%DNOMTABNAME, %TOPOSUBNAME, and %ID) to return the DNOM table name of the component, substation name, and component ID.

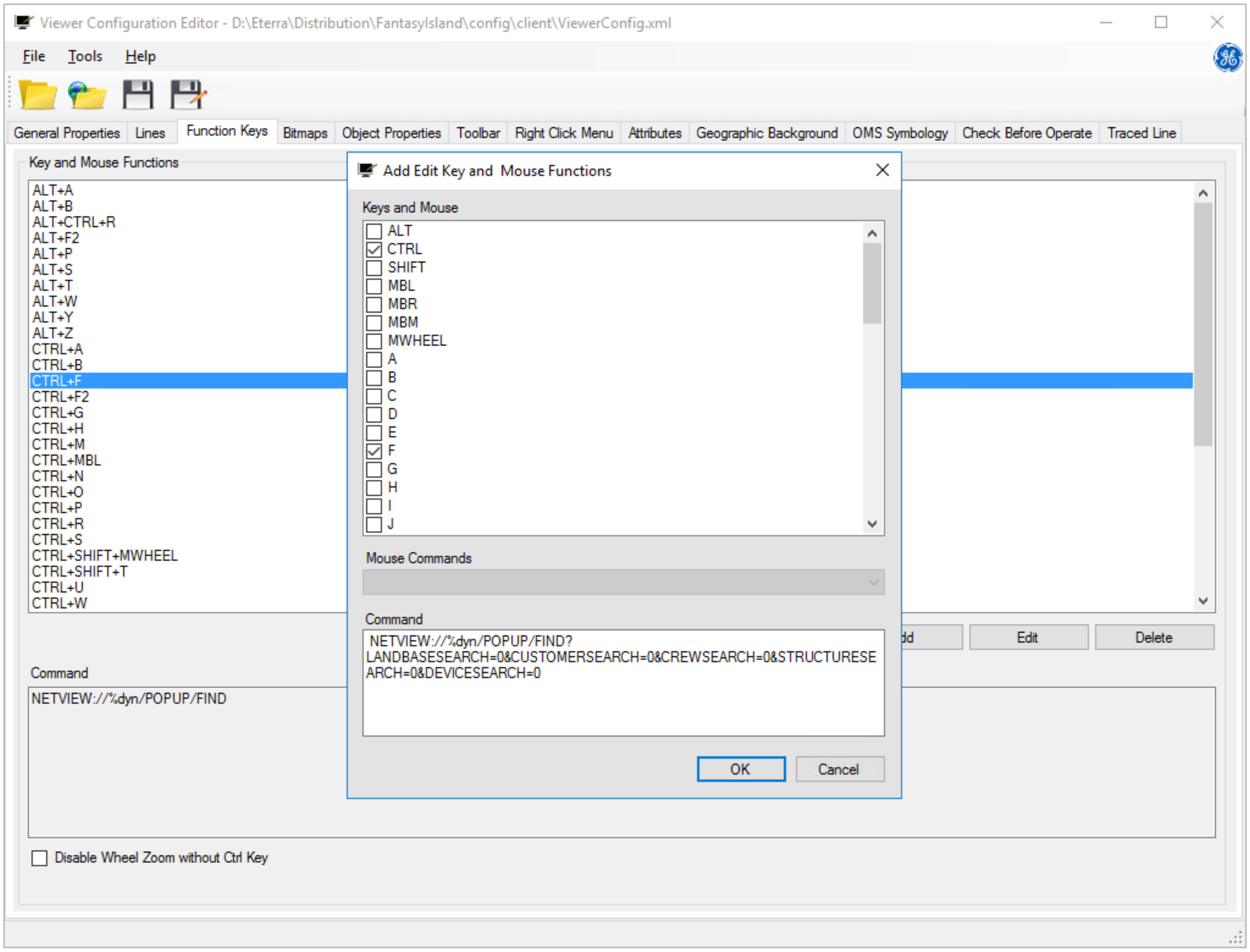
Similar to environment fields, DNOM query fields are prefixed with "%"to indicate that a substitution is needed. The substituted value is always a string.

For a list of DNOM query fields, refer to the *ADMS Modeling Reference Tables*.

16.5. Find Popup Parameters

This section describes command line parameters to configure the Find pop-up display. You can use these parameters to modify the default command for calling the Find pop-up display that is bound to the Ctrl+F shortcut on the Function Keys tab of the Viewer Configuration Editor. For example, use the following command to show only the Analyst Search tab on the Find pop-up display:

```
NETVIEW://%dyn/POPUP/FIND?LANDBASESEARCH=0&CUSTOMERSEARCH=0&CREWSEARCH=0&STRUCTURESEARCH=0&DEVICESEARC
H=0
```



The following command line parameters can configure the Find pop-up display:

- **DEVICESEARCH:** Controls the visibility of the Device Search tab.
- **LANDBASESEARCH:** Controls the visibility of the Landbase Search tab.
- **CUSTOMERSEARCH:** Controls the visibility of the Customer Search tab.
- **CREWSEARCH:** Controls the visibility of the Crew Search tab.
- **STRUCTURESEARCH:** Controls the visibility of the Structure Search tab.
- **CHECKANALYST:** Specifies that the Analyst Search tab is enabled. Deprecated. The Analyst Search tab cannot be disabled using the command string.
- **SEARCHIN:** Specifies whether to use global or current view search on the Analyst Search tab.
- **SEARCHFOR:** Specifies the device type to search for.
- **CHECKSTATIONFLAG:** Controls the visibility of the Station field in the Device list on the Device Search tab.
- **CHECKSTATIONANALYSTFLAG:** Controls the visibility of the Station field in the Device list on the Analyst Search tab.
- **DEVICESEARCHBUTTONSSET:** Specifies the visibility of buttons on the Device Search tab.

- **SEARCHTAB:** Specifies the tab that opens when the Find pop-up opens. (0=Not Specified, 1=Device Search, 2=Customer Search, 3=Landbase Search, 4=Analyst Search, 5=Crew Search)
- **DEVICESEARCHDEVICETYPE:** Value for the Device Type list on the Device Search tab. The value is the Coded Translation Editor device type key (integer value).
- **LANDBASESEARCHBKLAYER:** Value for the Background Label list on the Landbase Search tab. The layer name is a string.
- **ANALYSTSEARCHDEVICETYPE:** Value for the Device Type list on the Analyst Search tab. The value is the Coded Translation Editor device type key (integer value).
- **CREWSEARCHCREWTYPE:** Value for the Crew Type list on the Crew Search tab. The name is a string (Foreman or First Responder).
- **SEARCHSTRING:** Search argument. This can be any string.
- **SEARCHTEMPLATE:** Search type (EXACTMATCH, CONTAINS, BEGINSWITH, ENDSWITH, or DYNAMIC).

  When the DYNAMIC search type is used, the wildcard search is enabled. The asterisk symbol ("*") can be added to the search text to substitute the part before or after the search text.
- **STARTIMMEDIATELY:** If True, initiates the search immediately (recommended for using with SEARCHTEMPLATE, SEARCHSTRING, SEARCHTAB, and other commands).
- **PICPATH:** Picture path.
- **BMPPATH:** Picture (.bmp or .png) path.
- **DYNNAME:** Name of the dynamics.
- **GEONAME:** Name of the geographic geometries (by default, DETAIL).
- **USESHAPETILES:** Control using the shape tiles.
- **BACKGROUNDTILEPATH:** Specifies the path to the background tiles.
- **VIEWERCONFIGFILEPATH:** Specifies the path to the `viewerconfig.xml` file.
- **STREETSUFFIXES:** Specifies the street suffixes.
- **CONNECTED:** Checks the server connection status.